

Merged research report for the Surreal project

# Omnific Groups and Lattices

Algebraic groups, elementary quotients,  
and missing lattice minima over the omnific integers

Built from four manuscripts dated September 23, 2026

## Abstract

We study matrix groups and lattices over the omnific integers  $\mathbf{Oz}$  and the Gaussian omnific integers  $\mathbf{Oz}[i]$ . Three phenomena are treated, each in its own part and under its own hypotheses.

*Algebraic groups.* A real algebraic group acquires no nonconstant points over the ring of surreal normal forms with nonnegative exponents exactly when it has no nonidentity real unipotent element, equivalently when its identity component is reductive with compact real derived group. For a closed subgroup scheme of  $\mathrm{GL}_N$  over  $\mathbb{Z}$  this gives an exact dichotomy: either every omnific point is an ordinary integral point, or the constant-term kernel contains an injective copy of the additive class of purely infinite normal forms. Over the Gaussian omnific integers the criterion is that the complex identity component be a torus. This answers, for such group schemes, a question on rigid affine varieties left open in the omnific Diophantine report.

*Elementary groups.* For  $n \geq 3$  the kernel of constant-term reduction  $E_n(\mathbf{Oz}) \rightarrow \mathrm{SL}_n(\mathbb{Z})$ , and its Gaussian analogue, has no nontrivial homomorphism to any set-sized group, so  $\mathrm{SL}_n(\mathbb{Z})$ , respectively  $\mathrm{SL}_n(\mathbb{Z}[i])$ , is the universal set-sized quotient; this strengthens the universal ring-quotient theorem of the sibling report to arbitrary abstract group maps. Two independent proofs are given, and a third that rebuilds a ring from the root maps and so derives the group theorem from the ring theorem. The kernel is torsion-free and centerless, has no nontrivial set-sized normal subgroup, and its nonidentity conjugacy classes are proper classes; the Steinberg groups have the analogous universal quotient; and in the set-sized rings of supports of size below a regular uncountable  $\kappa$  every nonidentity kernel element has least detecting group of cardinality exactly  $\kappa^{<\kappa}$ . In rank two the answer reverses:  $E_2$  is an amalgam, its kernel is a free product of copies of the purely infinite ideal, and one fixed countable group separates all elements, while finite groups see nothing of the kernel. Explicit non-elementary  $2 \times 2$  unipotents become commutators after one stabilization.

*Lattices.* For finite sums of ordinary positive-semidefinite forms with successively infinitesimal weights we give exact criteria for shortest nonzero vectors, shortest unimodular vectors and closest points to every surreal target. When a shortest vector fails to exist the energy values have no surreal infimum, their lower bounds are an explicit cut, and one monomial vector improves every set-sized family at once; usual Gauss and LLL reduced bases then do not exist.

This report merges four independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source's limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 06 (the base text) and 10, pinned to repository commit 220784a, and 07, pinned to fb5c4b5; manuscript 11, added later and pinned to 9693b28; their titles are in [subsection 1.2](#). Results already printed elsewhere in the collection are cited, not reprinted. The provenance of every section is recorded in [Appendix A](#); non-claims are collected in [Appendix B](#). Finite checks accompany the report and check finite identities and examples only.

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# 1 Scope, results, and conventions

An omnific integer has an ordinary integer constant coefficient and an arbitrary, possibly transfinite, purely infinite part whose coefficients are arbitrary real numbers. This mixture makes linear algebra over  $\mathbf{Oz}$  quite different both from linear algebra over  $\mathbb{Z}$  and from linear algebra over the surreal field. This report studies three linear questions over  $\mathbf{Oz}$  and over the Gaussian omnific integers  $\mathbf{Oz}[i]$ .

- (I) Which algebraic groups acquire new omnific points ([Part I](#))?
- (II) Can the purely infinite directions of an elementary matrix group act on any ordinary set, when no algebraicity or continuity is imposed on the action ([Part II](#))?
- (III) Does a positive-definite form on  $\mathbf{Oz}^n$  have a shortest vector, and what replaces it when it does not ([Part III](#))?

The answers reveal three distinct mechanisms: geometric rigidity controlled by unipotent elements; an abstract representation obstruction controlled by the supply of surreal exponent scales and a third matrix index; and a competition between the constant-term retraction and monomial vectors lying on irrational kernels.

## 1.1 Principal results

Bracketed numbers name the source manuscripts ([subsection 1.2](#)). Here  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $\Pi_k$  is the class of normal forms with coefficients in  $k$  supported on strictly positive exponents,  $\mathcal{B}_k = k \oplus \Pi_k$ , and  $\mathcal{A}_{D,k} = D \oplus \Pi_k$ , so that  $\mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbf{Oz}$  and  $\mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbf{Oz}[i]$  ([subsection 1.4](#)).

- (1) **Real omnific rigidity** [06]. For a real algebraic group  $G$ :  $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$  iff  $G(\mathbb{R})$  has no unipotent element other than 1 iff  $G$  has no subgroup isomorphic to  $\mathbb{G}_a$  iff  $G^0$  is reductive with  $(G^0)^{\text{der}}(\mathbb{R})$  compact. For a closed subgroup scheme  $\mathcal{G} \subseteq \text{GL}_N, \mathbb{Z}$  these are equivalent to  $\mathcal{G}(\mathbf{Oz}) = \mathcal{G}(\mathbb{Z})$ ; otherwise  $(\Pi, +)$  embeds in the kernel of  $\text{ct} : \mathcal{G}(\mathbf{Oz}) \rightarrow \mathcal{G}(\mathbb{Z})$  ([theorem 4.4](#)).
- (2) **Gaussian omnific rigidity** [06]. For a complex algebraic group,  $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$  iff  $G^0$  is a torus; the same holds for  $\mathbb{Z}[i]$ -models and  $\mathbf{Oz}[i]$  ([theorem 4.5](#)). Thus  $\text{SO}_3(\mathbf{Oz})$  has 24 elements while  $\text{SO}_3(\mathbf{Oz}[i])$  contains injective unipotent families. Together, (1) and (2) answer the question `odg:q:affine` of the omnific Diophantine report for closed subgroup schemes of  $\text{GL}_N$  ([remark 4.6](#)).
- (3) **Bounded fibers, unipotent kernels, filtrations** [06]. Bounded real fibers over a torus give rigidity ([theorem 3.3](#)); the constant-term kernel of a unipotent group is  $\exp(\mathfrak{u} \otimes \Pi_k)$  ([proposition 6.1](#));  $\text{UT}_n(\Pi_k)$  has the exact filtration  $[\text{Fil}_r, \text{Fil}_s] = \text{Fil}_{r+s}$  and nontrivial characters ([theorem 6.2](#)).
- (4) **The elementary kernel for  $n \geq 3$**  [06, 07, 11]. Let  $P_n = \ker(\text{ct} : E_n(\mathcal{A}_{D,k}) \rightarrow \text{SL}_n(D))$ . It is generated by the elementary matrices with purely infinite parameters, it is perfect and nontrivial,  $E_n(\mathcal{A}_{D,k}) = P_n \rtimes \text{SL}_n(D)$ , and every homomorphism from  $P_n$  to a set-sized group is trivial. Hence composition with  $\text{ct}$  identifies the homomorphisms from  $E_n(\mathcal{A}_{D,k})$  and from  $\text{SL}_n(D)$  to every set-sized group ([theorem 9.1](#)). Two independent proofs are printed. The theorem implies, for  $k \in \{\mathbb{R}, \mathbb{C}\}$ , the universal ring-quotient theorem `osq:thm:universal` ([remark 9.6](#)); conversely a third proof (11) rebuilds a set-sized ring from the root maps of a group homomorphism and derives the theorem from the ring theorem ([theorem 9.2](#)).

- (5) **A congruence tower and the boundaries** [06, 07].  $P_n$  is not simple: a proper-class chain of congruence subgroups separates its elements, though no set-sized subfamily does (theorem 10.1). The theorem fails in every fixed set-sized Hahn workspace and for finite supports (section 12); an abstract collision criterion isolates the size input (proposition 9.7). On the finite side,  $\mathrm{SL}_n$  of the finite surreals has universal set-sized quotient  $\mathrm{SL}_n(k)$  for  $n \geq 3$  [merge] (proposition 11.1).
- (6) **Rank two** [07].  $E_2(\mathcal{A}_{D,k}) \cong \mathrm{SL}_2(D) *_{C_D} B_D$ ; the kernel  $P_2$  is a free product of copies of  $(\Pi_k, +)$  indexed by  $\mathbb{P}^1(\mathrm{Frac} D)$ ; homomorphisms into the fixed countable group  $\mathrm{SL}_2(\mathrm{Frac}(D)(t))$  separate every finite subset of  $E_2$ ; neither  $E_2$  nor  $P_2$  has a universal set-sized quotient (theorems 13.2, 14.1 and 15.3).
- (7) **The detection trichotomy** [07]. A nonidentity  $g \in E_n$  is detected by a finite group iff  $\mathrm{ct}(g) \neq 1$ ; a nonidentity element of  $P_2$  has least detecting cardinal  $\aleph_0$ ; a nonidentity element of  $P_n$ ,  $n \geq 3$ , has no set-sized detecting target (theorem 16.5).
- (8) **Characters and stabilization** [07].  $E_2(\mathbf{Oz})^{\mathrm{ab}} \cong \mathbb{Z}/12 \oplus \Pi$  (classical, Cohn; Mirzaii–Torres), while every abelian image of  $E_2(\mathbf{Oz}[i])$  kills  $P_2$  (propositions 17.1 and 17.2). A cusp-residue obstruction shows that  $I + \omega^{2a} \begin{pmatrix} \alpha & 1 \\ -\alpha^2 & -\alpha \end{pmatrix}$ ,  $\alpha \in k \setminus \mathrm{Frac} D$ , lies in  $\mathrm{SL}_2 \setminus E_2$ ; after one stabilization it is a commutator of two elements of  $P_3$  (theorems 18.2 and 18.4).
- (9) **Structure of the kernel, Steinberg groups and set-sized fragments** [11]. For  $n \geq 3$  the kernel  $P_n$  is torsion-free (so is the constant-term kernel of  $\mathrm{GL}_n(\mathcal{A}_{D,k})$  for every  $n$ ) and centerless, it has no nontrivial set-sized normal subgroup, every nonidentity element has a proper-class conjugacy class, and finite subgroups of  $E_n(\mathcal{A}_{D,k})$  inject under  $\mathrm{ct}$  (theorems 19.1 and 19.2, corollary 19.3, and proposition 19.4). The Steinberg group  $\mathrm{St}_n(\mathcal{A}_{D,k})$  has universal set-sized quotient  $\mathrm{St}_n(D)$  and a perfect kernel with no set-sized images (theorem 20.2). With the Bass–Milnor–Serre theorem the profinite completion of  $E_n(\mathbf{Oz})$  is  $\mathrm{SL}_n(\hat{\mathbb{Z}})$  (corollary 19.7). In the set-sized ring of normal forms with fewer than  $\kappa$  exponents in  $\bigoplus_{\alpha < \kappa} \mathbb{Q}$ ,  $\kappa$  regular uncountable, the kernel has no nontrivial image of cardinality below  $\kappa$  (theorem 21.3); combined with the sibling report’s field bound, each nonidentity kernel element has least detecting group of cardinality exactly  $\kappa^{<\kappa}$  [merge] (proposition 21.12).
- (10) **Shortest vectors and missing infima** [10]. For a separated positive pencil  $\mathcal{E} = \sum_j \varepsilon_j q_j$  on  $\mathcal{A}_{D,k}^n$ : a shortest nonzero vector exists iff the deepest kernel meets  $D^n$  nontrivially; otherwise the nonnegative lower bounds of the energy values are exactly  $\varepsilon_m(\mathcal{O}_{\mathrm{fin}})_{\geq 0}$  and one monomial vector improves every set-sized family (theorems 24.1 and 24.2).
- (11) **Unimodular vectors and closest points** [10]. A rational-span criterion decides unimodular attainment, and unimodular energies always have a countable cointial family (theorem 26.4); closest points exist for every surreal target iff every prefix kernel is rational, and then the closest points lie among at most  $2^n$  (real) or  $4^n$  (Gaussian) ordinary candidates after removing the target’s purely infinite part (theorems 27.2 and 27.5). The ordinary backbone of these statements is already in the Hahn–Tate report and is cited there (subsection 27.1).
- (12) **Reduction obstructions** [10]. In the nonattainment case no Gauss-reduced basis and no  $\delta$ -LLL basis with ordinary  $\delta \in (1/4, 1]$  exists, and a Gaussian variant holds for  $\delta \in (1/2, 1]$  (theorems 30.3 and 30.4).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. Each source names its own most distinctive contributions and credits its classical inputs (Appendix B); no exhaustive historical-priority claim is made.

## 1.2 Provenance of this report

This report was first built from three manuscripts written independently and dated September 23, 2026. They are items 06, 07 and 10 of the batch placed in commit `f4c9504`; their shipped file prefixes are `06-algebraic-groups-`, `07-matrix-dichotomy-` and `10-shortest-vectors-`. Sources 06 and 10 pin repository commit `220784a` (19 commits before that placement), and source 07 pins `fb5c4b5` (12 commits before it). Both pins precede the written omnific reports: at the pins the directories `docs/surreal/omnific-diophantine-geometry` and `docs/surreal/set-sized-quotients-of-omnific-integers` held only their placed base manuscripts, not the merged reports that now carry the labels cited below (Appendix A.3).

A fourth manuscript was added afterwards. It is item 05 of the batch placed in commit `66d7e55`, shipped under the file prefix `11-matrix-shadows-`, and is called source 11 here, after that prefix, as the first three sources are called after theirs. It is also dated September 23, 2026, and pins `9693b28` (24 commits before its placement), at which this report had already been written. Source 11 was written without knowledge of this report (Appendix A.3); its main theorem is [theorem 9.1](#) again, and its other results are new to the collection.

| Source | Manuscript                                                                    | Principal contribution to this report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06     | <i>Omnific Points of Algebraic Groups</i> (22 pp.)                            | Base text and structure; the real and Gaussian dichotomies and the bounded-fiber principle ( <a href="#">Part I</a> ); orthogonal examples; unipotent kernels and unitriangular filtrations; the $n \geq 3$ theorem with the proof by division by a difference of nearby monomials; normality of the root-generated subgroup for any constant ring; the congruence tower; fixed-workspace and finite-support boundaries.                                                                                                |
| 07     | <i>A Matrix-Size Dichotomy over the Omnific Integers</i> (25 pp.)             | Base of <a href="#">Part II</a> ; the $n \geq 3$ theorem with the proof by rescaling a large set field, including homomorphisms defined only on the kernel; rank two: amalgam, free-product kernel, countable detection; the finite/countable/no-set trichotomy; abelian characters; cusp residues, non-elementary unipotents and stabilization; the abstract collision criterion.                                                                                                                                      |
| 10     | <i>Shortest Vectors and Missing Infima over the Omnific Integers</i> (26 pp.) | All of <a href="#">Part III</a> : constant-term contraction, the shortest-vector dichotomy and exact lower cut, simultaneous improvement, the low-energy flag, the unimodular criterion, closest points for all surreal targets, the Pell comparison, the strict hierarchy, reduction obstructions, and the fixed-Hahn boundary.                                                                                                                                                                                        |
| 11     | <i>Arithmetic Shadows of Omnific Matrix Groups</i> (24 pp.)                   | A third proof of the $n \geq 3$ theorem, by reconstructing a ring from the root maps ( <a href="#">section 9</a> ); torsion-freeness, centerlessness, set-sized normal subgroups and conjugacy classes, finite subgroups, a principal congruence chain and common congruence elements, the profinite completion ( <a href="#">section 19</a> ); Steinberg groups ( <a href="#">section 20</a> ); set-sized fragments, cardinal and action thresholds, and the exact-threshold criterion ( <a href="#">section 21</a> ). |

### Where the merge had to choose.

- The theorem for  $n \geq 3$  was proved by both 06 and 07, independently and by different mechanisms. It is printed once ([theorem 9.1](#)), in 06's generality for the constant ring and 07's intrinsic form for the kernel, with both proofs kept.
- Part II takes 07 as its base, because 07 contains the rank-two theory that 06 left open; 06's normality, splitting, tower and boundary results are inserted where they belong.
- Background on normal forms, units, gaps and divisibility, proved by all four sources and by the omnific reports, is printed once ([section 2](#)).

- Results already printed in other reports are cited rather than reprinted:  $O_n(\mathbf{Oz})$  (`odg:cor:orthogonal`), the anisotropic orthogonal example (covered by `odg:cor:unipotent`), the lattice-projection lemma, the rational-flag minimum criterion, the  $2^n$  parity bound and the Pell example of the Hahn–Tate report ([subsection 27.1](#)). Source proofs are kept only where they prove more or proceed differently.
- Part III was placed with Parts I–II rather than in a separate report: it studies modules and bases over the same rings, its reduction theory changes bases in  $GL_n(\mathbf{Oz})$ , and its ordinary backbone belongs to a surcomplex report about theta series, where its omnific content would be foreign.
- Source 11 proves [theorem 9.1](#) a third time. The theorem is not reprinted: 11 is added to its sources, and 11’s proof, which rebuilds a ring from the root maps and invokes the ring theorem of the sibling report, is kept as a marked third route ([section 9](#)). 11’s finite-residual theorem and finite-support counterexample are credited at [theorem 16.4](#) and [proposition 12.2](#), and its action and representation corollaries at [corollary 9.5](#). Its remaining results form three new sections at the end of Part II ([sections 19 to 21](#)), with the label sub-prefix `ogl:ker:.`
- Source 11 declines to identify the root-generated subgroup with the kernel; [proposition 7.3](#) (06) proves the identity, so 11’s statements about both groups are printed as statements about  $P_n$ . 11’s set-sized ring is the ring  $R_\kappa^{11}$  of the sibling report (`osq:rel:lem:embedding`; the superscript there names that report’s manuscript 11, not source 11 here, and the ring is written  $R_{<\kappa}$  below); its ring-level facts are cited there, and the merge combines them with 11’s exact-threshold criterion ([proposition 21.12](#)).

Results added by the merge carry the tag [merge] and have complete proofs. [Appendix A](#) records, section by section, which source proved what and every symbol renamed from a source.

### 1.3 Class size and foundations

The surreal field  $\mathbf{No}$ , the rings  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , and most groups below are proper classes. Sources 06 and 07 work in Gödel–Bernays set theory with global choice, source 10 in NBG; 07 and 06 note that global choice is never used, and no proof printed here uses it. Source 11 allows Gödel–Bernays class theory with global choice and uses choice only for sets (a successor cardinal above a target, and the equivalence of reverse well ordering with the absence of increasing sequences); its set-sized fragments give ordinary ZFC statements. Each individual surreal, normal-form support, matrix, tuple and finite word is a set. A homomorphism from a class group is a class function satisfying the group identities; statements about all such homomorphisms are read one class function at a time, and no set of all such maps is formed. Residual properties are universally quantified kernel statements. A class quotient may be read as a kernel with its universal property, or through Scott’s trick (06).

Every cardinality contradiction is a contradiction about sets: for a target set  $H$  one produces, inside the source, a set of parameters that cannot inject into  $H$ . Sources 06 and 07 choose a cardinal  $\lambda > |H|$ ; the Hartogs ordinal of  $H$  serves equally and needs no choice, as in the sibling report (`osq:thm:universal`). No cardinal is assigned to a proper class. Source 07 adds that the class statements concern class homomorphisms existing in the chosen foundational setting, not externally described maps in a larger metatheory.

## 1.4 Notation and conventions

**Sign convention.** Exponents are those of Conway normal forms: supports are reverse well ordered and  $\omega^\gamma$  is *infinite* for  $\gamma > 0$ , even for infinitesimal  $\gamma > 0$ . *Purely infinite* refers to support:  $-\omega$  and  $\sqrt{2}\omega$  are purely infinite,  $\omega - 1$  is infinite but not purely infinite, and  $\omega^{-1}$  is not purely infinite. In the Hahn convention  $t = \omega^{-1}$  all exponent signs change. This agrees with the omnific reports, whose letter  $\Pi$  we use for the purely infinite ideal (`found:eq:omnific` in the foundations report).

**Two constant maps.** `ct` extracts the coefficient of  $\omega^0$  from an element of  $\mathcal{B}_k$ ; it is a ring homomorphism there, not on **No** (both  $\omega$  and  $\omega^{-1}$  have constant coefficient 0). `st` is the standard part of a *finite* surreal or surcomplex number. For example  $\text{ct}(\omega + 7) = 7$ , although  $\omega + 7$  is not finite. The two are never interchanged.

**Two notions of finite.** “Finite surreal” means bounded in modulus by an ordinary integer, not a normal form with finite support (10): the monomial  $\omega^{1/\omega}$  has one term and is infinite. Group ranks, matrix sizes, polynomial degrees and lattice ranks are always ordinary finite integers.

Table 1: Notation used throughout.

| Notation                                 | Meaning in this report                                                                                                                                     |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(D, k)$                                 | $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$ ; in some statements $D$ is any unital subring of $k$ .                                        |
| $F$                                      | The ambient field <b>No</b> or <b>No</b> $[i]$ .                                                                                                           |
| $\Pi_k, \Pi$                             | Normal forms with coefficients in $k$ supported on <b>No</b> $^{>0}$ , with 0; $\Pi = \Pi_{\mathbb{R}}, \Pi_{\mathbb{C}} = \Pi + i\Pi$ .                   |
| $\mathcal{B}_k = k \oplus \Pi_k$         | Nonnegative-support ring; $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$ as in the omnific Diophantine report.                                       |
| $\mathcal{A}_{D,k} = D \oplus \Pi_k$     | <b>Oz</b> = $\mathcal{A}_{\mathbb{Z},\mathbb{R}}, \mathbf{Oz}[i] = \mathcal{A}_{\mathbb{Z}[i],\mathbb{C}}$ . Direct sums of additive groups, not of rings. |
| <code>ct</code> , <code>st</code>        | Constant coefficient on $\mathcal{B}_k$ ; standard part on finite elements.                                                                                |
| $\deg_\omega f$                          | Largest exponent in $\text{supp } f$ ; not an ordinal and not a birthday.                                                                                  |
| $\mathcal{O}_{\text{fin}}, \mathfrak{m}$ | Finite surreals (or surcomplex numbers) and infinitesimals; $\mathfrak{m}_k$ for coefficients in $k$ .                                                     |
| $u \ll v$                                | $u/v \in \mathfrak{m}$ , for $u \geq 0, v > 0$ .                                                                                                           |
| $K_D = \text{Frac } D$                   | $\mathbb{Q}$ or $\mathbb{Q}(i)$ .                                                                                                                          |
| <i>Part I</i>                            |                                                                                                                                                            |
| $G, G^0, \mathcal{G}, T$                 | Algebraic group, its algebraic identity component, an integral model, a torus.                                                                             |
| $\text{UT}_n(\Pi_k), \text{Fil}_r$       | Unitriangular group with purely infinite entries and its superdiagonal filtration.                                                                         |
| <i>Part II</i>                           |                                                                                                                                                            |
| $e_{ij}(a), E_n(A)$                      | $1_n + aE_{ij}$ ; the subgroup generated by these (finite words).                                                                                          |

| Notation                                     | Meaning in this report                                                                                                                                                    |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_n$                                        | $\ker(\text{ct} : E_n(\mathcal{A}_{D,k}) \rightarrow E_n(D))$ ; generated by $e_{ij}(\Pi_k)$ for $n \geq 3$ .                                                             |
| $P_n^{(a)}$                                  | Congruence subgroup of $P_n$ at scale $\omega^a$ .                                                                                                                        |
| $[x, y]$                                     | $xyx^{-1}y^{-1}$ .                                                                                                                                                        |
| $G_D, C_D, B_D$                              | $\text{SL}_2(D)$ , its constant upper triangular subgroup, the upper triangular subgroup of $\text{SL}_2(\mathcal{A}_{D,k})$ .                                            |
| $U(x), \mathcal{T}$                          | $e_{12}(x)$ ; a transversal of $G_D/C_D$ .                                                                                                                                |
| $P_{\text{fin}}, P_{\infty}$                 | Finite points of $\mathbb{P}^1(F)$ ; the infinite points and $\infty$ .                                                                                                   |
| $\Phi_h$                                     | Exponent rescaling $\sum a_s \omega^s \mapsto \sum a_s \omega^{hs}$ .                                                                                                     |
| $\text{St}_n(B), x_{ij}(a), P_n^{\text{St}}$ | Steinberg group, its generating symbols, and the kernel of $\text{ct} : \text{St}_n(\mathcal{A}_{D,k}) \rightarrow \text{St}_n(D)$ ( <a href="#">section 20</a> ).        |
| $P_n \langle a \rangle$                      | $\{g \in P_n : g - 1_n \in M_n(aA)\}$ for $0 \neq a \in A$ ; $P_n^{(a)} = P_n \langle \omega^a \rangle$ .                                                                 |
| $\Gamma_{\kappa}, e_{\alpha}$                | $\bigoplus_{\alpha < \kappa} \mathbb{Q}e_{\alpha}$ , least index dominant; its basis vectors (one index, no argument; root elements $e_{ij}(a)$ have two).                |
| $R_{<\kappa}, \Pi_{<\kappa}, P_{n,<\kappa}$  | The sibling report's $R_{\kappa}^{11}$ and $\Pi_{\kappa}^{11}$ (supports of size $< \kappa$ in $\Gamma_{\kappa}$ ), and the kernel of $\text{ct}$ on $E_n(R_{<\kappa})$ . |
| $\kappa^{<\kappa}, \mathfrak{c}$             | $\sup_{\nu < \kappa} \kappa^{\nu}; 2^{\aleph_0}$ .                                                                                                                        |
| <i>Part III</i>                              |                                                                                                                                                                           |
| $q_j, \varepsilon_j, \mathcal{E}$            | Ordinary positive-semidefinite forms, separated weights, the energy $\sum \varepsilon_j q_j$ .                                                                            |
| $K_j, L_j, K_F$                              | Prefix kernels, their integral points $K_j \cap D^n$ , scalar extension of an ordinary subspace.                                                                          |
| $\text{Um}(\mathcal{A}^n)$                   | Unimodular vectors (coordinates generate the unit ideal).                                                                                                                 |
| $\text{Low}_j$                               | Vectors of energy $\ll \varepsilon_j$ .                                                                                                                                   |
| $\Delta_j$                                   | Gram–Schmidt energies.                                                                                                                                                    |

**Renamed symbols.** The sources' names are changed where they collided with each other or with the omnific reports; the complete table is [table 3](#). The most important changes: the purely infinite ideal is  $\Pi_k$  (06 wrote  $\mathcal{I}_k$ , 07  $\mathcal{J}_k$ , 10  $\mathcal{J}$ ); the constant term is  $\text{ct}$  (07 wrote  $c_0$ ); the kernel of reduction is  $P_n$  (06's  $P_n(k)$ , 07's  $N_n$ ); 06's congruence subgroups  $N_a$  are  $P_n^{(a)}$ , so that no letter  $N$  carries two meanings; 10's energy  $E$  is  $\mathcal{E}$ , because  $E_n$  is the elementary group; 10's word *primitive* is *unimodular*, because the omnific Diophantine report uses *primitive* for “no common nonunit divisor” (`odg:thm:realray`), a weaker property. Source 11's kernel  $N_n(A)$  is  $P_n$ , its Steinberg kernel  $K_n(A)$  is  $P_n^{\text{St}}$  (since  $K_D$  and  $K_j$  are taken), its congruence subgroups  $N^{(m)}$  modulo  $(1 + \omega)^m$  are  $P_n \langle (1 + \omega)^m \rangle$ , and its fragments  $A_{\kappa,D}, \Pi_{\kappa,k}$  are the sibling report's rings, written  $R_{<\kappa}, \Pi_{<\kappa}$  (so that no  $\Pi_{\kappa}$  is confused with  $\Pi_k$ ).

## 1.5 Relationship with the repository and the literature

The omnific integers are classical [5, 6]; L'Innocente and Mantova give a modern account and a factorization theory [8]. The collection now contains two omnific reports: the omnific Diophantine report `docs/surreal/omnific-diophantine-geometry` (prefix `odg:`) and the set-sized quotient report `docs/surreal/set-sized-quotients-of-omnific-integers` (prefix `osq:`). This report builds on both: Part I answers one of the former's questions,

and Part II strengthens the latter's main theorem and uses its two collision lemmas. The euclidean-three-space report proves a parallel standard-part theorem for surreal rotation groups, and the Hahn–Tate report already contains the ordinary lattice backbone of Part III. The precise relations are in [section 33](#). Classical inputs (Tarski transfer, the structure of reductive groups, Cartan's theorem, Cohn's two-row theory, the abelianization of  $E_2$ , the congruence subgroup theorem of Bass–Milnor–Serre, higher-rank Voronoi theory and LLL reduction) are attributed at their points of use, as is the precedent for recovering coefficient algebra from root commutators [\[15\]](#).

## 2 Normal forms, constant terms, and support gaps

This section collects the facts about normal forms used in all three parts. All four sources prove them (source 11 its units, gap and single-product facts), and so do the omnific reports (`osq:lem:ct`, `osq:lem:gap`, `osq:prop:common`, `odg:prop:units`); they are background, not new results.

### 2.1 Normal forms

Every surreal number has a unique Conway normal form

$$f = \sum_{s \in S} a_s \omega^s, \quad a_s \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where  $S \subseteq \mathbf{No}$  is a set and is reverse well ordered: every nonempty subset has a greatest element [\[5, 6, 3\]](#). Addition is coefficientwise; multiplication is Hahn convolution, and the coefficient of a product at a fixed exponent is a finite sum. The sign of a nonzero normal form is the sign of its leading coefficient, and  $\omega^{s+t} = \omega^s \omega^t$  is Conway's monomial law, not an analytic exponential. Complex coefficients give exactly  $\mathbf{No}[i]$ , by separating real and imaginary parts; a finite union of reverse-well-ordered supports is reverse well ordered, so no larger support convention is needed. For  $f \in \mathbf{No}[i]$ ,  $|f| = (f\bar{f})^{1/2}$ ; the leading coefficient of  $f\bar{f}$  is the squared modulus of the leading coefficient of  $f$  (07).

The Hahn support lemma is used freely: products are defined, and a series supported on strictly negative exponents has a formal geometric inverse near 1, so that for  $\gamma > 0$

$$(1 - \omega^{-\gamma})^{-1} = \sum_{m \geq 0} \omega^{-m\gamma}. \quad (2.2)$$

This is an identity of normal forms. It does not assert convergence of a sequence in the fine topology of  $\mathbf{No}$  (06, 07, 10).

### 2.2 The rings and the constant term

For  $k \in \{\mathbb{R}, \mathbb{C}\}$  and a unital subring  $D \subseteq k$  put

$$\Pi_k = \{f : \text{supp } f \subseteq \mathbf{No}^{>0}\}, \quad \mathcal{B}_k = k \oplus \Pi_k, \quad \mathcal{A}_{D,k} = D \oplus \Pi_k, \quad (2.3)$$

with coefficients in  $k$ . The zero series belongs to  $\Pi_k$ . Then  $\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}, \mathbb{R}}$  and  $\mathbf{Oz}[i] = \mathcal{A}_{\mathbb{Z}[i], \mathbb{C}}$ ;  $\mathcal{B}_{\mathbb{R}} = \mathcal{A}_{\mathbb{R}, \mathbb{R}}$  and  $\mathcal{B}_{\mathbb{C}} = \mathcal{A}_{\mathbb{C}, \mathbb{C}}$  are the nonnegative-support rings.

**Lemma 2.1** (Constant term, units and discreteness [\[06, 07, 10\]](#)).

1.  $\Pi_k$  is an ideal of  $\mathcal{B}_k$  and of  $\mathcal{A}_{D,k}$ , and  $\text{ct} : \mathcal{A}_{D,k} \rightarrow D$  is a surjective ring homomorphism with kernel  $\Pi_k$ , split by the inclusion of constants. It commutes with conjugation and with every ordinary finite matrix.
2.  $\deg_\omega(fg) = \deg_\omega f + \deg_\omega g$  for nonzero  $f, g \in \mathcal{B}_k$ , and  $\mathcal{B}_k^\times = k^\times$ ,  $\mathcal{A}_{D,k}^\times = D^\times$ . In particular  $\mathbf{Oz}^\times = \{\pm 1\}$  and  $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$ .
3. An element of  $\mathcal{B}_k$  with a nonzero positive-exponent part has infinite modulus. Hence  $\mathcal{B}_k$  has no nonzero infinitesimal, its finite elements are exactly  $k$ , the finite elements of  $\mathcal{A}_{D,k}$  are exactly  $D$ , every nonzero element of  $\mathcal{A}_{D,k}$  for  $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$  has modulus at least 1, and every vector of  $\mathcal{A}_{D,k}^n \setminus D^n$  has infinite Euclidean norm.

*Proof.* All exponents occurring are nonnegative, and a sum of two nonnegative exponents is zero only when both are zero; so the constant coefficient of a product is the product of the constant coefficients, and a product with a factor in  $\Pi_k$  has positive support. Additivity, the kernel and the section are immediate. Commutation with conjugation and ordinary matrices is coefficientwise.

Hahn multiplication has leading coefficient the product of the leading coefficients, which gives the degree formula. If  $fg = 1$  with  $f, g \in \mathcal{B}_k$ , both degrees are nonnegative and add to zero, so both are zero; a normal form with nonnegative support and leading exponent 0 is constant. Intersecting with  $\mathcal{A}_{D,k}$  and applying the argument to the inverse gives  $\mathcal{A}_{D,k}^\times = D^\times$ .

If the positive-exponent part is nonzero, it has a greatest exponent  $\gamma > 0$ , and the leading term of the absolute value (real case) or of the squared modulus (complex case) is  $|c|\omega^\gamma$  or  $|c|^2\omega^{2\gamma}$ , which dominates every ordinary constant; adding a constant does not change this. Nonzero integers and Gaussian integers have modulus at least one. For vectors apply this to one nonconstant coordinate.  $\square$

Negative exponents are what matter, not size:  $\omega + \omega^{-1}$  is infinite but does not lie in  $\mathcal{B}_\mathbb{R}$ , and  $\text{ct}$  is not order preserving on  $\mathcal{B}_\mathbb{R}$ , since  $\omega - 1 > 0$  while  $\text{ct}(\omega - 1) = -1$  (06, 10).

**Lemma 2.2** (Bounded coordinates are constant [06]). *If  $f \in \mathcal{B}_\mathbb{R}$  and  $|f| \leq M$  for an ordinary real  $M$ , then  $f \in \mathbb{R}$ ; hence  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$ . If  $z \in \mathcal{B}_\mathbb{C}$  has real and imaginary parts bounded by ordinary reals, then  $z \in \mathbb{C}$ .*

*Proof.* This is lemma 2.1(iii), applied coordinatewise.  $\square$

## 2.3 Gaps below sets of exponents

**Lemma 2.3** (Gaps [06, 07, 10]).

1. Every set of surreal numbers has a strict upper bound and a strict lower bound in  $\mathbf{No}$ .
2. For every set  $S \subseteq \mathbf{No}^{>0}$  there is  $\delta > 0$  with  $m\delta < s$  for all  $s \in S$  and all ordinary  $m \geq 1$ .
3. For every set-indexed family  $(h_s)_{s \in S}$  of positive infinite surreals there is  $\gamma > 0$  with  $\omega^{2\gamma} \ll h_s$  for every  $s$ .

*Proof.* For a set  $T$ , the cuts  $\{T \mid \emptyset\}$  and  $\{\emptyset \mid T\}$  give (i). For (ii) apply the set cut  $\{0 \mid T\}$  to  $T = \{s/m : s \in S, m \geq 1\}$ , and take  $\delta = 1$  for empty  $S$ . For (iii) apply (ii) to the set  $\{\deg_\omega h_s\}$  of positive exponents and take  $\gamma = \delta/2$ ; the leading exponent of  $\omega^{2\gamma}/h_s$  is  $2\gamma - \deg_\omega h_s < 0$ .  $\square$

The full-class hypothesis is substantive (06, 07, 10). The support  $\{1/m : m \geq 1\}$  of  $\sum_{m \geq 1} \omega^{1/m} \in \Pi$  has no positive *real* lower bound, but it has a positive surreal one, for instance  $1/\omega$ . Confusing the two assertions would invalidate the main proofs of Parts II and III; in a fixed exponent group such as  $\mathbb{R}$  they fail (sections 12 and 32).

**Proposition 2.4** (Single-product divisibility [06, 07]). *For every set-sized family  $\mathcal{F} \subseteq \Pi_k$  there is  $\delta > 0$  with  $\omega^{-\delta} f \in \Pi_k$  for all  $f \in \mathcal{F}$ . In particular every  $c \in \Pi_k$  is a product  $c = \omega^\delta (\omega^{-\delta} c)$  of two elements of  $\Pi_k$ , and  $\Pi_k^r = \Pi_k$  for every ordinary  $r \geq 1$ .*

*Proof.* The union of the supports is a set of positive exponents; choose  $\delta$  strictly below it by lemma 2.3. Translating supports by  $-\delta$  keeps them positive. Induction gives the powers.  $\square$

This is `osq:prop:common(i),(ii)` of the sibling report.

## Part I

# Omnific points of algebraic groups

This part is source 06's. An algebraic group is affine and of finite type over a field unless stated otherwise; for a real algebraic group  $G$ ,  $G^0$  is its *algebraic* identity component, not the identity component of the real Lie group  $G(\mathbb{R})$ . The main theorems are theorem 4.4 and theorem 4.5.

## 3 A bounded-fiber rigidity principle

### 3.1 Tori and finite schemes

A torus over  $k$  is an algebraic group that becomes a product of copies of  $\mathbb{G}_m$  over an algebraic closure of  $k$ .

**Lemma 3.1** (Torus freezing [06]). *For every real torus  $T$ ,  $T(\mathcal{B}_{\mathbb{R}}) = T(\mathbb{R})$ . For every complex torus  $T$ ,  $T(\mathcal{B}_{\mathbb{C}}) = T(\mathbb{C})$ .*

*Proof.* A complex torus is isomorphic over  $\mathbb{C}$  to  $\mathbb{G}_m^r$ , and its  $\mathcal{B}_{\mathbb{C}}$ -points are  $(\mathcal{B}_{\mathbb{C}}^\times)^r = (\mathbb{C}^\times)^r$  by lemma 2.1. For a real torus, extend scalars from  $\mathcal{B}_{\mathbb{R}}$  to  $\mathcal{B}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} = \mathcal{B}_{\mathbb{C}}$ . After a complex splitting the resulting point is constant by the complex case; hence the original point is a constant complex point fixed by the real conjugation descent datum, and such points are exactly  $T(\mathbb{R})$ . This is ordinary Galois descent for points;  $\mathcal{B}_{\mathbb{R}} \hookrightarrow \mathcal{B}_{\mathbb{C}}$  is injective.  $\square$

The complex case is the special case, for the coordinate ring of a torus, of the remark in the omnific Diophantine report that an affine complex variety whose coordinate ring is generated by units has only constant  $\mathcal{B}_{\mathbb{C}}$ -points (subsection “An algebraic-geometric formulation” of `odg:sec:rigidity`).

**Lemma 3.2** (Finite schemes do not acquire points [06]). *If  $X$  is a finite scheme over  $k = \mathbb{R}$  or  $\mathbb{C}$ , or more generally a zero-dimensional affine scheme of finite type, then  $X(\mathcal{B}_k) = X(k)$ .*

*Proof.* Every element of the finite-dimensional coordinate algebra  $k[X]$  is algebraic over  $k$ , so its image under a  $\mathcal{B}_k$ -point is algebraic over  $k$  in  $F$ . The only elements of  $\mathbf{No}$  algebraic over  $\mathbb{R}$  are real, because  $\mathbb{R}$  is real closed and  $\mathbf{No}$  is an ordered field extension; the complex case follows because  $\mathbb{C}$  is algebraically closed. Nilpotents map to 0. The coordinate algebra

of a zero-dimensional affine scheme of finite type is Artinian and finite-dimensional, so no reducedness is needed.  $\square$

### 3.2 Bounded real fibers

**Theorem 3.3** (Bounded fibers above a torus [06]). *Let  $X \hookrightarrow \mathbb{A}_{\mathbb{R}}^m$  be an affine real scheme of finite type,  $T$  a real torus and  $\pi : X \rightarrow T$  a morphism. Suppose that for every  $t \in T(\mathbb{R})$  the real points of the fiber  $X_t$  are bounded in the given affine coordinates. Then  $X(\mathcal{B}_{\mathbb{R}}) = X(\mathbb{R})$ . No bound uniform in  $t$  is required.*

*Proof.* Let  $x \in X(\mathcal{B}_{\mathbb{R}})$ . By lemma 3.1,  $\pi(x) = t \in T(\mathbb{R})$ . Applying  $\text{ct}$  gives a real point  $\text{ct}(x)$  of  $X_t$ , so that fiber is nonempty; choose an ordinary  $M > 0$  bounding all coordinates of  $X_t(\mathbb{R})$ . The statement that every solution of the finitely many polynomial equations of  $X_t$  satisfies  $-M \leq x_j \leq M$  is a first-order formula of ordered fields with real parameters. Every embedding of real closed fields is elementary (Tarski’s quantifier elimination; see [9]). Apply this to  $\mathbb{R} \subset K$ , where  $K \subset \mathbf{No}$  is the real closure inside  $\mathbf{No}$  of the field generated by  $\mathbb{R}$  and the coordinates of  $x$ ; it is a set. So the bounds hold for  $x$ , and each coordinate of  $x$  is real by lemma 2.2.  $\square$

The only transfer used is transfer between real closed *fields*. No elementary equivalence of  $\mathbb{Z}$  and  $\mathbf{Oz}$ , and no transfer in a language with an omnific-integer predicate, is asserted (06). The same transfer proves the bounded semialgebraic rigidity theorem `odg:thm:bounded`; for algebraic sets, the case of the trivial torus of theorem 3.3 gives the constant-point form of that theorem, while `odg:thm:bounded` covers bounded semialgebraic sets and theorem 3.3 covers unbounded algebraic sets with bounded fibers over a torus. Neither contains the other.

**Corollary 3.4** (Complex zero-dimensional fibers [06]). *If  $X$  is affine of finite type over  $\mathbb{C}$  and  $\pi : X \rightarrow T$  is a morphism to a complex torus whose fibers over complex points are zero-dimensional, then  $X(\mathcal{B}_{\mathbb{C}}) = X(\mathbb{C})$ .*

*Proof.* Freeze  $\pi(x)$  by lemma 3.1, then apply lemma 3.2 to that fiber.  $\square$

**Corollary 3.5** (Compact-by-torus groups [06]). *Let  $G$  be a connected real reductive group whose derived group  $H = G^{\text{der}}$  has compact real points. Then  $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$ .*

*Proof.* The quotient  $T = G/H$  is a torus, by the structure theory of connected reductive groups [10]. A fiber with a real point  $g$  has real points  $gH(\mathbb{R})$ , a compact set, so its coordinates are bounded in any affine embedding; empty real fibers are bounded too. Apply theorem 3.3.  $\square$

*Provenance.* The bounded-fiber formulation is 06’s bridge between normal-form rigidity and algebraic groups. Its inputs are unit rigidity, classical torus splitting and real-closed-field transfer; it generalizes the bounded-coordinate and constant-product arguments of the omnific Diophantine report without treating them as new.

## 4 Unipotents and the classification

### 4.1 A polynomial subgroup from one unipotent

A matrix  $u \in \text{GL}_N(k)$  is unipotent if  $u - 1_N$  is nilpotent; the notion does not depend on the faithful representation.

**Lemma 4.1** (Polynomial one-parameter subgroup [06]). *Let  $k$  have characteristic zero,  $G \subseteq \mathrm{GL}_{N,k}$  an algebraic group and  $1_N \neq u \in G(k)$  unipotent. Then*

$$N_0 = \log u = \sum_{j=1}^{N-1} \frac{(-1)^{j+1}}{j} (u - 1_N)^j \neq 0, \quad v(t) = \exp(tN_0) = \sum_{j=0}^{N-1} \frac{t^j N_0^j}{j!}$$

define a closed immersion  $v : \mathbb{G}_a \hookrightarrow G$ , with  $v(s+t) = v(s)v(t)$  and  $v(-t) = v(t)^{-1}$  over every  $k$ -algebra.

*Proof.* The exponential and logarithm identities used are identities modulo a fixed nilpotence power, hence polynomial identities over  $\mathbb{Q}$ ; so  $N_0$  is nilpotent,  $\exp N_0 = u$  and  $N_0 \neq 0$ . For each integer  $m \geq 0$ ,  $v(m) = u^m \in G(k)$ , and  $\det v(t) = 1$ . Each defining equation of  $G$  pulls back to a polynomial in  $k[t]$  vanishing at all nonnegative integers, hence to zero, so  $v$  lands in  $G$ . If  $(N_0)_{pq} \neq 0$ , the polynomial function  $g \mapsto (\log g)_{pq}/(N_0)_{pq}$  recovers  $t$  on the image, so  $k[G] \rightarrow k[t]$  is surjective:  $v$  is a closed immersion and is injective on points. The group laws follow from the commuting nilpotent exponential identities.  $\square$

**Corollary 4.2** (Purely infinite parameters [06]). *Let  $(D, k) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ . If the  $k$ -fiber of a closed subgroup scheme  $\mathcal{G} \subseteq \mathrm{GL}_{N,D}$  contains a nonidentity unipotent, then the map of lemma 4.1 restricts to an injective homomorphism*

$$v : (\Pi_k, +) \hookrightarrow \mathcal{G}(\mathcal{A}_{D,k}), \quad \mathrm{ct}(v(a)) = 1_N.$$

*Proof.* For  $a \in \Pi_k$  every positive-degree term of  $v(a)$  has entries in  $\Pi_k$ , which is closed under multiplication and under multiplication by  $k$ ; so  $v(a)$  and  $v(-a)$  have entries in  $D + \Pi_k$ . The  $D$ -equations of  $\mathcal{G}$  vanish, since their substitution into  $v(t)$  is the zero polynomial. The constant term is  $1_N$ , and  $\log v(a) = aN_0$  gives injectivity.  $\square$

Rationality of the unipotent is unnecessary: irrational coefficients of  $N_0$  multiply purely infinite terms and do not affect the integral constant coefficient.

## 4.2 The real structural input

A connected linear algebraic group in characteristic zero has a unipotent radical  $R_u(G)$  and is reductive when it is trivial; for connected reductive  $G$  the derived group is connected semisimple and  $G/G^{\mathrm{der}}$  is a torus; a nontrivial unipotent group over  $\mathbb{R}$  has nontrivial real points, and its exponential is an isomorphism of schemes from its Lie algebra [10, Proposition 14.32].

**Lemma 4.3** (Real semisimple compactness criterion [06]). *A connected semisimple real algebraic group  $H$  has no nonidentity real unipotent element if and only if  $H(\mathbb{R})$  is compact.*

*Proof.* If  $H(\mathbb{R})$  is compact, average an inner product in a faithful representation; every element is then orthogonal, and an orthogonal unipotent matrix is the identity because orthogonal matrices are diagonalizable over  $\mathbb{C}$ .

Conversely, by Cartan's theorem a faithful representation has a basis in which  $H$  is stable under transpose [12, Theorem 1.16 and Example 1.17(c)]; so its real Lie algebra  $\mathfrak{h}$  is stable under transpose. If every element of  $\mathfrak{h}$  is skew-symmetric, the real Lie identity component of  $H(\mathbb{R})$  is a closed subgroup of the orthogonal group, hence compact, and a real algebraic group has finitely many real connected components; so  $H(\mathbb{R})$  is compact. Otherwise there is a nonzero symmetric  $S \in \mathfrak{h}$ . As  $\mathfrak{h}$  is semisimple its center is zero, so  $\mathrm{ad} S \neq 0$ ; for  $\langle A, B \rangle = \mathrm{tr}(AB^T)$  the operator  $A \mapsto [S, A]$  is self-adjoint (permute the factors cyclically in the trace), so it has an

eigenvector  $Y \neq 0$  with real eigenvalue  $\lambda \neq 0$ . Diagonalize  $S$  orthogonally;  $[S, Y] = \lambda Y$  makes  $Y$  map the  $s$ -eigenspace of  $S$  into the  $(s + \lambda)$ -eigenspace, so  $Y$  is nilpotent. The Lie exponential of  $tY$ , here a finite polynomial, lies in  $H(\mathbb{R})$ , and at  $t = 1$  it is a nonidentity unipotent (the truncated logarithm recovers  $Y$ ).  $\square$

The proof uses ordinary real Lie theory only at the Cartan and exponential steps and no global exponential on  $\mathbf{No}[i]$ . One may also obtain the noncompact direction from real rank and relative root groups [11, Sections 22–23].

### 4.3 The two dichotomies

**Theorem 4.4** (Omnific rigidity and real unipotents [06]). *For a real algebraic group  $G$  the following are equivalent:*

1.  $G(\mathcal{B}_{\mathbb{R}}) = G(\mathbb{R})$ ;
2.  $G(\mathbb{R})$  has no unipotent element other than the identity;
3.  $G$  has no real algebraic subgroup isomorphic to  $\mathbb{G}_a$ ;
4.  $G^0$  is reductive and  $(G^0)^{\text{der}}(\mathbb{R})$  is compact.

Let  $\mathcal{G} \hookrightarrow \text{GL}_{N, \mathbb{Z}}$  be a closed subgroup scheme and  $G = \mathcal{G}_{\mathbb{R}}$ . The conditions are also equivalent to  $\mathcal{G}(\mathbf{Oz}) = \mathcal{G}(\mathbb{Z})$ . When they fail there is an injective homomorphism

$$(\Pi, +) \longrightarrow \ker(\text{ct} : \mathcal{G}(\mathbf{Oz}) \rightarrow \mathcal{G}(\mathbb{Z})), \quad a \longmapsto \exp(aN),$$

for a fixed nonzero real nilpotent matrix  $N$ , the exponential being a finite polynomial; and every ordinary integral point  $q$  lies on the injectively parametrized family  $a \mapsto q \exp(aN)$  of omnific points.

*Proof.* Assume (2). The unipotent radical of  $G^0$  is trivial, so  $G^0$  is reductive; its derived group has no real unipotents, hence compact real points by lemma 4.3; so (4) holds and  $G^0(\mathcal{B}_{\mathbb{R}}) = G^0(\mathbb{R})$  by corollary 3.5. For  $g \in G(\mathcal{B}_{\mathbb{R}})$  put  $g_0 = \text{ct}(g) \in G(\mathbb{R})$ . The component group  $\pi_0(G) = G/G^0$  is finite, so its  $\mathcal{B}_{\mathbb{R}}$ -points are constant (lemma 3.2);  $g$  and  $g_0$  have the same image there, hence  $g_0^{-1}g \in G^0(\mathcal{B}_{\mathbb{R}})$  is real, and applying  $\text{ct}$  shows it is 1. Thus (2) implies (1), and (4) implies (1) by the same component argument.

If  $G(\mathbb{R})$  has a nonidentity unipotent, lemma 4.1 (over  $\mathcal{B}_{\mathbb{R}}$ ) gives nonconstant points and a real  $\mathbb{G}_a$  subgroup; conversely a  $\mathbb{G}_a$  subgroup gives nonconstant points by substituting a purely infinite parameter into its nonconstant polynomial coordinates, and its real points are unipotent. So (1), (2), (3) are equivalent, and (2) implies (4) was shown.

For the integral model: in the rigid case an omnific point is a constant point of  $G$ , whose matrix entries and inverse-determinant coordinate lie in  $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$  (lemma 2.2); so it is an ordinary integral point. In the nonrigid case corollary 4.2 gives the injection, and left translation by  $q \in \mathcal{G}(\mathbb{Z})$  gives the family through  $q$ .  $\square$

Compactness of  $G(\mathbb{R})$  is not the criterion: every real torus, including the noncompact  $\mathbb{G}_m$ , is rigid, while a noncompact semisimple group is not. Only the characteristic-zero fiber is used; smoothness of the integral model at the primes is not needed. Keeping the model fixed matters: ordinary integral points depend on the model, whereas the criterion for new omnific points depends only on the real fiber (06).

**Theorem 4.5** (The Gaussian dichotomy [06]). *For a complex algebraic group  $G$ ,  $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$  if and only if  $G^0$  is a torus. For a closed subgroup scheme  $\mathcal{G} \hookrightarrow \mathrm{GL}_{N, \mathbb{Z}[i]}$ ,  $\mathcal{G}(\mathbf{Oz}[i]) = \mathcal{G}(\mathbb{Z}[i])$  if and only if  $(\mathcal{G}_{\mathbb{C}})^0$  is a torus. If rigidity fails, the constant-term kernel contains an injective image of  $(\Pi_{\mathbb{C}}, +)$  under a finite matrix exponential.*

*Proof.* Over an algebraically closed field of characteristic zero, a connected group with no nontrivial connected unipotent subgroup is a torus [10, Corollary 17.25]; so a connected group that is not a torus has a nonidentity unipotent. If  $G^0$  is a torus, lemma 3.1 freezes its points and the component argument of theorem 4.4 gives  $G(\mathcal{B}_{\mathbb{C}}) = G(\mathbb{C})$ . Otherwise lemma 4.1 and corollary 4.2 supply injective purely infinite families, also on the integral model; constant points of  $\mathbf{Oz}[i]$  have entries in  $\mathbb{Z}[i]$ .  $\square$

*Remark 4.6* (The rigid-affine-variety question for group schemes [merge]). The omnific Diophantine report asks for which affine varieties  $X$  over  $\mathbb{Z}$  one has  $X(\mathbf{Oz}) = X(\mathbb{Z})$  (odg:q:affine). For  $X$  a closed subgroup scheme of  $\mathrm{GL}_{N, \mathbb{Z}}$ , theorem 4.4 answers it completely: exactly when  $X_{\mathbb{R}}$  has no nonidentity real unipotent, equivalently when  $X_{\mathbb{R}}^0$  is reductive with compact real derived group; in the Gaussian setting,  $X(\mathbf{Oz}[i]) = X(\mathbb{Z}[i])$  exactly when  $(X_{\mathbb{C}})^0$  is a torus (theorem 4.5). *Status* (batch 31): the question remains open for general affine varieties of dimension at least two. For curves that report now answers it, after this remark was written. Let  $\mathbb{k}$  be a field of characteristic zero and call a point nonconstant if its coordinates lie in that report's Hahn ring over  $\mathbb{k}$  of series with nonnegative support (for a nonzero exponent group), but not all in  $\mathbb{k}$ . A smooth geometrically integral affine curve over  $\mathbb{k}$  has nonconstant points exactly when it is the affine line (odg:cr:thm:affine). A geometrically integral affine curve, singular or not, has them exactly when its normalization is the affine line (odg:sg:thm:main). Hence an affine model whose real fiber is geometrically integral with normalization other than  $\mathbb{A}_{\mathbb{R}}^1$  satisfies  $X(\mathbf{Oz}) = X(\mathbb{Z})$ , and a  $\mathbb{Z}[i]$ -model whose complex fiber is geometrically integral with normalization other than  $\mathbb{A}_{\mathbb{C}}^1$  satisfies  $X(\mathbf{Oz}[i]) = X(\mathbb{Z}[i])$  (odg:sg:thm:omnific). For such a  $\mathbb{Z}[i]$ -model, nonordinary  $\mathbf{Oz}[i]$ -points exist exactly when the complex normalization is  $\mathbb{A}_{\mathbb{C}}^1$  and  $X(\mathbb{Z}[i]) \neq \emptyset$  (odg:sg:cor:gaussian). Over  $\mathbf{Oz}$  one case stays open: a real curve with normalization  $\mathbb{A}_{\mathbb{R}}^1$  whose integer points all lack a real preimage under the normalization, such as the isolated real node of  $Y^2 = X^2(X - 1)$  (odg:sg:q:real). In the terms of that report, a satisfactory criterion should separate polynomial parametrization, torus rigidity and genuinely algebraic branches; for group schemes the first two decide rigidity: nonconstant points exist exactly when a polynomial unipotent family exists, and otherwise the identity component is compact-by-torus. In the nonrigid case the theorem supplies families through every integral point but does not describe all omnific points. The theorem also contains several results of that report as special cases: the unipotent subgroup  $(\Pi, +) \hookrightarrow \mathrm{SO}(q, \mathbf{Oz})$  for indefinite forms in at least three variables (odg:cor:unipotent;  $\mathrm{SO}(q)_{\mathbb{R}}$  is then isotropic, hence has real unipotents), the finiteness of  $\mathrm{SO}_n(\mathbf{Oz})$  (odg:cor:orthogonal;  $\mathrm{SO}_n$  is compact), and the torus and unit remark cited after lemma 3.1. The explicit matrices of those results remain useful: theorem 4.4 gives existence of a family, while odg:cor:unipotent writes it down.

## 5 Orthogonal examples

### 5.1 Compact real groups and Gaussian escape

For the standard positive-definite form,  $\mathrm{O}_n(\mathbf{Oz}) = \mathrm{O}_n(\mathbb{Z})$  is the group of signed permutation matrices, of order  $2^n n!$ , and  $\mathrm{SO}_n(\mathbf{Oz})$  consists of those of determinant one (odg:cor:orthogonal, proved there by sources 01 and 02 of that report and reproved by 06 through the column bound  $\sum_i g_{ij}^2 = 1$ ). In particular  $|\mathrm{SO}_3(\mathbf{Oz})| = 24$ .

Over  $\mathbb{C}$  the same group scheme escapes (06). Take  $v = (1, i, 0)^\top$ ,  $w = (0, 0, 1)^\top$  and

$$N = vw^\top - wv^\top = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & i \\ -1 & -i & 0 \end{pmatrix}, \quad N^\top = -N, \quad N^3 = 0, \quad N^2 \neq 0.$$

The finite exponential is

$$g(a) = 1_3 + aN + \frac{a^2}{2}N^2 = \begin{pmatrix} 1 - a^2/2 & -ia^2/2 & a \\ -ia^2/2 & 1 + a^2/2 & ia \\ -a & -ia & 1 \end{pmatrix}. \quad (5.1)$$

For every  $a \in \Pi_{\mathbb{C}}$  it lies in  $\mathrm{SO}_3(\mathbf{Oz}[i])$  ( $\exp(aN)^\top = \exp(-aN)$ , and  $\det = 1$  since  $N$  is nilpotent), has constant term  $1_3$ , and its  $(1, 3)$  entry recovers  $a$ . Neither compactness of the real form nor a positive-definite defining form prevents new Gaussian points: the equation  $z_1^2 + z_2^2 + z_3^2 = 1$  is not a positivity statement over  $\mathbb{C}$ . A conjugate-transpose condition, in contrast, defines a real algebraic unitary group, which must be analyzed as a real group by [theorem 4.4](#), not as this complex orthogonal group (06).

## 5.2 Real, not rational, unipotents

The form  $q = x^2 + y^2 - 3z^2$  is anisotropic over  $\mathbb{Q}$ , so  $\mathrm{SO}(q)(\mathbb{Q})$  has no nonidentity unipotent; but  $\mathrm{SO}(q)(\mathbf{Oz})$  has purely infinite unipotent families. The existence of such a family is a special case of `odg:cor:unipotent` (any nondegenerate indefinite integral form in at least three variables), whose construction already allows irrational real isotropic vectors; it is not reproved here. Source 06 adds two points that the earlier corollary leaves implicit.

*No rational unipotent* [06]. Anisotropy: a primitive integer solution of  $x^2 + y^2 = 3z^2$  would have  $3 \mid x, y$  (squares are 0, 1 modulo 3), hence  $3 \mid z$ . If  $u \neq 1$  were a unipotent in  $\mathrm{SO}(q)(\mathbb{Q})$  with  $N = \log u$ , then  $\exp(tN)$  preserves  $q$  for all  $t$  ([lemma 4.1](#)), so  $N^\top Q + QN = 0$  with  $Q = \mathrm{diag}(1, 1, -3)$ ; for  $r$  maximal with  $N^r \neq 0$  and rational  $z$  with  $N^r z \neq 0$ ,  $B(N^r z, N^r z) = (-1)^r B(z, N^{2r} z) = 0$ , contradicting anisotropy.

*An explicit family* [06]. With  $v = (\sqrt{3}, 0, 1)^\top$  and  $w = (0, 1, 0)^\top$  one has  $B(v, v) = B(v, w) = 0$ ,  $B(w, w) = 1$ , and

$$N = vw^\top - wv^\top = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ -\sqrt{3} & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}, \quad N^\top Q + QN = 0, \quad N^3 = 0,$$

$$\exp(aN) = \begin{pmatrix} 1 - 3a^2/2 & \sqrt{3}a & 3\sqrt{3}a^2/2 \\ -\sqrt{3}a & 1 & 3a \\ -\sqrt{3}a^2/2 & a & 1 + 3a^2/2 \end{pmatrix}. \quad (5.2)$$

For  $a \in \Pi$  this lies in  $\mathrm{SO}(q)(\mathbf{Oz})$  with constant term  $1_3$ , and its  $(3, 2)$  entry is  $a$ ;  $a = \omega$  gives an explicit nonconstant omnific matrix. The polynomial subgroup exists over  $\mathbb{R}$ , not over  $\mathbb{Q}$ , and that suffices for omnific integrality. There is no conflict with the rigidity of nonzero binary norm fibers in the omnific Diophantine report: the ambient group here is three-dimensional with noncompact real derived group.

### 5.3 A comparison table

| Group or model                  | Real omnific behavior    | Reason                               |
|---------------------------------|--------------------------|--------------------------------------|
| A real torus                    | All points constant      | Units freeze after complex splitting |
| Compact semisimple group        | All points constant      | Bounded real coordinates             |
| $\mathrm{SO}(x^2 - y^2)$        | All points constant      | A torus, although noncompact         |
| $\mathrm{SL}_n, n \geq 2$       | Purely infinite families | Elementary $\mathbb{G}_a$ subgroups  |
| $\mathrm{SO}(x^2 + y^2 - 3z^2)$ | Purely infinite families | Real, not rational, unipotents       |

The entries concern constant-term rigidity, not finite generation of an arithmetic subgroup or elementary generation of all omnific matrices (06). The table agrees with the quadratic-level dichotomy `odg:thm:quadclassification`; for instance  $\mathrm{SO}(x^2 - y^2)$  is a torus, matching its case of binary forms.

## 6 Unipotent kernels and unitriangular filtrations

Let  $U$  be a unipotent algebraic group over  $k \in \{\mathbb{R}, \mathbb{C}\}$ ; in a faithful representation it is conjugate over  $k$  to a group of upper unitriangular matrices, its Lie algebra  $\mathfrak{u}$  consists of nilpotent matrices with a common nilpotence bound, and exponential and logarithm are mutually inverse polynomial maps [10, Proposition 14.32].

**Proposition 6.1** (The constant-term kernel of a unipotent group [06]). *There is a polynomial-coordinate identification*

$$\ker(U(\mathcal{B}_k) \rightarrow U(k)) = \exp(\mathfrak{u}(k) \otimes_k \Pi_k),$$

where the right side means finite combinations of a fixed  $k$ -basis of  $\mathfrak{u}$  with coefficients in  $\Pi_k$ , and multiplication corresponds to the truncated Baker–Campbell–Hausdorff polynomial. For an integral model with unipotent  $k$ -fiber, its constant-term identity fiber is the same group inside the faithful representation.

*Proof.* Exponential and logarithm are polynomials over  $k$ , so they commute with ct. A Lie-algebra point has zero constant term exactly when its coordinates lie in  $\Pi_k$ , and then its exponential has constant term the identity. The Baker–Campbell–Hausdorff expression truncates, and its brackets have coefficients in  $\Pi_k$ . For an integral model these points and their inverses satisfy the defining equations as in [corollary 4.2](#); the converse follows by embedding into the  $k$ -fiber and taking logarithms.  $\square$

No convergence of an exponential series at an infinite argument is asserted; every displayed matrix function is a finite polynomial.

Put  $U_n = \mathrm{UT}_n(\Pi_k)$ , the upper unitriangular matrices with entries above the diagonal in  $\Pi_k$ , and for  $r \geq 1$

$$\mathrm{Fil}_r = \{g \in U_n : g_{ij} = 0 \text{ whenever } 0 < j - i < r\}, \quad \mathrm{Fil}_1 = U_n, \quad \mathrm{Fil}_n = 1.$$

**Theorem 6.2** (Exact commutator filtration [06]). *For all  $r, s \geq 1$ ,  $[\mathrm{Fil}_r, \mathrm{Fil}_s] = \mathrm{Fil}_{r+s}$ . Consequently the lower central series is  $\gamma_r(U_n) = \mathrm{Fil}_r$ , the derived series is  $U_n^{(j)} = \mathrm{Fil}_{2^j}$ , and for  $n \geq 2$  the nilpotency class is  $n - 1$  and the derived length is  $\lceil \log_2 n \rceil$ . The first superdiagonal induces  $U_n/[U_n, U_n] \cong (\Pi_k, +)^{n-1}$ .*

*Proof.* Write  $\text{Fil}_r = 1 + J_r$  with  $J_r$  the corresponding strictly upper triangular matrices;  $J_r J_s \subseteq J_{r+s}$  and finite geometric inversion of  $1 + X$  give  $[\text{Fil}_r, \text{Fil}_s] \subseteq \text{Fil}_{r+s}$ . Elimination of successive superdiagonals writes every member of  $\text{Fil}_t$  as a finite product of  $e_{ij}(a)$  with  $j - i \geq t$  and  $a \in \Pi_k$ . For a generator  $e_{ij}(a)$  of  $\text{Fil}_{r+s}$  with  $r + s < n$  choose  $h = i + r$ , so  $h - i \geq r$  and  $j - h \geq s$ , write  $a = bc$  with  $b, c \in \Pi_k$  ([proposition 2.4](#)), and use  $[e_{ih}(b), e_{hj}(c)] = e_{ij}(bc)$ . If  $r + s \geq n$  both sides are trivial. Induction gives the two series;  $\text{Fil}_t \neq 1$  exactly when  $t < n$ , witnessed by  $e_{1n}(a)$ ,  $a \neq 0$ . The first superdiagonal is additive, onto  $(\Pi_k)^{n-1}$ , with kernel  $\text{Fil}_2$ .  $\square$

For every positive surreal exponent  $\rho$ , coefficient extraction  $f \mapsto [\omega^\rho]f$  is a nonzero additive map  $\Pi_k \rightarrow k$ ; composed with a first-superdiagonal coordinate it is a nontrivial homomorphism  $U_n \rightarrow (k, +)$  for  $n \geq 2$ . A purely infinite unipotent group can therefore be visible to small groups. The opposite-root commutators of [Part II](#) destroy all such visibility; the invisibility theorem there is not a statement about all purely infinite matrix groups (06).

*Provenance.* The exponential correspondence and the unitriangular commutator calculus are classical specializations; the exact formulas use the single-product property of  $\Pi_k$ . They describe the nonrigid alternative and are not presented as independent discoveries (06).

## Part II

# Elementary groups and their set-sized images

This part asks how much of an omnific matrix a set-sized group can detect. A group representation of matrices need not come from a homomorphism of their coefficient rings, so the question is genuinely stronger than the ring question answered in the sibling report (`osq:thm:universal`). The answer depends sharply on whether a third row and column are available: two rows admit nonmultiplicative coefficient specializations, and three rows provide the commutator relation that recovers multiplication, which the supply of surreal scales turns into a cardinality obstruction (07). The part follows source 07, with source 06's normality, splitting, tower and boundary results inserted. Source 11's third proof of the main theorem is in [section 9](#); its structural, Steinberg and cardinal results form the last three sections of the part ([sections 19 to 21](#)).

## 7 Setting

For a commutative unital ring  $A$ ,  $E_n(A)$  is the subgroup of  $\text{SL}_n(A)$  generated by the elementary matrices  $e_{ij}(a) = 1_n + aE_{ij}$ ,  $a \in A$ ,  $i \neq j$ . Generation always means finite words. *We never identify  $E_n(A)$  with  $\text{SL}_n(A)$  for an omnific ring*; for  $n = 2$  they differ ([theorem 18.2](#)). Direct multiplication of matrix units gives, for distinct  $i, h, j$ ,

$$e_{ij}(a)e_{ij}(b) = e_{ij}(a + b), \quad [e_{ih}(a), e_{hj}(b)] = e_{ij}(ab), \quad (7.1)$$

finite polynomial identities over any commutative ring [[16](#), Chapter III, Section 1].

Throughout,  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  is a unital subring (for most statements  $D = \mathbb{Z}$  or  $\mathbb{Z}[i]$ ),  $A = \mathcal{A}_{D,k}$ , and

$$P_n = \ker(\text{ct} : E_n(A) \rightarrow E_n(D)), \quad (7.2)$$

where  $\text{ct}$  is applied entrywise; the inclusion of constant matrices is a section. For  $D = \mathbb{Z}, \mathbb{Z}[i]$  Euclidean row reduction gives  $E_n(D) = \text{SL}_n(D)$  (the entries of the first column of a determinant-one matrix generate the unit ideal; reduce it to a unit times  $e_1$ , remove the unit by an elementary diagonal block, and induct; over  $\mathbb{Z}[i]$  use the Gaussian norm). These are finite ordinary matrix arguments; no Euclidean algorithm in  $\mathbf{Oz}$  is assumed (06, 07).

**Lemma 7.1** (The split relative kernel [07, 11]). *For every  $n \geq 2$ ,  $P_n$  is the normal closure in  $E_n(A)$  of the matrices  $e_{ij}(x)$ ,  $x \in \Pi_k$ .*

*Proof.* Let  $N'$  be that normal closure;  $N' \subseteq P_n$ . Modulo  $N'$ ,  $e_{ij}(d+x) = e_{ij}(d)e_{ij}(x)$  is congruent to the constant  $e_{ij}(d)$ , so every word is congruent to a matrix of  $E_n(D)$ ; if the word lies in  $P_n$ , its constant representative is the identity. So  $P_n = N'$ . Only the split inclusion of  $D$  is used, not a relative exactness theorem.  $\square$

Source 11 proves the same lemma for every split unital ring  $D \oplus I$  with ideal  $I$  (its Lemma 5.1), by the same finite-word replacement, and stresses that it uses neither normality of the root-generated subgroup, nor a presentation of  $E_n$ , nor  $E_n = \text{SL}_n$ .

For  $n \geq 3$  the normal closure is not needed: the root subgroups already generate a normal subgroup (06).

**Proposition 7.2** (An abstract idempotent-ideal lemma [06]). *Let  $A$  be a commutative unital ring,  $n \geq 3$ , and  $I$  an ideal in which every element is a product of two elements of  $I$ . Then  $P = \langle e_{ij}(a) : a \in I, i \neq j \rangle$  is perfect and normal in  $E_n(A)$ .*

*Proof.* For a generator  $e_{ij}(a)$  choose a third index  $h$  and  $a = bc$  with  $b, c \in I$ ; (7.1) writes it as a commutator of two elements of  $P$ , so  $P = [P, P]$ . Conjugating a generator by  $z = e_{kl}(r)$ ,  $r \in A$ , gives, except in the opposite-root case  $(k, l) = (j, i)$ , either the generator or a product of root elements with parameters  $a$  and  $\pm ra$ , all in  $I$ . For the opposite root write  $e_{ij}(a) = [e_{ih}(b), e_{hj}(c)]$  and use

$$e_{ji}(r)e_{ih}(b)e_{ji}(-r) = e_{jh}(rb)e_{ih}(b), \quad (7.3)$$

$$e_{ji}(r)e_{hj}(c)e_{ji}(-r) = e_{hi}(-cr)e_{hj}(c); \quad (7.4)$$

both conjugated factors lie in  $P$ , hence so does their commutator.  $\square$

A version with  $I^2 = I$  and finite sums of products follows by splitting  $e_{ij}(a)$  along (7.1); the single-product hypothesis holds for  $\Pi_k$  by proposition 2.4. The perfectness half, in this  $I^2 = I$  form and for any group generated by root maps, is also source 11's (corollary 9.3(iii)).

**Proposition 7.3** (Exact split kernel [06]). *For  $n \geq 3$  and every unital subring  $D \subseteq k$ ,*

$$E_n(A) = P_n \rtimes E_n(D), \quad P_n = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle.$$

*Proof.* Let  $P'$  be the subgroup on the right; it is normal by propositions 2.4 and 7.2. Writing  $e_{ij}(d+b) = e_{ij}(b)e_{ij}(d)$  and moving purely infinite factors to the left using normality writes every  $g \in E_n(A)$  as  $g = pq$  with  $p \in P'$ ,  $q \in E_n(D)$ ; applying  $\text{ct}$  gives  $\text{ct}(g) = q$ . Hence  $P_n = P'$ ,  $P' \cap E_n(D) = 1$ , and the constant inclusion is a section.  $\square$

## 8 Two collision mechanisms

A set-sized target identifies two of any sufficiently large set of root elements. To turn such a collision into the death of an arbitrary root element one needs a divisor: a purely infinite quotient of the prescribed parameter by the difference of the two colliding parameters. The two sources find the divisor in different ways, and both ways are already lemmas of the sibling report, which is where the sources found them.

### 8.1 Division by a difference of nearby monomials

**Lemma 8.1** (Absorption [06]). *Fix  $0 \neq c \in \Pi_k$ . There is  $\varepsilon > 0$  such that for all  $0 < \beta < \alpha < \varepsilon$ ,*

$$\frac{c}{\omega^\alpha - \omega^\beta} = \sum_{m \geq 0} c \omega^{-\alpha - m(\alpha - \beta)} \in \Pi_k. \quad (8.1)$$

*For every ordinal  $\Lambda$  the interval  $(0, \varepsilon)$  contains  $\Lambda$  distinct exponents, for instance  $\alpha_\xi = \varepsilon/(2 + \xi)$ ,  $\xi < \Lambda$ , with surreal arithmetic on the embedded ordinals.*

*Proof.* Choose  $\delta > 0$  strictly below  $\text{supp } c$  (lemma 2.3) and put  $\varepsilon = \delta/(4\omega)$  in the exponent field, so that  $m\varepsilon < \delta$  for every ordinary  $m \geq 1$ . With  $\gamma = \alpha - \beta > 0$ ,  $(\omega^\alpha - \omega^\beta)^{-1} = \omega^{-\alpha}(1 - \omega^{-\gamma})^{-1} = \sum_{m \geq 0} \omega^{-\alpha - m\gamma}$  by (2.2). Every exponent of the product with  $c$  has the form  $s - \alpha - m\gamma$  with  $s \in \text{supp } c$ , and  $s - \alpha - m\gamma > \delta - (m + 1)\varepsilon > 0$ ; cancellation cannot create a nonpositive exponent. The  $\alpha_\xi$  are distinct and lie in  $(0, \varepsilon)$ .  $\square$

This is the subordinate division lemma `osq:lem:division` of the sibling report (sources 09, 10, 11 there), with the admissible scales  $(0, \varepsilon)$  inside its class  $U_{\mathbf{No}}(\text{supp } c)$ ; source 06 cites it as the geometric collision construction of the quotient report's base manuscript, adapted to produce an arbitrary prescribed root parameter.

*Remark 8.2* (No series is mapped term by term [06]). If  $d = \omega^\alpha - \omega^\beta$  and  $b$  is the whole series in (8.1), the only identity used is  $db = c$ . A target homomorphism is applied to a finite matrix commutator involving  $b$ , never to the summands. For the truncations  $b_M$  one has

$$d b_M = c(1 - \omega^{-(M+1)(\alpha - \beta)}), \quad (8.2)$$

with nonzero remainder; replacing  $b$  by a truncation would not prove the theorem (compare `osq:rem:nolimit`).

### 8.2 Rescaling a large set field

**Lemma 8.3** (Exponent scaling [07]). *For every  $h \in \mathbf{No}^{>0}$ ,  $\Phi_h(\sum_s a_s \omega^s) = \sum_s a_s \omega^{hs}$  is an ordered field automorphism of  $\mathbf{No}$  fixing  $\mathbb{R}$ , with inverse  $\Phi_{1/h}$ ; it extends coefficientwise to  $\mathbf{No}[i]$ .*

*Proof.* Multiplication by  $h$  is a strictly increasing additive bijection of the exponent group; it preserves reverse well ordering, set size, exponent sums and the finite convolution formulas, and the leading coefficient.  $\square$

This is the dilation  $S_h$  of the surcomplex automorphism report (`saut:eq:dilation`, with  $t = \omega^{-1}$ ) and of `odg:def:ex:dilation`. No compatibility with the surreal exponential, simplicity order or birthday is asserted (07).

**Lemma 8.4** (Set-field compression [07]). *For every  $\delta > 0$  and every infinite set cardinal  $\lambda$  there is a set subfield  $L \subset \mathbf{No}$  with  $|L| \geq \lambda$  and  $\text{supp } a \subset (-\delta, \delta)$  for every  $a \in L$ ; in particular  $(a - b)^{-1}$  has the same support bound for distinct  $a, b \in L$ .*

*Proof.* Let  $L_0$  be the subfield generated by  $\lambda$  distinct surreals (for instance embedded ordinals); it is a set with at least  $\lambda$  elements. The union  $T$  of the supports of its members is a set; choose  $\beta > 0$  with  $|s| < \beta$  for  $s \in T$  (lemma 2.3), put  $h = \delta/\beta$  and  $L = \Phi_h(L_0)$ . Then  $|hs| < \delta$  for every exponent of every member of  $L$ , and  $(a - b)^{-1} \in L$ .  $\square$

The quantifiers matter:  $L$  is chosen after  $\delta$  and  $\lambda$ , and the supports of the whole field, not only of its generators, are compressed (07).

**Lemma 8.5** (A commutator-ready family [07]). *Let  $0 \neq x \in \Pi_k$  and  $H$  a set. There are  $\delta > 0$  and a set subfield  $L \subset \mathbf{No}$  with  $|L| > |H|$  such that, with  $\tau = \omega^{2\delta}$ ,*

$$\tau a \in \Pi_k \quad (a \in L), \quad \frac{x}{\tau(a - b)} \in \Pi_k \quad (a, b \in L, a \neq b).$$

*Proof.* Choose  $\delta > 0$  with  $4\delta < s$  for all  $s \in \text{supp } x$  and apply lemma 8.4 with a cardinal larger than  $|H|$ . Every exponent of  $\tau a$  lies in  $(\delta, 3\delta)$ , and every exponent of  $x\omega^{-2\delta}(a - b)^{-1}$  exceeds  $4\delta - 2\delta - \delta = \delta$ . The auxiliary field may stay real when  $x$  has complex coefficients.  $\square$

This is the scaled-field collision lemma `osq:lem:collision` of the sibling report (source 06 there, with 03) in the form needed for root parameters: there  $m = X^{g/2}$  and  $mL \subseteq \Pi_K$  give  $(mx - my) \cdot m/(x - y) = m^2$ ; here  $\tau a$  plays the role of  $mx$  and the prescribed  $x$  that of  $m^2$ . Source 07 credits the scaled-field collision to the quotient report's base manuscript.

## 9 Three or more rows: no set-sized images

**Theorem 9.1** (The purely infinite elementary kernel is invisible to sets [06, 07, 11]). *Let  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  a unital subring,  $A = \mathcal{A}_{D,k}$  and  $n \geq 3$ .*

1.  $P_n = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle$ , and  $E_n(A) = P_n \rtimes E_n(D)$ .
2.  $P_n$  is perfect and nontrivial.
3. Every class homomorphism from  $P_n$  to a set-sized group is trivial.
4. For every set-sized group  $H$ , composition with `ct` is a bijection  $\text{Hom}(E_n(D), H) \rightarrow \text{Hom}(E_n(A), H)$ . Thus  $E_n(D)$  is the universal set-sized group quotient of  $E_n(A)$ ; it is  $\text{SL}_n(D)$  for  $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ , so for  $E_n(\mathbf{Oz})$  and  $E_n(\mathbf{Oz}[i])$ , and  $\text{SL}_n(k)$  for  $D = k$ .

*No continuity, preservation of infinite sums, definability or algebraicity of a homomorphism is assumed.*

Assertion (iii) is stronger than the factorization (iv): a homomorphism defined only on the kernel need not extend to  $E_n(A)$ , and all three proofs below treat such maps directly (06, 07, 11). Source 06 states (i)–(iv) for  $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$  and, in its corollary on other constant rings, for every set-sized unital  $D \subseteq k$ ; source 07 states them for  $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ , and its proof uses  $D$  only through the split inclusion of lemma 7.1, so it applies verbatim to every unital  $D \subseteq k$ . Source 11 states them for  $D \in \{\mathbb{Z}, \mathbb{Z}[i]\}$  (its Theorem A, proved as its Theorem 5.2 with Lemma 5.3), also for maps defined only on the kernel; its proof uses  $D$  only through the split retraction, so it too applies to every unital  $D \subseteq k$ . Every  $D \subseteq k$  is a set.

*Proof of (i), (ii) and (iv) given (iii).* (i) is [proposition 7.3](#). (ii): perfectness is [proposition 7.2](#), and  $e_{12}(\omega) \neq 1$ . (iv): a homomorphism  $\psi : E_n(A) \rightarrow H$  restricts to the trivial map on  $P_n$  by (iii), so it factors uniquely through  $\text{ct}$  in the split decomposition; restriction to the constant subgroup gives the factor, and for every  $g$  the matrix  $g \text{ct}(g)^{-1}$  lies in  $P_n$ . Conversely every homomorphism  $E_n(D) \rightarrow H$  composes with  $\text{ct}$ . Over the field  $k$  ordinary elimination gives  $E_n(k) = \text{SL}_n(k)$ .  $\square$

*First proof of (iii): rescaled set fields* [07]. By [lemma 7.1](#) it suffices to kill each conjugate  $g e_{ih}(x)g^{-1}$  with  $g \in E_n(A)$ ,  $0 \neq x \in \Pi_k$ ,  $i \neq h$ . Let  $\psi : P_n \rightarrow H$ . Choose an index  $j \notin \{i, h\}$  and take  $\tau, L$  from [lemma 8.5](#). Since  $P_n$  is normal,  $a \mapsto \psi(g e_{ij}(\tau a)g^{-1})$  is defined on  $L$ , and  $|L| > |H|$  gives distinct  $a, b \in L$  with equal images; by (7.1)  $\psi(g e_{ij}(\tau(a-b))g^{-1}) = 1$ . Put  $d = x/(\tau(a-b)) \in \Pi_k$ . The conjugated commutator identity

$$[g e_{ij}(\tau(a-b))g^{-1}, g e_{jh}(d)g^{-1}] = g e_{ih}(\tau(a-b)d)g^{-1} = g e_{ih}(x)g^{-1}$$

is an identity between elements of  $P_n$ , and applying  $\psi$  kills the normal generator. The conjugator  $g$  need not lie in  $P_n$ : only the two conjugated root elements and their commutator, all in the domain, are fed to  $\psi$ .  $\square$

*Second proof of (iii): division by nearby monomials* [06]. By (i),  $P_n$  is generated by the  $e_{ij}(c)$ ,  $c \in \Pi_k$ , without conjugates. Let  $\varphi : P_n \rightarrow H$ ,  $c \neq 0$ , and fix a third index  $h$ . Take  $\varepsilon$  from [lemma 8.1](#) and a set  $T \subset (0, \varepsilon)$  that does not inject into  $H$  (its Hartogs ordinal, or a cardinal  $> |H|$ , indexes such a set). The elements  $e_{ih}(\omega^\alpha)$ ,  $\alpha \in T$ , lie in  $P_n$  and cannot have distinct images, so for some  $\alpha > \beta$  in  $T$ ,  $\varphi(e_{ih}(\omega^\alpha - \omega^\beta)) = 1$ . With  $d = \omega^\alpha - \omega^\beta$  and  $b = c/d \in \Pi_k$ ,

$$\varphi(e_{ij}(c)) = \varphi([e_{ih}(d), e_{hj}(b)]) = [1_H, \varphi(e_{hj}(b))] = 1_H.$$

Every generator, hence every finite word, is killed.  $\square$

The third route turns a group homomorphism into a ring homomorphism. It rests on an abstract lemma about groups generated by root maps. For a commutative ring  $I$ , possibly without unit, write  $x_{ij}$  ( $1 \leq i \neq j \leq n$ ) for maps from  $I$  into a group; the letters  $x_{ij}$  always carry two indices and are not the elements  $x \in \Pi_k$  of the proofs above.

**Theorem 9.2** (Root-kernel reconstruction [11]). *Let  $n \geq 3$ , let  $I$  be a commutative ring, possibly without unit and possibly a proper class, and let  $\Sigma$  be a group generated by additive homomorphisms  $x_{ij} : (I, +) \rightarrow \Sigma$ ,  $i \neq j$ , with*

$$[x_{ih}(a), x_{hj}(b)] = x_{ij}(ab) \quad (i, h, j \text{ distinct}, a, b \in I). \quad (9.1)$$

*For a homomorphism  $\varphi : \Sigma \rightarrow H$  to a set-sized group put  $J_\varphi = \{a \in I : \varphi(x_{ij}(a)) = 1 \text{ for all } i \neq j\}$ . Then  $J_\varphi$  is an ideal of  $I$ , and there are a set-sized ring  $R_\varphi$ , possibly without unit, and a surjective additive and multiplicative map  $\vartheta_\varphi : I \rightarrow R_\varphi$  with kernel  $J_\varphi$  and*

$$|R_\varphi| \leq |H|^{n(n-1)}.$$

*Moreover  $R_\varphi = 0$  if and only if  $\varphi$  is trivial.*

*Proof.* Each condition defining  $J_\varphi$  is the kernel of an additive homomorphism, so  $J_\varphi$  is an additive subgroup. For  $a \in J_\varphi$ ,  $b \in I$  and  $i \neq j$  choose  $h \notin \{i, j\}$ ; then  $\varphi(x_{ij}(ab)) = [\varphi(x_{ih}(a)), \varphi(x_{hj}(b))] = [1, \varphi(x_{hj}(b))] = 1$ . This holds for every ordered pair, so  $ab \in J_\varphi$ , and commutativity gives a two-sided ideal. Put  $\vartheta_\varphi(a) = (\varphi(x_{ij}(a)))_{i \neq j} \in H^{n(n-1)}$  and  $R_\varphi = \vartheta_\varphi(I)$ , a subclass of a set and hence a set. By additivity,  $\vartheta_\varphi(a) = \vartheta_\varphi(b)$  if and only if  $a - b \in J_\varphi$ . Define  $\vartheta_\varphi(a) + \vartheta_\varphi(b) = \vartheta_\varphi(a + b)$  and  $\vartheta_\varphi(a)\vartheta_\varphi(b) = \vartheta_\varphi(ab)$ ; these are well defined because  $J_\varphi$  is a subgroup and an ideal, and the ring axioms are inherited from  $I$ . If  $R_\varphi = 0$ , every generator of  $\Sigma$  is killed, so  $\varphi$  is trivial; the converse is clear.  $\square$

The ring structure on  $R_\varphi$  is transported from  $I/J_\varphi$ ; it is not the componentwise product in the group  $H^{n(n-1)}$ , which carries no ring structure. No set of class-sized cosets is formed: the tuple image is the carrier (11).

**Corollary 9.3** (Transfer from rings to groups [11]). *In the setting of [theorem 9.2](#):*

1. *if every additive multiplicative map from  $I$  to a set-sized ring is zero, every homomorphism from  $\Sigma$  to a set-sized group is trivial;*
2. *if  $\kappa$  is an uncountable cardinal and every additive multiplicative map from  $I$  to a ring of cardinality  $< \kappa$  is zero, every homomorphism from  $\Sigma$  to a group of cardinality  $< \kappa$  is trivial;*
3. *if  $I^2 = I$  then  $\Sigma$  is perfect, and if every element of  $I$  is a single product  $bc$ , each  $x_{ij}(a)$  is a single commutator of root elements.*

*Proof.* (i) Apply the hypothesis to  $\vartheta_\varphi$ . (ii) If  $|H| < \kappa$  then  $|H|^{n(n-1)} < \kappa$ : it is finite for finite  $H$  and equals  $|H|$  for infinite  $H$ . (iii) For  $a = \sum_{\ell=1}^r b_\ell c_\ell$  and  $h \notin \{i, j\}$ ,  $x_{ij}(a) = \prod_\ell [x_{ih}(b_\ell), x_{hj}(c_\ell)]$ .  $\square$

**Lemma 9.4** (Normal generation [11]). *Let  $U$  be a subgroup of a group  $G$  and  $N$  its normal closure. If every homomorphism  $U \rightarrow H$  is trivial for a given group  $H$ , so is every homomorphism  $N \rightarrow H$ . If  $U$  is perfect, so is  $N$ .*

*Proof.*  $N$  is generated by the conjugates  $gUg^{-1}$ ,  $g \in G$ . For  $f : N \rightarrow H$ , the map  $u \mapsto f(gug^{-1})$  is a homomorphism  $U \rightarrow H$ , hence trivial. Each  $gUg^{-1}$  is perfect when  $U$  is, so it lies in  $[N, N]$ .  $\square$

*Third proof of (iii): a ring rebuilt from the root maps [11].* Let  $U = \langle e_{ij}(a) : a \in \Pi_k, i \neq j \rangle$ . The restrictions  $x_{ij} = e_{ij}|_{\Pi_k}$  are additive and satisfy (9.1) by (7.1), so [theorem 9.2](#) applies with  $I = \Pi_k$  and  $\Sigma = U$ . Every additive multiplicative map from  $\Pi_k$  to a set-sized ring is zero: this is part (iii) of the ring theorem `osq:thm:universal` of the sibling report, which source 11 reproves (its Theorem 3.3) with the division identity of [lemma 8.1](#) in the form  $c/(\omega^\alpha - \omega^\beta) \in \Pi_k$  for  $0 < \beta < \alpha < \delta$ , where  $m\delta < s$  for all  $s \in \text{supp } c$  and ordinary  $m \geq 1$ , proving strong summability of the quotient as well as positivity, and with the exponents  $\delta/(\xi + 2)$ ,  $\xi < |R|^+$ , to force a collision in the target ring  $R$ . By [corollary 9.3](#)(i) every homomorphism from  $U$  to a set-sized group is trivial.  $P_n$  is the normal closure of  $U$  ([lemma 7.1](#)), and [lemma 9.4](#) kills every homomorphism from  $P_n$  to  $H$ . By (i)  $U = P_n$ , so the last step is not needed here; source 11 does not use (i), and kills a map on the kernel separately on each conjugate of  $U$ .  $\square$

The first two proofs do not construct a ring map from  $A$  to  $H$  or to an endomorphism ring; both apply to arbitrary abstract group maps on  $P_n$ , and for  $k = \mathbb{C}$  neither needs an order on coefficients, only positivity of exponents (06, 07). The third proof does build a ring, but from the root maps of the given group homomorphism, not from any ring map on  $A$ , so it too applies to arbitrary abstract group maps defined only on the kernel (11). Neither of the first two proofs uses the other's stronger ingredients: 06's needs only single-product divisibility, gaps and the geometric inverse, and 07's needs the field automorphism  $\Phi_h$  and a set field closed under inverses. The third is not independent of them in mechanism: it imports the ring theorem, whose proofs use the same collision identities ([section 8](#)); what it adds is the general transfer [corollary 9.3](#), which also gives the Steinberg and cardinal versions below ([theorems 20.2](#) and [21.3](#)). The recovery of coefficient algebra from root commutators has substantial precedent [15]; source 11 claims neither that idea nor the normal-generation

machinery, only its elementary finite-tuple form for cardinality and nonunital parameter rings. Perfectness alone would kill only abelian images; the cardinal collision kills images in finite simple groups and in arbitrary nonabelian groups as well. The theorem does not make  $P_n$  trivial: its tautological representation on the proper-class space  $F^n$  is faithful and has no set-sized target (06, 07).

**Corollary 9.5** (Actions and linear representations [06, 07, 11]). *For  $n \geq 3$ , every action of  $P_n$  on a set is trivial, and every action of  $E_n(A)$  on a set factors uniquely through  $\text{ct}$ . Every linear representation of  $P_n$  on a set-sized vector space over a set-sized field, of any dimension, is trivial; the same holds for representations on set-sized abelian groups or on any structure whose automorphism group is a set.*

*Proof.* An action on a set  $X$  is a homomorphism to the set-sized group  $\text{Sym}(X)$ ; for linear representations use the invertible linear maps of the space.  $\square$

Source 11 proves the same (its Theorem 7.1 and Corollary 7.2) and adds two points.  $P_n$  has no nontrivial transitive action on a set with more than one point. Restriction to constant matrices and pullback by  $\text{ct}$  identify the representations of  $E_n(A)$  on set-sized vector spaces with those of  $E_n(D)$ , intertwiners included: a linear map intertwines two pulled-back representations exactly when it intertwines their restrictions to the constant subgroup, since every matrix acts through its constant term. Here “arithmetic” means only this factorization; the representations of  $\text{SL}_n(D)$  are not classified, nor asserted to be algebraic (11).

*Remark 9.6* (The ring theorem as a consequence [merge]). **Theorem 9.1** implies the universal set-sized quotient theorem `osq:thm:universal` of the sibling report for coefficient fields  $K \in \{\mathbb{R}, \mathbb{C}\}$  and every unital  $D \subseteq K$ . *Proof.* Let  $S$  be a set-sized unital ring, possibly noncommutative, and  $\psi : A \rightarrow S$  a unital ring homomorphism. Entrywise application of  $\psi$  is multiplicative on matrices and sends  $e_{ij}(a)$  to  $1 + \psi(a)E_{ij} \in \text{GL}_n(S)$ , so it restricts to a homomorphism  $E_n(A) \rightarrow \text{GL}_n(S)$  into a set-sized group, with  $n = 3$ . By **theorem 9.1**(iii) it kills  $e_{12}(x)$  for  $x \in \Pi_k$ , so  $\psi(x) = 0$  and  $\psi = \psi|_D \circ \text{ct}$ ; this is part (ii) of the ring theorem. For an additive multiplicative map  $\varphi : A \rightarrow S$  into a possibly nonunital set-sized ring, let  $S^+ = \mathbb{Z} \times S$  be the unitalization, with  $(m, s)(m', s') = (mm', ms' + m's + ss')$ ; then  $m + x \mapsto (m, \varphi(x))$  ( $m \in \mathbb{Z}, x \in \Pi_k$ ) is a unital ring homomorphism from  $\mathcal{A}_{\mathbb{Z},k} = \mathbb{Z} \oplus \Pi_k$  to the set-sized ring  $S^+$ , so  $\varphi(\Pi_k) = 0$  by the unital case; this gives parts (i) and (iii). A set-sized unital module  $M$  gives a unital map  $A \rightarrow \text{End}_{\mathbb{Z}}(M)$ , which gives part (iv).  $\square$

The implication is logical, not an independent proof: both proofs of **theorem 9.1** use the same collision identities as the ring theorem ([section 8](#)). The group theorem is strictly stronger in scope: an abstract group homomorphism on  $E_n(A)$  or on  $P_n$  need not be induced by a ring or module map, so the theorem also strengthens `osq:cor:faithful` and `osq:prop:matrices`, which concern representations coming from ring maps. The ring theorem is proved in the sibling report for every set field  $K$ ; the group theorem is stated here only for  $k \in \{\mathbb{R}, \mathbb{C}\}$ , as in the sources.

The converse implication is source 11’s: by [corollary 9.3](#)(i) and [lemma 9.4](#), part (iii) of the ring theorem implies **theorem 9.1**(iii) (the third proof above). For  $k \in \{\mathbb{R}, \mathbb{C}\}$  the group theorem (iii) and the nonunital ring theorem `osq:thm:universal`(iii) therefore imply each other through finite reductions. Neither implication is an independent proof of its conclusion.

**Proposition 9.7** (An abstract collision criterion [07]). *Let  $A$  be a commutative ring,  $I \triangleleft A$ ,  $n \geq 3$ , and  $E_n(A, I)$  the normal closure of  $e_{ij}(I)$  in  $E_n(A)$ . Fix a set-sized group  $H$ . Suppose that for each  $0 \neq x \in I$  there are a set  $\mathcal{P}$  with  $|\mathcal{P}| > |H|$  and elements  $u_p \in I$  ( $p \in \mathcal{P}$ ) and  $v_{pq} \in I$  ( $p \neq q$ ) with*

$$(u_p - u_q)v_{pq} = x. \quad (9.2)$$

Then every homomorphism  $E_n(A, I) \rightarrow H$  is trivial.

*Proof.* For a normal generator  $g e_{ih}(x)g^{-1}$  choose a third index  $j$ . The images of  $g e_{ij}(u_p)g^{-1}$  cannot all be distinct; a collision at  $p \neq q$  kills  $g e_{ij}(u_p - u_q)g^{-1}$ , and its commutator with  $g e_{jh}(v_{pq})g^{-1}$  is  $g e_{ih}(x)g^{-1}$ .  $\square$

The first proof supplies the criterion with  $u_a = \tau a$  and  $v_{ab} = x/(\tau(a - b))$ , the second with  $u_\alpha = \omega^\alpha$  and  $v_{\alpha\beta} = x/(\omega^\alpha - \omega^\beta)$ . The criterion does not require  $A$  to be a field,  $A/I$  to split, or  $H$  to be linear; the splitting was used only to identify  $E_n(A, I)$  with the whole reduction kernel. Source 07 proposes it as the first target for a formalization: it isolates all size and support input in (9.2).

## 10 A separating proper-class congruence tower

A group with no small images might be mistaken for a simple group; the kernels give an explicit counterexample to that inference (06). Fix  $A = \mathbf{Oz}$  or  $\mathbf{Oz}[i]$  and  $n \geq 3$ . For each positive surreal  $a$  put

$$P_n^{(a)} = \{g \in P_n : g - 1_n \in M_n(\omega^a A)\}, \quad (10.1)$$

the kernel on  $P_n$  of reduction modulo the principal ideal  $\omega^a A$ , used only as a congruence relation.

**Theorem 10.1** (A non-simple invisible group with a large residual tower [06]). *For every  $a > 0$ ,  $P_n^{(a)}$  is a nontrivial proper normal subgroup of  $P_n$ , and  $P_n^{(b)} \subsetneq P_n^{(a)}$  for  $0 < a < b$ . Moreover*

$$\bigcap_{a \in \mathbf{No}^{>0}} P_n^{(a)} = 1, \quad \bigcap_{a \in S} P_n^{(a)} \neq 1 \quad \text{for every set } S \subset \mathbf{No}^{>0}.$$

*Each quotient  $P_n/P_n^{(a)}$  is nontrivial and perfect, has no nontrivial homomorphism to a set-sized group, and cannot itself be set-sized.*

*Proof.* Matrices congruent to 1 modulo an ideal form a normal subgroup of  $\mathrm{GL}_n(A)$ ; intersecting with  $P_n$  gives normality.  $e_{12}(\omega^a) \in P_n^{(a)}$ , while  $e_{12}(\omega^{a/2}) \notin P_n^{(a)}$  since  $\omega^{-a/2} \notin A$ . For  $b > a$ ,  $\omega^{b-a} \in A$  gives the inclusion, and  $e_{12}(\omega^a)$  witnesses strictness. If  $g \neq 1$ , pick a nonzero entry  $f$  of  $g - 1$ ; for  $a > \deg_\omega f$ ,  $\omega^{-a}f$  has negative leading exponent, so  $g \notin P_n^{(a)}$ . For a set  $S$  choose  $b$  above  $S$  (lemma 2.3); then  $e_{12}(\omega^b)$  lies in every  $P_n^{(a)}$ ,  $a \in S$ , including  $S = \emptyset$ . A homomorphism from  $P_n/P_n^{(a)}$  to a set-sized group composes to one from  $P_n$ , hence is trivial; a nontrivial set-sized quotient would contradict this through its identity map. Perfectness passes to quotients.  $\square$

The tower separates individual matrices only when all surreal scales are allowed. This is a set/class distinction, not a numerical size assigned to  $P_n$  (06).

## 11 The finite side and the rotation groups

The same mechanism applies on the side of infinitesimals. Let  $\mathcal{O}_{\mathrm{fin}}(k)$  be the ring of finite elements of  $F$  ( $k = \mathbb{R}$  for  $\mathbf{No}$ ,  $k = \mathbb{C}$  for  $\mathbf{No}[i]$ ),  $\mathfrak{m}_k$  its infinitesimals, and  $\mathrm{st} : \mathcal{O}_{\mathrm{fin}}(k) \rightarrow k$  the standard part, a surjective ring homomorphism with kernel  $\mathfrak{m}_k$  and section  $k \subset \mathcal{O}_{\mathrm{fin}}(k)$ . No source states the following group analogue of the finite-side ring theorem `osq:thm:finite`; it was observed in the placement dossier and is added here with a complete proof.

**Proposition 11.1** (Universal set-sized quotient of  $\mathrm{SL}_n$  of the finite surreals [merge]). *Let  $n \geq 3$  and  $Q_n = \ker(\mathrm{st} : \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) \rightarrow \mathrm{SL}_n(k))$ . Every class homomorphism from  $Q_n$  to a set-sized group is trivial, and every class homomorphism  $\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) \rightarrow H$  to a set-sized group factors uniquely through  $\mathrm{st}$ . Thus  $\mathrm{SL}_n(k)$  is the universal set-sized quotient of  $\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$ .*

*Proof.*  $\mathcal{O}_{\mathrm{fin}}(k)$  is local with maximal ideal  $\mathfrak{m}_k$ . If  $x$  is finite and  $\mathrm{st} x \neq 0$ , then  $|x| \geq |\mathrm{st} x|/2$ , so  $x^{-1} \in F$  is finite; elements of  $\mathfrak{m}_k$  are not units, and  $\mathfrak{m}_k$  is an ideal.

$\mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) = E_n(\mathcal{O}_{\mathrm{fin}}(k))$ . Let  $g \in \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$ . Since  $\det g \notin \mathfrak{m}_k$ , some entry of the first column is a unit; elementary row operations move a unit to position  $(1, 1)$  (add that row to the first if needed: if  $g_{11} \in \mathfrak{m}_k$  and  $g_{r1}$  is a unit, the new  $(1, 1)$  entry  $g_{11} + g_{r1}$  is a unit), then clear the first column and the first row. By induction  $g$  becomes  $\mathrm{diag}(u_1, \dots, u_n)$  with  $\prod u_i = 1$ , and  $\mathrm{diag}(u, u^{-1}) = e_{12}(u)e_{21}(-u^{-1})e_{12}(u) (e_{12}(1)e_{21}(-1)e_{12}(1))^{-1}$  (Whitehead's identity, valid over any commutative ring) makes the diagonal matrix elementary.

$Q_n$  is the normal closure of the  $e_{ij}(\mathfrak{m}_k)$ . This is the argument of lemma 7.1 with the split surjection  $\mathrm{st}$  in place of  $\mathrm{ct}$  and  $E_n(\mathcal{O}_{\mathrm{fin}}(k)) = \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$ ,  $E_n(k) = \mathrm{SL}_n(k)$ .

*Killing.* Let  $\psi : Q_n \rightarrow H$ , and consider a normal generator  $g e_{ih}(\varepsilon)g^{-1}$  with  $g \in \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k))$  and  $0 \neq \varepsilon \in \mathfrak{m}_k$ ; let  $a = -\deg_\omega \varepsilon > 0$ . For  $0 < \gamma < \delta < a$ ,  $\varepsilon/(\omega^{-\gamma} - \omega^{-\delta}) \in \mathfrak{m}_k$  (`osq:lem:finitediff`: the denominator has leading exponent  $-\gamma$ , so the quotient has leading exponent  $\gamma - a < 0$ ). Choose a third index  $j$  and a set  $T \subset (0, a)$  that does not inject into  $H$  (for an ordinal  $\Lambda$  that does not inject into  $H$ , the exponents  $a/\omega^{\xi+1}$ ,  $\xi < \Lambda$ , are distinct and lie in  $(0, a)$ ). The elements  $g e_{ij}(\omega^{-\gamma})g^{-1}$ ,  $\gamma \in T$ , lie in  $Q_n$ , so two of them, for  $\gamma < \delta$ , have equal images, and  $\psi(g e_{ij}(d)g^{-1}) = 1$  for  $d = \omega^{-\gamma} - \omega^{-\delta} \in \mathfrak{m}_k$ . With  $b = \varepsilon/d \in \mathfrak{m}_k$ ,  $[g e_{ij}(d)g^{-1}, g e_{jh}(b)g^{-1}] = g e_{ih}(\varepsilon)g^{-1}$  is an identity in  $Q_n$ , so the generator is killed.

*Factorization.* A homomorphism  $\varphi : \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(k)) \rightarrow H$  kills  $Q_n$ , and  $g \mathrm{st}(g)^{-1} \in Q_n$  gives  $\varphi = \varphi|_{\mathrm{SL}_n(k)} \circ \mathrm{st}$ ; uniqueness follows from surjectivity of  $\mathrm{st}$ .  $\square$

*Remark 11.2* (Comparison with the surreal rotation groups). The euclidean-three-space report proves that every homomorphism from  $\mathrm{SO}(3, \mathbf{No})$  to a set-sized group factors through standard part (`e3:cut:thm:smallquotient`), and the same for  $\mathrm{SO}(n, \mathbf{No})$ ,  $n \geq 3$ , and  $\mathrm{SU}(n, \mathbf{No}[i])$ ,  $n \geq 2$  (`e3:cut:thm:allranks`), while it fails for the circle (`e3:cut:prop:circlecounter`). Those theorems and theorem 9.1 are parallel instances of one phenomenon, a universal set-sized quotient obtained by killing a kernel through a supply of scales, but neither implies the other. The rings differ (all of  $\mathbf{No}$ , whose compact groups have finite entries, against  $\mathbf{Oz}$ ); the kernels differ (infinitesimal matrices under  $\mathrm{st}$  against purely infinite matrices under  $\mathrm{ct}$ ); and the mechanisms differ (normal closure at valuation cuts in a compact group followed by Givens elimination, against Steinberg commutators with collisions).  $\mathrm{SO}(3, \mathbf{Oz})$  is finite (section 5), so the rotation group has trivial omnific kernel, and a homomorphism on  $E_n(\mathbf{Oz})$  says nothing about  $\mathrm{SO}(n, \mathbf{No})$ . Proposition 11.1 sits between the two: it concerns the finite side like the rotation theorems but uses the elementary mechanism. It does not imply the rotation theorems either, since a homomorphism defined on the subgroup  $\mathrm{SO}(n, \mathbf{No}) \subseteq \mathrm{SL}_n(\mathcal{O}_{\mathrm{fin}}(\mathbb{R}))$  need not extend. The abelian boundaries also correspond: the circle characters of `e3:cut:prop:circlecounter`, the characters of  $\mathrm{UT}_n(\Pi_k)$  in section 6, and the rank-two countable detection of section 15.

## 12 Sharp boundaries of the argument

### 12.1 A fixed Hahn workspace is not the full class

Let  $\Gamma$  be a nonzero set-sized divisible ordered abelian group, write large monomials  $X^\gamma$ , and put  $\Pi_k(\Gamma) = \{f : \text{supp } f \subseteq \Gamma^{>0}\}$  and  $\mathcal{A}_{D,k}(\Gamma) = D \oplus \Pi_k(\Gamma)$  inside the Hahn field with set-sized reverse-well-ordered supports; these rings are sets.

The geometric dichotomy of [Part I](#) persists (06): the real Hahn field over a divisible group is real closed and the complex one is algebraically closed [[14](#), Corollary 4]; unit and bounded-coordinate rigidity are unchanged; and a positive exponent gives a nonconstant parameter. So the proofs of [theorems 4.4](#) and [4.5](#) apply to the Hahn-cone rings in place of  $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$ , with integral constant rings as before.

Invisibility cannot persist: the purely infinite elementary group of a nontrivial set-sized workspace is itself a set-sized nontrivial group, and its identity homomorphism is faithful. The full-class proofs need arbitrarily many scales below a support gap (06), and the compression of [lemma 8.4](#) produces a field that need not lie in any preassigned set-sized subfield or birthday-bounded fragment (07).

**Proposition 12.1** (Local cardinal form [06]). *Fix  $0 \neq c \in \Pi_k(\Gamma)$  and  $0 < \delta < \text{supp } c$ . Let  $T \subseteq \Gamma^{>0}$  satisfy  $m\alpha < \delta$  for all  $\alpha \in T$  and ordinary  $m \geq 1$ . For every homomorphism from the group generated by the purely infinite elementary matrices of rank  $n \geq 3$  over this workspace to a group  $H$  with  $|T| > |H|$ , every root element  $e_{ij}(c)$  maps to the identity.*

*Proof.* The second proof of [theorem 9.1](#) applies verbatim: for the two colliding exponents  $\alpha > \beta$ ,  $\alpha + m(\alpha - \beta) \leq (m+1)\alpha < \delta$ , so  $c/(X^\alpha - X^\beta)$  lies in the workspace with strictly positive support. No normality or perfectness in the workspace is needed.  $\square$

This is a conditional lower bound on target size, not an assertion that every exponent group supplies such a  $T$ : for an Archimedean group no positive  $\alpha$  has all  $m\alpha < \delta$  (06). [Proposition 9.7](#) can be applied at a set-sized stage for targets smaller than an available parameter family; an optimal threshold there requires further information about support bounds and the exponent group (07; [question 34.4](#)). For the rings of supports of size  $< \kappa$  over  $\bigoplus_{\alpha < \kappa} \mathbb{Q}$  and for the sibling report's other set-sized models this information is available, and the optimal threshold is determined in [proposition 21.12](#) and [corollary 21.13](#) [merge].

### 12.2 Finite supports change the answer

Let  $\Pi_k^{\text{fin}}$  consist of the purely infinite normal forms with finite support and  $A^{\text{fin}} = D + \Pi_k^{\text{fin}}$ ; the exponent class is still all of **No**.

**Proposition 12.2** (A visible finite-support kernel [06, 11]). *The coefficient sum  $\sum_s a_s \omega^s \mapsto \sum_s a_s$  is a ring homomorphism  $A^{\text{fin}} \rightarrow k$ , and for  $n \geq 3$  it maps the group generated by the purely infinite elementary matrices onto the set-sized group  $\text{SL}_n(k)$ .*

*Proof.* Sums and products involve finite supports, so expanding a product and summing coefficients gives multiplicativity; it is evaluation  $\omega \mapsto 1$  on the group ring  $k[\mathbf{No}]$ . For  $r \in k$  and  $a > 0$  the purely infinite  $r\omega^a$  maps to  $r$ , so every  $e_{ij}(r)$  is in the image, and these generate  $\text{SL}_n(k)$ .  $\square$

The finite-support ideal still has the single-product property, so its elementary kernel is perfect and normal; nevertheless it has a large ordinary image. Infinite supports supply

the exact divisor (8.1), which finite truncations cannot replace (06). Source 11 proves the nontriviality part (its Proposition 11.3: the same evaluation maps the finite-support kernel nontrivially to  $E_n(k)$ , since  $e_{12}(\omega) \mapsto e_{12}(1)$ ) and notes that the evaluation does not extend to the full ring, by the ring theorem. This is the group form of the finite-support augmentation in the sibling report's support threshold (`osq:thm:support`).

### 12.3 Rank two and the full special linear group

All three proofs need a third index to convert a killed additive root parameter into an arbitrary multiplicative one (the third needs it to make the common root kernel an ideal), so none says anything about rank two; and the theorem concerns  $E_n(A)$ , not  $\mathrm{SL}_n(A)$ , and uses no stable-range theorem for an omnific ring (06, 07, 11). Source 06 left both issues open, and so does source 11 (its Questions 11.1 and 11.2), whose pin already contained 07's answer in this report; 11 states  $n \geq 3$  as a proved range, not as a claim that  $n = 2$  is a sharp negative boundary. Rank two is settled by source 07 in the next sections: invisibility fails there (theorem 15.3). Whether the universal set-sized quotient of  $\mathrm{SL}_n(\mathbf{Oz})$ ,  $n \geq 3$ , is  $\mathrm{SL}_n(\mathbb{Z})$  remains open (question 34.1); for  $n = 2$ ,  $\mathrm{SL}_2(\mathbf{Oz}) \neq E_2(\mathbf{Oz})$  (theorem 18.2).

## 13 Two rows: an amalgam detected by finite and infinite points

From here to the end of section 18 the source is 07, and  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $A = \mathcal{A}_{D,k}$ ,  $K_D = \mathrm{Frac} D$ . The proof of theorem 9.1 cannot be repeated with two rows, because it uses a third index. The two-row group is identified explicitly instead. The argument is a specialized elementary form of Cohn's classical two-row normal-form method [4]; it also gives the Gaussian presentation needed later.

### 13.1 The upper triangular subgroup

Write  $G_D = \mathrm{SL}_2(D)$ ,

$$C_D = \left\{ \begin{pmatrix} u & d \\ 0 & u^{-1} \end{pmatrix} : u \in D^\times, d \in D \right\}, \quad B_D = \left\{ \begin{pmatrix} u & r \\ 0 & u^{-1} \end{pmatrix} : u \in D^\times, r \in A \right\},$$

$U(x) = e_{12}(x)$  and  $U(\Pi_k) = \{U(x) : x \in \Pi_k\}$ . By lemma 2.1,  $B_D$  is the whole upper triangular subgroup of  $\mathrm{SL}_2(A)$ ; each member is uniquely  $U(x)c$  with  $x \in \Pi_k$  and  $c = \mathrm{ct}$  of it in  $C_D$ . For  $c$  with diagonal  $u, u^{-1}$ ,

$$cU(x)c^{-1} = U(u^2x), \quad \text{so} \quad B_D = U(\Pi_k) \rtimes C_D. \quad (13.1)$$

The action is trivial for  $D = \mathbb{Z}$  and by the signs  $\pm 1$  for  $D = \mathbb{Z}[i]$ . Both  $G_D$  and  $B_D$  lie in  $E_2(A)$ , they generate it (every upper elementary matrix is in  $B_D$ , and  $SU(x)S^{-1} = e_{21}(-x)$  with  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in G_D$ ), and  $G_D \cap B_D = C_D$ .

### 13.2 A projective separation argument

In  $\mathbb{P}^1(F) = F \cup \{\infty\}$  let  $P_{\mathrm{fin}} = \{z \in F : |z| < m \text{ for some ordinary } m\}$  be the finite points and  $P_\infty$  the rest (infinite elements and  $\infty$ ), with the fractional-linear action of determinant-one matrices.

**Lemma 13.1** (Alternation of finite and infinite points [07]).

1. Every  $b \in B_D \setminus C_D$  maps  $P_{\text{fin}}$  into  $P_\infty$  and fixes  $\infty$ .
2. Every  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G_D \setminus C_D$  maps  $P_\infty$  into  $P_{\text{fin}}$ , has  $c \neq 0$ , and  $\text{st}(gz) = a/c \in K_D$  for  $z \in P_\infty$  (exactly  $a/c$  at  $z = \infty$ ).

*Proof.* (i) Write  $b = U(x)c'$  with  $0 \neq x \in \Pi_k$ ,  $c' \in C_D$ :  $c'$  preserves finite points and fixes  $\infty$ , and adding the nonzero purely infinite  $x$  makes a finite point infinite. (ii)  $c \neq 0$  since  $g$  is not upper triangular; its pole  $-d/c$  is constant, hence not in  $P_\infty$ . For infinite  $z$ ,

$$\frac{az + b}{cz + d} - \frac{a}{c} = \frac{bc - ad}{c(cz + d)} = -\frac{1}{c(cz + d)}, \quad (13.2)$$

which is infinitesimal.  $\square$

**Theorem 13.2** (The two-row amalgam [07]). *The inclusions induce an isomorphism  $E_2(A) \cong G_D *_{C_D} B_D$ .*

*Proof.* Surjectivity is generation. For injectivity use reduced words of the amalgam: a nontrivial reduced word is conjugate to a nonidentity element of one factor or to a cyclically reduced word  $g_1 b_1 g_2 b_2 \cdots g_m b_m$  with  $g_j \in G_D \setminus C_D$ ,  $b_j \in B_D \setminus C_D$ . These finite-word rules are valid for class groups: words are finite, coset representatives are sets, and each shortening step reduces the finite length. Apply the matrix of the cyclic word to  $\infty$ :  $b_m$  fixes it,  $g_m$  sends it into  $P_{\text{fin}}$ ,  $b_{m-1}$  back into  $P_\infty$ , and so on (lemma 13.1); the final image is finite, so the matrix is not the identity. Nonidentity elements of the factors are nonidentity matrices, and conjugation preserves this.  $\square$

No convergence, continued infinite fraction or action on a set-sized tree is involved, and the Gaussian case needs no order on the coefficients (07). The proof used only that the constant subgroup is  $\text{SL}_2(D)$ , that nonconstant upper triangular translations are infinite, that all units are constant, and that a non-upper-triangular constant matrix sends infinite points to finite ones. So it applies to the subgroup of  $\text{SL}_2(K_D(t))$  generated by  $G_D$  and  $B'_D = \{U(tq)c : q \in \mathbb{Q}, c \in C_D\}$ , using degree at infinity or the injective embedding  $K_D(t) \rightarrow F$ ,  $t \mapsto \omega$ :

$$\langle G_D, B'_D \rangle \cong G_D *_{C_D} B'_D. \quad (13.3)$$

## 14 The two-row kernel as a free product

$G_D$  acts transitively on  $\mathbb{P}^1(K_D)$  with stabilizer  $C_D$  at  $\infty$  (a point is represented by a coprime pair  $(a, c) \in D^2$ , which Bézout completes to a first column in  $\text{SL}_2(D)$ ), so  $G_D/C_D \cong \mathbb{P}^1(K_D)$ , a countable set. Choose a transversal  $\mathcal{T} \subset G_D$  of these left cosets containing 1.

**Theorem 14.1** (Kernel decomposition [07]). *There is an isomorphism  $P_2 \cong *_{t \in \mathcal{T}} \Pi_k^{(t)}$ , each  $\Pi_k^{(t)}$  a copy of  $(\Pi_k, +)$  embedded by  $x \mapsto tU(x)t^{-1}$ , and*

$$E_2(A) \cong \left( *_{t \in \mathcal{T}} \Pi_k^{(t)} \right) \rtimes G_D,$$

where, if  $gt = t'c$  with  $t' \in \mathcal{T}$ ,  $c \in C_D$  of first diagonal entry  $u$ , the action sends  $x$  in factor  $t$  to  $u^2x$  in factor  $t'$ .

*Proof.* Let  $\mathcal{F}$  be the free product with inclusions  $\iota_t$ , and let  $G_D$  act by  $g \cdot \iota_t(x) = \iota_{t'}(u^2x)$  when  $gt = t'c$ . Each  $u^2 = \pm 1$  gives an automorphism of the factor; coset factorization is unique, and

if  $g_2t = t'c_2$ ,  $g_1t' = t''c_1$  then  $g_1g_2t = t''c_1c_2$  with multiplier the product, so this is an action. In  $\mathcal{F} \rtimes G_D$ ,  $\iota_1(\Pi_k)$  and  $C_D$  form  $\Pi_k \rtimes C_D$  with the action (13.1), which gives a homomorphism from  $B_D$  agreeing with  $G_D$  on  $C_D$ , hence  $G_D *_C B_D \rightarrow \mathcal{F} \rtimes G_D$ . Conversely send  $G_D$  to its constant matrices and  $\iota_t(x)$  to  $tU(x)t^{-1}$ ; the defining relation of the action is (13.1). The two maps are inverse on generators; under the isomorphism, constant reduction is the projection to  $G_D$ , whose kernel is  $\mathcal{F}$ . Combine with theorem 13.2.  $\square$

For  $D = \mathbb{Z}$  every  $u^2 = 1$  and the action permutes factors indexed by  $\mathbb{P}^1(\mathbb{Q})$ . In the Gaussian case a change of representative can act on a factor by a sign; omitting it would misdescribe the semidirect product (07).

**Corollary 14.2** (Internal abelianization and torsion [07]).  $P_2^{\text{ab}} \cong \bigoplus_{q \in \mathbb{P}^1(K_D)} (\Pi_k, +)$ . The group  $P_2$  is torsion-free and not perfect.

*Proof.* A free product abelianizes to the direct sum of the abelianized factors, here abelian and nonzero. A cyclically reduced word with one syllable lies in a torsion-free factor; with several syllables its powers concatenate without cancellation. Conjugation preserves finite order.  $\square$

Torsion-freeness also follows, by a different route, from source 11's trace argument, which applies to the constant-term kernel of  $\text{GL}_n$  for every  $n$  (theorem 19.1).

This internal abelianization must not be confused with the image of  $P_2$  in  $E_2(A)^{\text{ab}}$ , where the constant group can identify or kill classes (section 17).

## 15 Countable representations separating finite sets

The needed additive functionals can be built without a basis of the proper-class vector space  $\Pi_k$  (07).

**Lemma 15.1** (A set-based additive functional [07]). *For finitely many nonzero  $x_1, \dots, x_m \in \Pi_k$  there is a  $\mathbb{Q}$ -linear class map  $\ell : \Pi_k \rightarrow \mathbb{Q}$  with  $\ell(x_j) \neq 0$  for all  $j$ , factoring through finitely many coefficient extractions into the set  $k^r$ .*

*Proof.* Select for each  $j$  one exponent where  $x_j$  has a nonzero coefficient; list the selected exponents  $s_1, \dots, s_r$  and put  $\mathbf{p}(x) = ([\omega^{s_1}]x, \dots, [\omega^{s_r}]x) \in k^r$ . Each  $\mathbf{p}(x_j) \neq 0$ . Let  $W \subseteq k^r$  be the finite-dimensional  $\mathbb{Q}$ -span of these vectors with basis  $w_1, \dots, w_d$ , and  $f_M(w_h) = M^{h-1}$ ;  $f_M(\mathbf{p}(x_j))$  is a nonzero polynomial in  $M$ , so some integer  $M$  avoids the finitely many roots. Extend  $f_M$   $\mathbb{Q}$ -linearly to  $k^r$  by extending a basis of the set  $W$ , and set  $\ell = f_M \circ \mathbf{p}$ .  $\square$

This is an existence construction with set choice, not a computability claim about arbitrary real or complex coefficients (07).

**Proposition 15.2** (Additive specialization [07]). *For every additive  $\ell : \Pi_k \rightarrow \mathbb{Q}$  there is a unique homomorphism  $\rho_\ell : E_2(A) \rightarrow \text{SL}_2(K_D(t))$  with  $\rho_\ell(g) = g$  for  $g \in G_D$  and  $\rho_\ell(U(x)) = U(t\ell(x))$  for  $x \in \Pi_k$ .*

*Proof.* On  $B_D = U(\Pi_k) \rtimes C_D$  send  $U(x)c \mapsto U(t\ell(x))c$ . Root addition is preserved, and compatibility with the  $C_D$ -action is  $\ell(u^2x) = u^2\ell(x)$ , true since  $u^2 = \pm 1$ . This agrees with the identity on  $C_D$ ; theorem 13.2 extends it, and generation gives uniqueness.  $\square$

In the Gaussian case  $\ell$  need not be complex-linear or satisfy  $\ell(ix) = i\ell(x)$ : diagonal conjugation only involves  $i^2 = -1$ . This is why Gaussian countable representations survive although the Gaussian abelian characters behave differently (07).

**Theorem 15.3** (One countable target separates finite patterns [07]). *For every finite  $\mathcal{S} \subset E_2(A)$  some  $\rho_\ell$  is injective on  $\mathcal{S}$ . In particular every nonidentity element of  $E_2(A)$  is detected by a homomorphism into the fixed countable group  $\mathrm{SL}_2(K_D(t))$  that is the identity on  $\mathrm{SL}_2(D)$ .*

*Proof.* Keep nontrivial each  $a^{-1}b$ ,  $a \neq b$  in  $\mathcal{S}$ . Write them in reduced amalgam form; the  $B_D \setminus C_D$  syllables are  $U(x)c$  with  $x \neq 0$ , finitely many. Choose  $\ell$  nonzero on all these  $x$  (lemma 15.1). Under  $\rho_\ell$ ,  $G_D \setminus C_D$  syllables are unchanged and  $B_D \setminus C_D$  syllables go to  $B'_D \setminus C_D$ , so reduced words stay reduced in (13.3) and are nontrivial; one-factor words are preserved as well.  $K_D(t)$  is countable.  $\square$

**Example 15.4** (A commutator needing a nonabelian detector [07]). Let  $\ell$  extract the coefficient of  $\omega^1$  (followed by a  $\mathbb{Q}$ -linear map taking 1 to 1). Then

$$\rho_\ell([U(\omega), e_{21}(\omega)]) = [U(t), e_{21}(t)] = \begin{pmatrix} 1 + t^2 + t^4 & -t^3 \\ t^3 & 1 - t^2 \end{pmatrix} \neq I_2,$$

a countable nonabelian image of a commutator in  $P_2$ , killed of course by every abelian image; valid over both coefficient rings.

**Corollary 15.5** (No universal set-sized quotient in rank two [07]). *Neither  $E_2(A)$  nor  $P_2$  has a universal set-sized group quotient.*

*Proof.* If  $q : E_2(A) \rightarrow H$  were one, every  $\rho_\ell$  would factor through  $q$ , so  $\ker q = 1$  by theorem 15.3; but  $U(\Pi_k)$  contains more than  $|H|$  distinct monomials. Restricting the  $\rho_\ell$  to  $P_2$  gives the same for  $P_2$ .  $\square$

There is no contradiction between residual countability and proper-class size: the separating maps are not a set-indexed family whose product is a faithful set-sized representation, and no single faithful countable representation is asserted (07).

## 16 Finite quotients and the detection trichotomy

**Lemma 16.1** (Divisibility kills finite root images [07]). *Every homomorphism from  $(\Pi_k, +)$  to a finite group is trivial.*

*Proof.* With  $m = |H|$ ,  $x/m \in \Pi_k$  and  $f(x) = f(x/m)^m = 1$ .  $\square$

**Proposition 16.2** (All finite images are arithmetic [07]). *For every  $n \geq 2$  every homomorphism from  $E_n(A)$  to a finite group factors uniquely through  $\mathrm{ct}$ , and every homomorphism from  $P_2$  to a finite group is trivial.*

*Proof.* A finite-target map kills each  $e_{ij}(\Pi_k)$  (lemma 16.1) and hence its normal closure  $P_n$  (lemma 7.1). For  $P_2$  use the free product (theorem 14.1).  $\square$

**Lemma 16.3** (Finite detection in the constant group [07]).  *$\mathrm{SL}_n(D)$  is residually finite for  $n \geq 2$ .*

*Proof.* If  $g \neq 1$ , a nonzero entry  $a$  of  $g - 1$  is not in  $mD$  for a suitable ordinary  $m \geq 2$  (for Gaussian  $a$ , take  $m$  larger than both integer coordinates), and reduction to  $\mathrm{SL}_n(D/mD)$  does not kill  $g$ .  $\square$

**Theorem 16.4** (Finite and set residuals [07]). *For every  $n \geq 2$ ,  $\text{Res}_{\text{fin}} E_n(A) = P_n$ . The set residual (the class of elements killed by every homomorphism to a set-sized group) satisfies  $\text{Res}_{\text{set}} E_2(A) = 1$  and  $\text{Res}_{\text{set}} E_n(A) = P_n$  for  $n \geq 3$ . For the kernels,  $\text{Res}_{\text{fin}} P_2 = P_2$ ,  $\text{Res}_{\text{set}} P_2 = 1$  and  $\text{Res}_{\text{set}} P_n = P_n$  for  $n \geq 3$ .*

*Proof.* Finite maps kill  $P_n$  (proposition 16.2), and elements outside  $P_n$  survive a finite map (lemma 16.3). The set residuals follow from theorems 9.1 and 15.3, and those of the kernels also from the last clause of proposition 16.2.  $\square$

The finite quotient theory of  $E_n(A)$  is exactly that of  $\text{SL}_n(D)$ ; its profinite completion, defined by this set-indexed system, is that of  $\text{SL}_n(D)$ . No congruence subgroup property is involved: arbitrary finite quotients are identified, and congruence reduction was used only for lemma 16.3 (07). For  $n \geq 3$  source 11 proves the finite and set residual statements again (its Theorem 9.1, by the same reduction modulo  $m$ ), and for  $\mathbf{Oz}$  it then identifies the completion with  $\text{SL}_n(\widehat{\mathbb{Z}})$  by importing the congruence subgroup theorem (corollary 19.7); that corollary is the only place in the report where a congruence subgroup property enters.

**Theorem 16.5** (The detection trichotomy [07]). *Let  $n \geq 2$  and  $1 \neq g \in E_n(A)$ . Exactly the applicable alternative holds:*

1. if  $\text{ct}(g) \neq 1$ , some finite group homomorphism detects  $g$ ;
2. if  $n = 2$  and  $g \in P_2$ , no finite group homomorphism detects  $g$ , a countable one does, and the least cardinality of a detecting target is  $\aleph_0$ , also for homomorphisms defined only on  $P_2$ ;
3. if  $n \geq 3$  and  $g \in P_n$ , no homomorphism to any set-sized group detects  $g$ , also for homomorphisms defined only on  $P_n$ .

*Proof.* Lemma 16.3, proposition 16.2, and theorems 9.1 and 15.3. The last alternative is the nonexistence of any set-sized detecting target, not an “infinite cardinal larger than all cardinals”.  $\square$

| Element                      | Smallest detecting target | Reason                      |
|------------------------------|---------------------------|-----------------------------|
| $\text{ct}(g) \neq 1$        | Some finite group         | Reduction modulo $m$        |
| $1 \neq g \in P_2$           | Countably infinite        | Additive coefficient maps   |
| $1 \neq g \in P_n, n \geq 3$ | No set-sized group        | Collision and a third index |

## 17 Abelian characters: a real–Gaussian contrast

**Proposition 17.1** (The real abelianization, a classical specialization [07]).  $E_2(\mathbf{Oz})^{\text{ab}} \cong \mathbb{Z}/12 \oplus (\Pi, +)$ . *For every  $s \in \mathbf{No}^{>0}$  there is a surjective homomorphism  $\chi_s : E_2(\mathbf{Oz}) \rightarrow (\mathbb{R}, +)$  with  $\chi_s(\text{SL}_2(\mathbb{Z})) = 0$  and  $\chi_s(U(x)) = [\omega^s]x$  for  $x \in \Pi$ .*

*Proof.* For a split group  $\mathcal{F} \rtimes G$ , the abelianization is  $G^{\text{ab}}$  plus the  $G$ -coinvariants of  $\mathcal{F}^{\text{ab}}$ . By theorem 14.1 and corollary 14.2,  $\mathcal{F}^{\text{ab}}$  is a direct sum of copies of  $\Pi$  on which  $\text{SL}_2(\mathbb{Z})$  acts by permuting the indices transitively without twist; the coinvariants are one copy of  $\Pi$  (sum of the finitely many nonzero coordinates). And  $\text{SL}_2(\mathbb{Z})^{\text{ab}} = \mathbb{Z}/12$ : from the presentation with generators  $S, U(1)$  and relations  $S^4 = 1, S^2 = (SU(1))^3$ , an abelian quotient has  $[S] = -3[U(1)]$

and  $12[U(1)] = 0$  and nothing more. Project the second summand to its  $\omega^s$ -coefficient for  $\chi_s$ ;  $\chi_s(U(r\omega^s)) = r$ .  $\square$

The formula is not a novelty claim (07). It is the omnific case of Cohn's abelianization theorem [4] in the formulation of Mirzaii and Torres Pérez [13]: their Corollary 3.3 (credited to Cohn) gives  $E_2(R)^{\text{ab}} \cong R/M$  for rings universal for  $\text{GE}_2$ , with  $M$  the additive subgroup generated by  $x(a^2 - 1)$  and  $3(b+1)(c+1)$  ( $x \in R$ ,  $a, b, c \in R^\times$ ); their Example 3.4(i) gives  $R/M = R/12\mathbb{Z}$  when  $R^\times = \{\pm 1\}$ ; and their Example 1.1(vi) records Cohn's theorem that discretely ordered rings are universal for  $\text{GE}_2$ . These numbers were checked against the arXiv version 3 of [13] for this report. Those statements concern rings that are sets; for the class ring  $\mathbf{Oz}$ , discretely ordered with units  $\pm 1$ , the proof above is self-contained. The quotient is by the additive subgroup  $12\mathbb{Z}$ , not by the ideal  $12\mathbf{Oz} = 12\mathbb{Z} + \Pi$ , which would wrongly discard the purely infinite summand (07).

**Proposition 17.2** (Gaussian ambient abelianization [07]). *Constant reduction induces  $E_2(\mathbf{Oz}[i])^{\text{ab}} \cong \text{SL}_2(\mathbb{Z}[i])^{\text{ab}}$ . Every homomorphism from  $E_2(\mathbf{Oz}[i])$  to an abelian group kills  $P_2$ .*

*Proof.* With  $d(i) = \text{diag}(i, -i)$ ,  $d(i)U(y)d(i)^{-1} = U(-y)$ , and since  $x/2 \in \Pi_{\mathbb{C}}$ ,

$$[d(i), U(-x/2)] = U(x) \quad (x \in \Pi_{\mathbb{C}}). \quad (17.1)$$

Conjugating by  $S$  handles lower roots, and normal closure gives  $P_2 \subseteq [E_2, E_2]$ . The constant inclusion is an inverse on abelianizations.  $\square$

No numerical description of  $\text{SL}_2(\mathbb{Z}[i])^{\text{ab}}$  is needed or claimed. The comparison is new in 07: purely infinite Gaussian information is absent from every abelian image but visible in countable nonabelian images (theorem 15.3). This does not make  $P_2$  perfect:  $d(i) \notin P_2$ , and  $P_2^{\text{ab}}$  is still the large sum of corollary 14.2; for  $n \geq 3$  the commutators of proposition 7.2 have both entries in  $P_n$ .

## 18 Cusp residues, non-elementary unipotents, and stabilization

**Proposition 18.1** (The finite cusp-residue obstruction [07]). *If  $h \in E_2(A)$  and  $h \cdot \infty$  is finite, then  $\text{st}(h \cdot \infty) \in K_D$ .*

*Proof.* By theorem 13.2 write  $h = a_0 U(x_1) a_1 \cdots a_{m-1} U(x_m) a_m$  with nonzero  $x_j \in \Pi_k$ , interior  $a_j \in G_D \setminus C_D$  and endpoint  $a_0, a_m \in G_D$  (absorb triangular constant parts into neighbours, move interior triangular constants through upper roots by (13.1), merge adjacent roots, and shorten when a parameter vanishes; the process is finite). If  $m = 0$  the claim is clear. If  $m \geq 1$ ,  $a_m$  sends  $\infty$  to  $\infty$  or to a point of  $K_D$ ,  $U(x_m)$  sends that point into  $P_\infty$ , each interior  $a_j$  sends  $P_\infty$  into  $P_{\text{fin}}$  with standard part in  $K_D$ , and the next root sends it back into  $P_\infty$ . If  $a_0 \in C_D$  the final point is in  $P_\infty$ , contrary to finiteness; otherwise lemma 13.1(ii) applies.  $\square$

The obstruction is stronger than looking at constant reduction: a matrix can reduce to the identity and still fail it, because a ratio of its large entries has a nonarithmetic finite standard part. The polynomial pattern  $\begin{pmatrix} 1+xy & x^2 \\ -y^2 & 1-xy \end{pmatrix}$  is Cohn's example of a non-elementary matrix over  $k[x, y]$  [4, Section 7, p. 26]; neither it nor the block identity below is claimed as new (07).

**Theorem 18.2** (A non-elementary omnific unipotent [07]). *Let  $\alpha \in k \setminus K_D$ ,  $a > 0$ , and*

$$Q_\alpha = \begin{pmatrix} \alpha & 1 \\ -\alpha^2 & -\alpha \end{pmatrix}, \quad M_\alpha = I_2 + \omega^{2a} Q_\alpha.$$

Then  $Q_\alpha^2 = 0$ ,  $\det M_\alpha = 1$ ,  $\text{ct}(M_\alpha) = I_2$ ,  $M_\alpha$  has infinite order, and  $M_\alpha \in \text{SL}_2(A) \setminus E_2(A)$ .

*Proof.* The square and determinant are polynomial identities; the nonconstant entries have exponent  $2a > 0$  and coefficients in  $k$ . With  $T = \omega^{2a}$ ,  $M_\alpha \cdot \infty = (1 + \alpha T)/(-\alpha^2 T) = -1/\alpha - 1/(\alpha^2 T)$  is finite with standard part  $-1/\alpha \notin K_D$ , so [proposition 18.1](#) excludes  $E_2(A)$ . Finally  $M_\alpha^m = I_2 + m\omega^{2a}Q_\alpha \neq I_2$ .  $\square$

For  $a = 1/2$  and  $\alpha = \sqrt{2}$ ,

$$M = \begin{pmatrix} 1 + \sqrt{2}\omega & \omega \\ -2\omega & 1 - \sqrt{2}\omega \end{pmatrix} \in \text{SL}_2(\mathbf{Oz}) \setminus E_2(\mathbf{Oz}), \quad (18.1)$$

and the same matrix is non-elementary over  $\mathbf{Oz}[i]$  since  $\sqrt{2} \notin \mathbb{Q}(i)$ . This needs no factorization theory of omnific integers. It is consistent with the omnific Diophantine report's non-Euclidean phenomena (pairs without a gcd, `odg:thm:nogcd`), and with its rationality theorem for primitive constant directions (`odg:thm:realray`), which concerns constant real projective points; the first column of  $M_\alpha$  is not one.

**Lemma 18.3** (A finite block identity [07]). *Over any commutative ring, for a column  $v$  and a row  $w$  of length two with  $wv = 0$ ,*

$$\left[ \begin{pmatrix} I_2 & v \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} I_2 & 0 \\ w & 1 \end{pmatrix} \right] = \begin{pmatrix} I_2 + vw & 0 \\ 0 & 1 \end{pmatrix}.$$

*Proof.* The inverses replace  $v, w$  by their negatives; multiply the four blocks, and the terms containing  $wv$  vanish.  $\square$

**Theorem 18.4** (One-step stabilization [07]). *With  $z = \omega^a$ ,  $v = z(1, -\alpha)^\top$ ,  $w = z(\alpha, 1)$ ,  $X = e_{13}(z)e_{23}(-\alpha z)$  and  $Y = e_{31}(\alpha z)e_{32}(z)$ , one has  $X, Y \in P_3$  and  $[X, Y] = \text{diag}(M_\alpha, 1)$ . Hence this nonidentity matrix is killed by every homomorphism from  $P_3$  or  $E_3(A)$  to a set-sized group.*

*Proof.* All parameters lie in  $\Pi_k$ .  $X$  and  $Y$  are the block matrices of [lemma 18.3](#),  $wv = z^2(\alpha - \alpha) = 0$  and  $vw = z^2Q_\alpha$ . Apply [theorem 9.1](#).  $\square$

The commutator is an eight-factor elementary word. The identity itself does not depend on surreals; the surreal content is the coefficient ring, the residue obstruction before stabilization and the invisibility after it (07).

**Corollary 18.5** (A strict failure of extension [07]). *Let  $j : E_2(A) \rightarrow E_3(A)$ ,  $g \mapsto \text{diag}(g, 1)$ . If  $\rho : E_2(A) \rightarrow H$ ,  $H$  set-sized, detects a nonidentity element of  $P_2$ , then  $\rho$  does not extend along  $j$  to a homomorphism into  $H$ , nor into any set-sized group containing  $H$ .*

*Proof.*  $j(P_2) \subseteq P_3$ , which every set-target homomorphism kills ([theorem 9.1](#)); an injective enlargement of  $H$  preserves  $\rho(g) \neq 1$ .  $\square$

For example  $U(\omega)$  is detected in two rows by a map to  $U(t)$ , but its block inclusion into three rows is killed by every set-sized representation. The embedding is injective; what changes is the class of homomorphisms permitted by the extra relations (07).

The two parts of this report meet at one point. In [section 30](#) bases of  $\mathcal{A}^n$  are changed by  $\text{GL}_n(\mathcal{A})$ ; [theorem 18.2](#) shows that already  $\text{SL}_2(\mathbf{Oz})$  is not reached by elementary moves, while [theorem 30.3](#) obstructs the *existence* of reduced bases whatever group is used.

## 19 The kernel for $n \geq 3$ : torsion, center and normal subgroups

This section and the next two are source 11's, with two merge results at the end of [section 21](#). Throughout them  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $A = \mathcal{A}_{D,k}$ , and  $P_n$  is the kernel [\(7.2\)](#); for  $n \geq 3$ ,  $P_n = \langle e_{ij}(\Pi_k) \rangle$  and  $E_n(A) = P_n \rtimes \mathrm{SL}_n(D)$  ([theorem 9.1](#)). Source 11 writes  $U_n(\Pi_k)$  for the root-generated subgroup and  $N_n(A)$  for the kernel, deliberately never identifies them, and proves its structural statements for both. By [proposition 7.3](#) (06) they are one group, so the statements are printed for  $P_n$ ; none of 11's proofs uses the identification.

### 19.1 Torsion and center

**Theorem 19.1** (Torsion-free constant-term kernels [11]). *For every  $n \geq 1$  the group  $\ker(\mathrm{ct} : \mathrm{GL}_n(A) \rightarrow \mathrm{GL}_n(D))$  is torsion-free. In particular  $P_n$  is torsion-free for every  $n \geq 2$ .*

*Proof.* Let  $g^r = 1_n$  with  $r \geq 1$  and  $\mathrm{ct}(g) = 1_n$ . Work in the algebraically closed field  $\mathbf{No}[i]$ , which contains both coefficient cases. The polynomial  $X^r - 1$  has the  $r$  distinct ordinary complex  $r$ th roots of unity as its roots, and the minimal polynomial of  $g$  divides it; so  $g$  is diagonalizable over  $\mathbf{No}[i]$  with eigenvalues  $\zeta_1, \dots, \zeta_n \in \mathbb{C}$ . Hence  $\mathrm{tr} g = \sum_j \zeta_j$  is an ordinary complex number, in particular finite; it lies in  $A$ , whose finite elements are those of  $D$  ([lemma 2.1](#)(iii)). So  $\mathrm{tr} g = \mathrm{ct}(\mathrm{tr} g) = \mathrm{tr} \mathrm{ct}(g) = n$ . Each  $\zeta_j$  has modulus one, so its real part is at most 1, and the real parts sum to  $n$ ; hence every  $\zeta_j = 1$ , and the diagonalizable  $g$  is  $1_n$ .  $\square$

For  $n = 2$  this is a second route to the torsion-freeness of  $P_2$  in [corollary 14.2](#), where 07 derives it from the free product. The theorem concerns the constant-term kernel only; the constant subgroup  $\mathrm{SL}_n(D)$  has torsion.

**Theorem 19.2** (Centerlessness [11]). *For  $n \geq 3$ ,  $P_n$  has trivial center.*

*Proof.* Let  $g$  be central and fix  $0 \neq p \in \Pi_k$ . Since  $e_{ij}(p) = 1_n + pE_{ij} \in P_n$  and  $p \neq 0$  in the field  $F$ ,  $gE_{ij} = E_{ij}g$  for all  $i \neq j$ . Comparing entries,  $g_{ai}\delta_{jb} = \delta_{ai}g_{jb}$ : with  $b = j$  and  $a \neq i$  every off-diagonal entry vanishes, and with  $a = i$ ,  $b = j$  one gets  $g_{ii} = g_{jj}$ . So  $g = \lambda 1_n$  with  $\lambda, \lambda^{-1} \in A$ , hence  $\lambda \in D^\times$  ([lemma 2.1](#)(ii)), and  $\mathrm{ct}(g) = 1_n$  forces  $\lambda = 1$ .  $\square$

$P_n$  is nonabelian:  $[e_{12}(\omega), e_{23}(\omega)] = e_{13}(\omega^2) \neq 1_n$ .

**Corollary 19.3** (Set-sized normal subgroups and conjugacy classes [11]). *For  $n \geq 3$ ,  $P_n$  has no nontrivial set-sized normal subgroup, and the conjugacy class in  $P_n$  of every nonidentity element is a proper class.*

*Proof.* If  $S \triangleleft P_n$  is a set, conjugation is a homomorphism  $P_n \rightarrow \mathrm{Aut}(S)$  into a set-sized group, trivial by [theorem 9.1](#)(iii); so  $S$  is central and  $S = 1$  by [theorem 19.2](#). If the conjugacy class of  $x \neq 1$  were a set  $X$ , conjugation would give a homomorphism  $P_n \rightarrow \mathrm{Sym}(X)$ , again trivial; then  $x$  would be central.  $\square$

This is consistent with the tower of [theorem 10.1](#): its members  $P_n^{(a)}$  are nontrivial proper normal subgroups, and so are proper classes; invisibility does not make  $P_n$  simple.

**Proposition 19.4** (Finite subgroups are detected by constant terms [11]). *Let  $n \geq 3$ . Every finite subgroup  $\Theta \leq E_n(A)$  is mapped injectively into  $\mathrm{SL}_n(D)$  by  $\mathrm{ct}$ . The isomorphism types of finite subgroups of  $E_n(A)$  are exactly those of  $\mathrm{SL}_n(D)$ .*

*Proof.* The kernel of  $\text{ct}|_{\Theta}$  is a finite subgroup of the torsion-free group  $P_n$  ([theorem 19.1](#)), hence trivial. Conversely the constant inclusion embeds every finite subgroup of  $\text{SL}_n(D)$ .  $\square$

The statement concerns isomorphism types and injectivity. It is not claimed that every finite subgroup of  $E_n(A)$  is conjugate to a constant subgroup; that would need a separate analysis of splittings over finite groups (11).

## 19.2 A principal congruence chain and common congruence elements

For  $0 \neq a \in A$  put  $P_n\langle a \rangle = \{g \in P_n : g - 1_n \in M_n(aA)\}$ , so that  $P_n^{(a)} = P_n\langle \omega^a \rangle$  in (10.1). As there, the ideal  $aA$  is used only as a congruence relation; no quotient ring is asserted to be a set.

**Theorem 19.5** (A chain at  $1 + \omega$  [11]). *Let  $n \geq 3$ . The subgroups  $P_n\langle (1 + \omega)^m \rangle$ ,  $m \geq 0$ , are normal in  $P_n$ ,  $P_n\langle 1 \rangle = P_n$ , and*

$$P_n\langle (1 + \omega)^m \rangle \supsetneq P_n\langle (1 + \omega)^{m+1} \rangle \quad (m \geq 0),$$

*while  $e_{12}(\omega^\omega)$  lies in every one of them.*

*Proof.* Matrices congruent to  $1_n$  modulo an ideal form a normal subgroup of  $\text{GL}_n(A)$ , which gives normality and the inclusions. The parameter  $p_m = (1 + \omega)^m \omega^{1/2}$  has support  $\{j + 1/2 : 0 \leq j \leq m\}$ , so  $e_{12}(p_m) \in P_n\langle (1 + \omega)^m \rangle$ . If it lay in the next subgroup,  $\omega^{1/2}/(1 + \omega)$  would lie in  $A$ ; its leading exponent is  $-1/2$ , which is impossible for a nonzero element of  $A$ . For  $m \geq 1$ ,

$$\frac{\omega^\omega}{(1 + \omega)^m} = \sum_{r \geq 0} (-1)^r \binom{m + r - 1}{r} \omega^{\omega - m - r} \in \Pi_k, \quad (19.1)$$

since  $(1 + \omega)^{-m} = \omega^{-m}(1 + \omega^{-1})^{-m}$ ; the exponents  $\omega - m - r$  are positive and strictly decreasing, and multiplying by  $(1 + \omega)^m$  checks the identity coefficientwise. So  $\omega^\omega \in (1 + \omega)^m A$  for every  $m$ .  $\square$

The comparison with [theorem 10.1](#) is instructive. The tower uses the monomial ideals  $\omega^a A$  over the whole class of scales  $a > 0$  and separates the elements of  $P_n$ ; the chain above is countable, uses powers of a non-monomial element, and does not separate. The second phenomenon is general for sets of principal congruences:

**Proposition 19.6** (Common root elements for a set of congruences [11]). *Let  $(a_s)_{s \in S}$  be a set-indexed family of nonzero elements of  $A$ . There is  $h_0 > 0$  such that  $\omega^h/a_s \in \Pi_k$  for all  $s \in S$  and all  $h > h_0$ . Consequently, for  $n \geq 3$ ,  $i \neq j$  and  $h > h_0$ ,  $e_{ij}(\omega^h) \in \bigcap_{s \in S} P_n\langle a_s \rangle$ .*

*Proof.* Each  $a_s^{-1} \in F$  has a set support, and the union  $T$  of these supports is a set. Choose  $h_0 > -t$  for all  $t \in T$  ([lemma 2.3](#)); then every exponent  $h + t$  of  $\omega^h/a_s$  is positive for  $h > h_0$ . Hence  $\omega^h = a_s(\omega^h/a_s) \in a_s A$ , and  $e_{ij}(\omega^h) \in P_n$ .  $\square$

This generalizes the set clause of [theorem 10.1](#) from the monomials  $\omega^a$  to arbitrary nonzero elements. It says nothing about the intersection over the proper class of all principal ideals, and makes no claim that the quotient rings  $A/a_s A$  are sets (11).

### 19.3 Finite quotients and the profinite completion

By [theorem 16.4](#) (07; for  $n \geq 3$  also 11),  $\text{Res}_{\text{fin}} E_n(A) = P_n$ , and by [theorem 9.1\(iv\)](#) every finite quotient of  $E_n(A)$  is a finite quotient of  $\text{SL}_n(D)$  composed with ct. The source is a class group, so its profinite completion needs a convention; following 11 it is the inverse limit of its finite quotients, and since the whole diagram of finite quotients is that of the set group  $\text{SL}_n(D)$ , this is an ordinary profinite group.

**Corollary 19.7** (The profinite completion [11]). *For  $n \geq 3$  there are canonical isomorphisms*

$$\widehat{E_n(\mathbf{Oz})} \cong \widehat{\text{SL}_n(\mathbb{Z})} \cong \text{SL}_n(\widehat{\mathbb{Z}}), \quad \widehat{E_n(\mathbf{Oz}[i])} \cong \widehat{\text{SL}_n(\mathbb{Z}[i])}.$$

*Proof.* The first isomorphism in each case identifies the finite quotients with those of the constant subgroup. For the second real isomorphism, the congruence subgroup theorem of Bass, Milnor and Serre [2] says that for  $n \geq 3$  the principal congruence subgroups of  $\text{SL}_n(\mathbb{Z})$  are cofinal among its subgroups of finite index. Reduction  $\text{SL}_n(\mathbb{Z}) \rightarrow \text{SL}_n(\mathbb{Z}/m\mathbb{Z})$  is surjective, since elementary elimination works over each local factor of  $\mathbb{Z}/m\mathbb{Z}$  and elementary parameters lift to integers. So the completion is the inverse limit of the  $\text{SL}_n(\mathbb{Z}/m\mathbb{Z})$ , which is  $\text{SL}_n(\widehat{\mathbb{Z}})$ .  $\square$

The last step is a deep imported theorem, not reproved here; the preceding factorization is proved. No isomorphism of  $\widehat{E_n(\mathbf{Oz}[i])}$  with  $\widehat{\text{SL}_n(\mathbb{Z}[i])}$  is asserted: congruence kernels need care in the totally imaginary case, which the classical source treats separately from fields with a real embedding (11).

## 20 Steinberg groups

No other statement about Steinberg groups or  $K_2$  appears in the collection. For a commutative unital ring  $B$  and  $n \geq 3$ ,  $\text{St}_n(B)$  is the group with generating symbols  $x_{ij}(a)$ ,  $a \in B$ ,  $i \neq j$ , and the Steinberg relations

$$x_{ij}(a)x_{ij}(b) = x_{ij}(a+b), \tag{20.1}$$

$$[x_{ij}(a), x_{lm}(b)] = 1 \quad (i \neq m, j \neq l), \tag{20.2}$$

$$[x_{ih}(a), x_{hj}(b)] = x_{ij}(ab) \quad (i, h, j \text{ distinct}). \tag{20.3}$$

$\pi_B : \text{St}_n(B) \rightarrow E_n(B)$ ,  $x_{ij}(a) \mapsto e_{ij}(a)$ , is a surjective homomorphism. The symbols  $x_{ij}$  are the universal instance of the root maps of [theorem 9.2](#). For a proper-class ring, 11 uses finite words and finite derivations from the relations, and represents an equivalence class of words by Scott's trick, as the set of equivalent words of least rank; the carrier is a class of sets with well-defined operations, and no class of proper classes is formed. Only the presentation and its elementary universal property are used; no theorem on the kernel of  $\pi_B$  or on universal central extensions is needed.

**Lemma 20.1** (The split Steinberg kernel [11]). *Let  $B = D_0 \oplus I$  be a unital commutative ring with a ring retraction  $c : B \rightarrow D_0$  and  $I = \ker c$ . Then  $c$  induces a split surjection  $\text{St}_n(B) \rightarrow \text{St}_n(D_0)$  whose kernel is the normal closure of the  $x_{ij}(a)$ ,  $a \in I$ .*

*Proof.* The inclusion  $D_0 \hookrightarrow B$  and  $c$  induce maps of presentations whose composite on  $\text{St}_n(D_0)$  is the identity. The normal closure  $N'$  lies in the kernel. Modulo  $N'$ , (20.1) gives  $x_{ij}(d+a) \equiv x_{ij}(d)$ , so every word is congruent to the image of its constant-term word; if that word is the identity in  $\text{St}_n(D_0)$ , so is its image in  $\text{St}_n(B)$ .  $\square$

**Theorem 20.2** (Universal set-sized quotient of the Steinberg group [11]). *Let  $n \geq 3$  and  $P_n^{\text{St}} = \ker(\text{ct} : \text{St}_n(A) \rightarrow \text{St}_n(D))$ . The sequence  $1 \rightarrow P_n^{\text{St}} \rightarrow \text{St}_n(A) \rightarrow \text{St}_n(D) \rightarrow 1$  is split exact,  $P_n^{\text{St}}$  is perfect, every homomorphism from  $P_n^{\text{St}}$  to a set-sized group is trivial, and every homomorphism from  $\text{St}_n(A)$  to a set-sized group factors uniquely through  $\text{St}_n(D)$ . Moreover  $\pi_A(P_n^{\text{St}}) = P_n$ .*

*Proof.* Let  $V \leq \text{St}_n(A)$  be generated by the  $x_{ij}(a)$ ,  $a \in \Pi_k$ . Relations (20.1) and (20.3) are the hypotheses of theorem 9.2; with the ring theorem `osq:thm:universal(iii)`, corollary 9.3 kills every set-sized image of  $V$  and, with proposition 2.4, makes  $V$  perfect. By lemma 20.1  $P_n^{\text{St}}$  is the normal closure of  $V$ , and lemma 9.4 transfers both properties. Factorization is as in theorem 9.1(iv). Finally  $\pi_A$  maps the generators of  $V$  onto the root elements  $e_{ij}(\Pi_k)$  and is onto  $E_n(A)$ , so it maps the normal closure of  $V$  onto the normal closure of those root elements, which is  $P_n$  (lemma 7.1).  $\square$

*Remark 20.3* (No  $K$ -theoretic conclusion [11]). The quotient  $\text{St}_n(D)$  is a set (countable here: its alphabet is countable and its words are finite), and it must not be replaced by  $\text{SL}_n(D)$  without accounting for the kernel of  $\pi_D$ .  $P_n^{\text{St}}$  is the kernel of the constant term, not of  $\pi_A : \text{St}_n(A) \rightarrow E_n(A)$ , which is not computed. The theorem does not calculate  $K_2$ , assert a universal central extension in any unstable rank, establish an excision theorem, or claim anything about algebraic  $K$ -theory in general; nor does it assert that  $P_n^{\text{St}}$  is centerless or torsion-free: the proofs of theorems 19.1 and 19.2 are about matrices and are not transferred.

## 21 Set-sized fragments and exact thresholds

In a fixed set-sized ring the identity map is a faithful set-sized representation, so only cardinal bounds can survive (section 12). Source 11 replaces the class of scales by a regular cardinal. Fix an uncountable regular cardinal  $\kappa$  and let  $\Gamma_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$ , ordered by the sign of the coefficient at the least index where it is nonzero; so  $e_\alpha > 0$  and  $me_\beta < e_\alpha$  for  $\alpha < \beta$  and ordinary  $m \geq 1$ . Let  $\Pi_{<\kappa}$  be the set of Hahn series in  $X$  over  $\Gamma_\kappa$  with coefficients in  $k$  whose support is a reverse well-ordered subset of  $\Gamma_\kappa^{>0}$  of cardinality  $< \kappa$ , and  $R_{<\kappa} = D \oplus \Pi_{<\kappa}$ . The map  $X^g \mapsto \omega^{\iota(g)}$ , with  $\iota(e_\alpha) = \omega^{-\alpha}$  for the ordinal  $\alpha$ , embeds  $R_{<\kappa}$  in  $A$ ; there are two levels of exponents, and  $X^{e_\alpha} = \omega^{\omega^{-\alpha}}$ , for instance  $X^{e_0} = \omega$  and  $X^{e_1} = \omega^{1/\omega}$ . This is the ring  $R_\kappa^{11}$  of the sibling report, with its ideal  $\Pi_\kappa^{11}$ , the same group (`osq:prop:lex`) and the same embedding (`osq:rel:lem:embedding`); source 11 describes it only as related to that report's constructions and proves the facts it needs directly. Put  $P_{n,<\kappa} = \ker(\text{ct} : E_n(R_{<\kappa}) \rightarrow E_n(D))$  (11's  $N_{n,\kappa,D}$ ).

### 21.1 Cardinal rigidity below $\kappa$

**Lemma 21.1** (Closure and small scales [11]).

1.  $R_{<\kappa}$  is a set-sized subring of  $A$ ,  $\text{ct}$  retracts it onto  $D$ , and its units are  $D^\times$ .
2. For every  $S \subseteq \Gamma_\kappa^{>0}$  with  $|S| < \kappa$  there is  $\beta < \kappa$  with  $me_\beta < s$  for all  $s \in S$  and ordinary  $m \geq 1$ .
3. Every element of  $\Pi_{<\kappa}$  is a product of two elements of  $\Pi_{<\kappa}$ .

*Proof.* (i) Unions and sumsets of fewer than  $\kappa$  exponents have fewer than  $\kappa$  elements, so Hahn multiplication preserves the support bound; positivity and the multiplicativity of  $\text{ct}$  are unchanged, the coefficient field and exponent group are sets, and the leading-exponent argument of lemma 2.1(ii) gives the units. (ii) The least nonzero coordinate indices of the

members of  $S$  are fewer than  $\kappa$  ordinals below  $\kappa$ , hence bounded below  $\kappa$  by regularity; for  $\beta$  above all of them, the leading coordinate of each  $s \in S$  is positive and occurs before  $\beta$ , so  $s$  dominates every  $me_\beta$ . (iii) For  $p \neq 0$  take  $\beta$  from (ii) for  $S = \text{supp } p$  and write  $p = X^{e_\beta}(X^{-e_\beta}p)$ .  $\square$

Parts (i) and (ii) are also `osq:rel:lem:embedding`(iii) and (ii).

**Lemma 21.2** (No small ring image of the fragment ideal [11]). *Every additive multiplicative map from  $\Pi_{<\kappa}$  to a ring of cardinality  $< \kappa$  is zero.*

*Proof.* Let  $\rho$  be such a map and  $0 \neq p \in \Pi_{<\kappa}$ ; take  $\beta$  from [lemma 21.1](#)(ii) for  $\text{supp } p$  and put  $\delta = e_\beta$ . The  $\kappa$  exponents  $e_\gamma$ ,  $\beta < \gamma < \kappa$ , lie in  $(0, \delta)$ , so two of the monomials  $X^{e_\gamma}$  have the same image; call their exponents  $\eta < \varepsilon$ . By the division lemma `osq:lem:division` (11's Lemma 3.1, whose proof uses only the ordered-group inequalities  $\varepsilon + r(\varepsilon - \eta) < (r + 1)\delta < s$ , reverse well ordering and Hahn multiplication),  $p = (X^\varepsilon - X^\eta)q$  with  $q$  of positive support of cardinality at most  $\max(\aleph_0, |\text{supp } p|) < \kappa$ , since  $\kappa$  is uncountable. So  $q \in \Pi_{<\kappa}$  and  $\rho(p) = \rho(X^\varepsilon - X^\eta)\rho(q) = 0$ .  $\square$

Regularity bounds the leading indices in [lemma 21.1](#)(ii); uncountability keeps the countable geometric tail within the support bound. 11 does not extend the lemma to singular cardinals or to finite supports. It is the nonunital form, for this ring, of the sibling report's coinitality theorem `osq:thm:coinital` (through `osq:prop:lex`), and it is weaker than that report's exact ring threshold `osq:prop:card11`; the merge uses the latter in [proposition 21.12](#).

**Theorem 21.3** (Cardinal elementary rigidity [11]). *Let  $n \geq 3$ . Every homomorphism  $P_{n,<\kappa} \rightarrow H$  with  $|H| < \kappa$  is trivial, and every homomorphism  $E_n(R_{<\kappa}) \rightarrow H$  with  $|H| < \kappa$  factors uniquely through  $\text{ct} : E_n(R_{<\kappa}) \rightarrow \text{SL}_n(D)$ . The group  $P_{n,<\kappa}$  is nontrivial, nonabelian, torsion-free, centerless and perfect.*

*Proof.* Let  $U_{<\kappa} = \langle e_{ij}(\Pi_{<\kappa}) \rangle$ . By [corollary 9.3](#)(ii) and [lemma 21.2](#) it has no nontrivial image of cardinality  $< \kappa$ , and by [corollary 9.3](#)(iii) and [lemma 21.1](#)(iii) it is perfect.  $P_{n,<\kappa}$  is its normal closure, by the split-kernel argument of [lemma 7.1](#) (11's Lemma 5.1, for any split ring), and [lemma 9.4](#) transfers both properties; factorization is as in [theorem 9.1](#)(iv). (By [proposition 7.2](#) and [lemma 21.1](#)(iii), in fact  $U_{<\kappa} = P_{n,<\kappa}$ .) The monomial  $X^{e_0}$  gives  $e_{12}(X^{e_0}) \neq 1_n$  and  $[e_{12}(X^{e_0}), e_{23}(X^{e_0})] = e_{13}(X^{2e_0}) \neq 1_n$ . Torsion-freeness holds because  $P_{n,<\kappa}$  lies in the kernel of [theorem 19.1](#). The proof of [theorem 19.2](#) applies inside  $R_{<\kappa}$ , whose root matrices are present and whose units are  $D^\times$ .  $\square$

**Corollary 21.4** (Cardinal Steinberg rigidity [11]). *Let  $n \geq 3$ . Every homomorphism from  $\text{St}_n(R_{<\kappa})$  to a group of cardinality  $< \kappa$  factors through  $\text{St}_n(D)$ . Its constant-term kernel  $P_{n,<\kappa}^{\text{St}}$  is perfect and has no nontrivial image in a group of cardinality  $< \kappa$ .*

*Proof.* Apply [corollary 9.3](#)(ii),(iii) and [lemma 21.2](#) to the subgroup generated by the  $x_{ij}(\Pi_{<\kappa})$ , and take its normal closure by [lemmas 9.4](#) and [20.1](#).  $\square$

**Proposition 21.5** (Non-simple fragments [11]). *Let  $n \geq 3$ ,  $t_0 = X^{e_0}$  and  $t_1 = X^{e_1}$ . The subgroups  $\{g \in P_{n,<\kappa} : g - 1_n \in M_n((1 + t_1)^m R_{<\kappa})\}$ ,  $m \geq 0$ , are normal and strictly decreasing, and  $e_{12}(t_0)$  lies in all of them. The group  $P_{n,<\kappa}$  is linear over the set field  $\text{Frac}(R_{<\kappa})$ .*

*Proof.* Normality follows from reduction modulo an ideal. The parameter  $(1 + t_1)^m X^{e_1/2}$  witnesses strictness, since  $X^{e_1/2}/(1 + t_1)$  has leading exponent  $-e_1/2 < 0$ . For  $m \geq 1$ ,  $t_0/(1 + t_1)^m = \sum_{r \geq 0} (-1)^r \binom{m+r-1}{r} X^{e_0 - (m+r)e_1} \in \Pi_{<\kappa}$ : the exponents are positive in the

lexicographic order and the support is countable. Finally  $R_{<\kappa}$  is a set-sized integral domain, and the matrix inclusion into its fraction field is faithful.  $\square$

## 21.2 Concrete cardinals, actions and representations

**Proposition 21.6** (The countable-support case [11]). *For  $\kappa = \aleph_1$  and  $n \geq 3$ ,  $|R_{<\aleph_1}| = |P_{n,<\aleph_1}| = \mathfrak{c}$ , and  $P_{n,<\aleph_1}$  has no nontrivial homomorphism to a countable group.*

*Proof.*  $|\Gamma_{\aleph_1}| = \aleph_1 \leq \mathfrak{c}$ , and a countable support with a countable list of coefficients is coded in a set of size  $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ . The series  $rX^{e_0}$ ,  $r \in \mathbb{R}$ , give  $\mathfrak{c}$  distinct root matrices in  $P_{n,<\aleph_1}$ , and finite matrices over a ring of size  $\mathfrak{c}$  are at most  $\mathfrak{c}$  in number. The last assertion is [theorem 21.3](#).  $\square$

**Proposition 21.7** (Action thresholds [11]). *Let  $n \geq 3$ . Every action of  $P_{n,<\kappa}$  on a set  $X$  with  $|\text{Sym}(X)| < \kappa$  is trivial, in particular for finite  $X$  and for infinite  $X$  with  $2^{|X|} < \kappa$ ; every action of  $E_n(R_{<\kappa})$  on such a set factors through  $\text{SL}_n(D)$ .*

*Proof.* Apply [theorem 21.3](#) to the target  $\text{Sym}(X)$ . For infinite  $|X| = \lambda$ ,  $|\text{Sym}(X)| = 2^\lambda$ :  $\lambda^\lambda \leq (2^\lambda)^\lambda = 2^\lambda$ , and swapping the members of any subfamily of  $\lambda$  disjoint pairs gives  $2^\lambda$  permutations.  $\square$

A target of cardinality  $< \kappa$  and an action on a set of cardinality  $< \kappa$  are different conditions; the action threshold uses the cardinality of the symmetric group (11).

**Corollary 21.8** (Ordinary groups invisible to bounded actions [11]). *For every infinite cardinal  $\lambda$  there is an explicit set-sized nontrivial, torsion-free, centerless, perfect group with no nontrivial action on any set of cardinality at most  $\lambda$ :  $P_{3,<\kappa}$  with  $D = \mathbb{Z}$  and  $\kappa = (2^\lambda)^+$ .*

*Proof.* A successor cardinal  $\mu^+$  is regular: a cofinal family of at most  $\mu$  ordinals below  $\mu^+$  would have union of size at most  $\mu \cdot \mu = \mu$ . For  $|X| \leq \lambda$ ,  $2^{|X|} \leq 2^\lambda < \kappa$ . Apply [theorem 21.3](#) and [proposition 21.7](#).  $\square$

**Corollary 21.9** (Finite-dimensional complex representations [11]). *For  $\kappa = \mathfrak{c}^+$  and  $n \geq 3$ , every finite-dimensional complex representation of  $P_{n,<\kappa}$  is trivial, and every such representation of  $E_n(R_{<\kappa})$  factors through  $\text{SL}_n(D)$ .*

*Proof.*  $|\text{GL}_m(\mathbb{C})| = \mathfrak{c} < \kappa$  for  $m \geq 1$ ; the case  $m = 0$  is trivial.  $\square$

*Remark 21.10* (No overstatement at  $\aleph_1$  [11]). The absence of a countable image in [proposition 21.6](#) does not by itself exclude an action on a countably infinite set, since  $|\text{Sym}(\mathbb{N})| = \mathfrak{c}$ , nor every finite-dimensional complex representation. The larger cardinals in [corollaries 21.8](#) and [21.9](#) are intentional. The exact threshold  $\mathfrak{c}$  at  $\aleph_1$  obtained in [proposition 21.12](#) does not change this:  $\text{Sym}(\mathbb{N})$  and  $\text{GL}_m(\mathbb{C})$  have cardinality exactly  $\mathfrak{c}$ .

## 21.3 Exact thresholds

**Proposition 21.11** (An exact-threshold criterion [11]). *Let  $I$  be a set-sized commutative ring, possibly without unit, of infinite cardinality  $\theta$ , such that every nonzero additive multiplicative map from  $I$  to a ring has image of cardinality at least  $\theta$ . Let  $A_0 = D_0 \oplus I$  be a split unital commutative ring with  $I \neq 0$  an ideal and  $|D_0| \leq \theta$ , and let  $n \geq 3$  and  $N = \ker(E_n(A_0) \rightarrow E_n(D_0))$ . Then  $|N| = \theta$ , every homomorphism from  $N$  that does not kill a given nonidentity element has a target of cardinality at least  $\theta$ , and the identity of  $N$  attains this bound for*

every nonidentity element. In particular the least cardinality of a group receiving a nontrivial homomorphic image of  $N$  is  $\theta$ .

*Proof.*  $N$  is the normal closure of  $U = \langle e_{ij}(I) \rangle$  by the split-kernel argument (lemma 7.1; 11's Lemma 5.1). If  $f : N \rightarrow H$  is nontrivial, it is nontrivial on some conjugate  $gUg^{-1}$ , and  $u \mapsto f(gug^{-1})$  is a nontrivial homomorphism  $U \rightarrow H$ . By theorem 9.2 there is a nonzero ring image  $R_\varphi$  of  $I$  with  $\theta \leq |R_\varphi| \leq |H|^{n(n-1)}$ ; as  $\theta$  is infinite,  $H$  is infinite and  $|H| \geq \theta$ . The root matrices inject  $I$  into  $N$ , and  $N \subseteq M_n(A_0)$  has at most  $\theta$  elements, so  $|N| = \theta$ .  $\square$

Source 11 notes that the hypothesis is stronger than lemma 21.2 and does not claim that its fragments have exact threshold  $\kappa$  or  $\kappa^{<\kappa}$ : any application must verify the ring hypothesis. The sibling report's field bound verifies it for  $R_{<\kappa}$ . The next two results are the merge's.

**Proposition 21.12** (Exact thresholds for the fragments [merge]). *Let  $\kappa$  be uncountable and regular and  $n \geq 3$ .*

1. *Every nonzero additive multiplicative map from  $\Pi_{<\kappa}$  to a ring (not necessarily unital or commutative) has image of cardinality at least  $\kappa^{<\kappa}$ , and  $|\Pi_{<\kappa}| = |R_{<\kappa}| = \kappa^{<\kappa}$ .*
2. *For every  $1 \neq g \in P_{n,<\kappa}$  the least cardinality of a group  $H$  admitting a homomorphism  $P_{n,<\kappa} \rightarrow H$  that does not kill  $g$  is exactly  $\kappa^{<\kappa}$ ; the same holds for homomorphisms defined on  $E_n(R_{<\kappa})$ . Every homomorphism from  $E_n(R_{<\kappa})$  to a group of cardinality  $< \kappa^{<\kappa}$  factors uniquely through  $\text{ct}$ , and every action of  $P_{n,<\kappa}$  on a finite set, or on an infinite set  $X$  with  $2^{|X|} < \kappa^{<\kappa}$ , is trivial.*
3. *The least cardinality of a group detecting a nonidentity element of  $P_{n,<\kappa}^{\text{St}}$ , by a homomorphism defined on  $P_{n,<\kappa}^{\text{St}}$ , is also exactly  $\kappa^{<\kappa}$ .*
4. *For  $\kappa = \aleph_1$  the value is  $\mathfrak{c}$ :  $P_{n,<\aleph_1}$  has no nontrivial homomorphic image of cardinality below  $\mathfrak{c}$ .*

*Proof.* (i) Let  $\rho : \Pi_{<\kappa} \rightarrow S$  be additive and multiplicative with  $\rho(x) \neq 0$ . By lemma 21.1(ii) there is  $\gamma = e_\beta > 0$  strictly below  $\text{supp } x$ ; put  $c = \gamma/2 \in \Gamma_\kappa$ . Then  $X^\gamma \in \Pi_{<\kappa}$ , and  $y = X^{-\gamma}x$  has positive support of the same cardinality, so  $y \in \Pi_{<\kappa}$ ; since  $\rho(X^\gamma)\rho(y) = \rho(x) \neq 0$ ,  $\rho(X^{2c}) \neq 0$ . Let  $\Gamma_\kappa^{\leq c} = \{h \in \Gamma_\kappa : m|h| < c \text{ for all ordinary } m \geq 1\}$  (the sibling report's  $H_c(\Gamma)$ ) and let  $\mathbb{L}_c$  be the set of Hahn series over  $\Gamma_\kappa$  with coefficients in  $k$  and fewer than  $\kappa$  exponents, all in  $\Gamma_\kappa^{\leq c}$ . By `osq:prop:card11` (with  $K = k$ ) and its proof,  $\mathbb{L}_c$  is a field with at least  $\kappa^{<\kappa}$  elements. For  $f \in \mathbb{L}_c$  every exponent  $c+h$  of  $X^c f$  is positive, since  $|h| < c$ , and there are fewer than  $\kappa$  of them; so  $X^c \mathbb{L}_c \subseteq \Pi_{<\kappa}$ . For distinct  $f, f' \in \mathbb{L}_c$  the element  $X^c/(f - f')$  lies in  $X^c \mathbb{L}_c \subseteq \Pi_{<\kappa}$ , and in  $\Pi_{<\kappa}$

$$(X^c f - X^c f') \cdot \frac{X^c}{f - f'} = X^{2c}.$$

If  $\rho(X^c f) = \rho(X^c f')$ , additivity and multiplicativity give  $\rho(X^{2c}) = 0$ , a contradiction. So  $f \mapsto \rho(X^c f)$  is injective on  $\mathbb{L}_c$  and  $|\rho(\Pi_{<\kappa})| \geq \kappa^{<\kappa}$ . This is the scaled-field collision lemma `osq:lem:collision(i)`, whose proof, as that report notes, uses only that the source is closed under addition and multiplication and contains  $X^c \mathbb{L}_c$ ; it therefore applies to maps defined on  $\Pi_{<\kappa}$  alone. Since  $|k| = \mathfrak{c} \leq \max(\kappa, \mathfrak{c})$ , `osq:prop:card11` gives  $|R_{<\kappa}| = \kappa^{<\kappa}$ ; and  $|\Pi_{<\kappa}| \geq \kappa^{<\kappa}$  by the case  $\rho = \text{id}$ .

(ii) Apply proposition 21.11 with  $I = \Pi_{<\kappa}$ ,  $\theta = \kappa^{<\kappa}$  and  $D_0 = D$ , which is countable. A homomorphism on  $E_n(R_{<\kappa})$  not killing  $g$  restricts to one on  $P_{n,<\kappa}$ , and the inclusion  $E_n(R_{<\kappa}) \subseteq \text{GL}_n(R_{<\kappa})$ , a group of cardinality  $\kappa^{<\kappa}$ , detects  $g$ . A homomorphism from  $E_n(R_{<\kappa})$

to a group of cardinality  $< \kappa^{<\kappa}$  therefore kills  $P_{n,<\kappa}$ , and factors uniquely through  $\text{ct}$  by the split decomposition  $E_n(R_{<\kappa}) = P_{n,<\kappa} \rtimes \text{SL}_n(D)$  as in [theorem 9.1\(iv\)](#). For actions,  $|\text{Sym}(X)|$  is finite or  $2^{|X|}$  (proof of [proposition 21.7](#)).

(iii) Let  $f : P_{n,<\kappa}^{\text{St}} \rightarrow H$  not kill  $g$ . By [lemma 20.1](#),  $P_{n,<\kappa}^{\text{St}}$  is the normal closure of the subgroup  $V$  generated by the  $x_{ij}(\Pi_{<\kappa})$ , so  $f$  is nontrivial on a conjugate of  $V$ , and composing with conjugation gives a nontrivial homomorphism  $V \rightarrow H$ . By (20.1), (20.3) and [theorem 9.2](#),  $\Pi_{<\kappa}$  has a nonzero ring image of cardinality at most  $|H|^{n(n-1)}$ , so  $|H| \geq \kappa^{<\kappa}$  by (i). Conversely  $\text{St}_n(R_{<\kappa})$  is a quotient of the free group on  $n(n-1)|R_{<\kappa}| = \kappa^{<\kappa}$  symbols, so its kernel has at most  $\kappa^{<\kappa}$  elements, and its identity map detects  $g$ .

(iv)  $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0}$ , and  $2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ .  $\square$

When  $\kappa^{<\kappa} > \kappa$ , for instance for  $\kappa = \aleph_1$  when the continuum hypothesis fails, the exact value is strictly larger than the bound  $\kappa$  of [theorem 21.3](#). The value agrees with the size  $\mathfrak{c}$  of [proposition 21.6](#). 11's non-claim about exact thresholds remains a true description of 11; the value is obtained here by combining 11's criterion with `osq:prop:card11` and the scaled-field collision, which 11 did not use.

The same argument applies to the other set-sized rings of the sibling report.

**Corollary 21.13** (Exact thresholds in the sibling report's set-sized models [merge]). *Let  $A_0$  be one of the five set-sized rings of `osq:thm:thresholds`, or one of the Gaussian versions named there, with constant ring  $D_0$ , and let  $n \geq 3$ . For every nonidentity  $g$  in  $\ker(\text{ct} : E_n(A_0) \rightarrow E_n(D_0))$  the least cardinality of a group detecting  $g$ , by a homomorphism defined on this kernel or on  $E_n(A_0)$ , is exactly  $|A_0|$ .*

*Proof.* In the notation of the sibling report's `osq:sec:fixed`,  $A_0 = D_0 + \Pi(\mathbb{F})$  for a subfield  $\mathbb{F}$  of a Hahn field over an ordered set group  $\Gamma$  containing the coefficients and every monomial, with  $\Pi(\mathbb{F})$  the elements of positive support (the report writes  $F$ ; here  $F$  is the ambient field `No` or `No[i]`). The proof of `osq:thm:thresholds` establishes, for each of these rings, that  $\Gamma$  is divisible, that every nonzero  $x \in \Pi(\mathbb{F})$  has a positive strict lower bound  $\gamma \in \Gamma$  of its support, and that for every  $c > 0$  the field  $L_c(\mathbb{F})$  of members of  $\mathbb{F}$  supported in  $\Gamma^{\ll c} = \{h \in \Gamma : m|h| < c \text{ for all ordinary } m \geq 1\}$  has exactly  $|A_0|$  elements, while  $X^c L_c(\mathbb{F}) \subseteq \Pi(\mathbb{F})$  (`osq:prop:fieldbound(i)`). With  $c = \gamma/2$ ,  $X^\gamma \in \Pi(\mathbb{F})$  and  $X^{-\gamma}x \in \Pi(\mathbb{F})$ , the argument of [proposition 21.12\(i\)](#) shows that every nonzero additive multiplicative map from  $\Pi(\mathbb{F})$  to a ring has image of cardinality at least  $|A_0|$ . Moreover  $|\Pi(\mathbb{F})| = |A_0|$ , which is infinite, since  $X^c L_c(\mathbb{F}) \subseteq \Pi(\mathbb{F})$ , and  $|D_0| \leq |A_0|$ . Apply [proposition 21.11](#) with  $\theta = |A_0|$ , and use the inclusion into  $\text{GL}_n(A_0)$  for homomorphisms defined on  $E_n(A_0)$ .  $\square$

Together, [proposition 21.12](#) and [corollary 21.13](#) answer [question 34.4](#) for  $n \geq 3$  in these rings: the least detecting target cardinal of every nonidentity relative elementary matrix is the cardinality of the ring, as for ring and module targets in `osq:thm:thresholds`. Other exponent groups and support restrictions remain open.

## Part III

# Shortest vectors and missing infima

This part is source 10's. An ordinary positive-definite quadratic form on  $\mathbb{Z}^n$  has a shortest nonzero vector: a positive lower bound by the Euclidean norm combines with the finiteness of

bounded subsets of  $\mathbb{Z}^n$ . Over  $\mathbf{Oz}$  neither positive definiteness nor uniform separation settles the question. The mechanism is visible in one example: for infinitesimal  $\varepsilon > 0$  let

$$\mathcal{E}_\varepsilon(x, y) = (x - \sqrt{2}y)^2 + \varepsilon(x^2 + y^2), \quad (x, y) \in \mathbf{Oz}^2. \quad (21.1)$$

No nonzero ordinary integer vector lies on the kernel  $x = \sqrt{2}y$  of the dominant form, but for every  $\gamma > 0$  the vector  $\omega^\gamma(\sqrt{2}, 1)/\sqrt{3}$  has omnific coordinates, lies on that line and has energy  $\varepsilon\omega^{2\gamma}$ ; making  $\gamma$  smaller never produces a finite vector. This creates a missing infimum. Unimodular vectors behave differently:  $\text{ct}$  sends a unimodular omnific vector to a nonzero unimodular ordinary vector, an admissible competitor, whereas the constant term of a nonzero vector may be zero.

Throughout,  $(k, D, F, \mathcal{A})$  is  $(\mathbb{R}, \mathbb{Z}, \mathbf{No}, \mathbf{Oz})$  or  $(\mathbb{C}, \mathbb{Z}[i], \mathbf{No}[i], \mathbf{Oz}[i])$ ; conjugation is trivial in the real row and fixes  $\mathbf{No}$  in the complex row, and  $\|x\|^2 = \sum_\ell \bar{x}_\ell x_\ell$ . *Terminology.* A vector is *unimodular* if its coordinates generate the unit ideal. Source 10 calls such vectors “primitive”; the omnific Diophantine report uses *primitive* for the weaker “no common nonunit divisor” (`odg:thm:realray`), so the word is not used for 10’s notion here. The energy is written  $\mathcal{E}$ , since  $E_n$  is the elementary group.

## 22 Positive forms contract under constant-term descent

For an ordinary symmetric or Hermitian positive-semidefinite matrix  $Q$  write  $q(x) = x^*Qx$ , on  $k^n$  and on  $F^n$ . The ordinary spectral theorem gives an ordinary  $B$  with  $Q = B^*B$ , so  $q(x) = \|Bx\|^2$ ; nonnegativity and the kernel persist after scalar extension to  $F$ .

**Theorem 22.1** (Constant-term contraction [10]). *For  $x \in \mathcal{A}^n$  and ordinary  $a \in k^n$ ,  $q(x - a) \geq q(\text{ct}(x) - a)$ , with equality exactly when  $x - \text{ct}(x)$  lies in the scalar-extended kernel of  $Q$ ; otherwise  $q(x - a)$  is positive infinite while  $q(\text{ct}(x) - a)$  is an ordinary real.*

*Proof.* Write  $x = z + j$  with  $z \in D^n$ ,  $j \in \Pi_k^n$  (lemma 2.1). Then  $B(x - a) = B(z - a) + Bj$  with  $B(z - a)$  ordinary and  $Bj$  purely infinite. If  $Bj \neq 0$  one coordinate is infinite (lemma 2.1(iii)), its squared modulus is positive infinite and the others are nonnegative. Finally  $Bj = 0$  iff  $Qj = 0$ .  $\square$

**Corollary 22.2** (Strict descent for a positive family [10]). *Let  $Q_0, \dots, Q_m$  be ordinary positive-semidefinite with common kernel 0 and  $\lambda_0, \dots, \lambda_m$  positive surreals. For  $\mathcal{E}(x) = \sum_i \lambda_i x^* Q_i x$  and  $a \in k^n$ ,  $\mathcal{E}(x - a) \geq \mathcal{E}(\text{ct}(x) - a)$  with equality exactly when  $x \in D^n$ .*

*Proof.* Each difference in theorem 22.1 is nonnegative, and for  $j = x - \text{ct}(x) \neq 0$  at least one is positive.  $\square$

**Remark 22.3** (What the argument does not assert [10]).  $\text{ct}$  is not order preserving on  $\mathcal{B}_k$ , the contraction is not a Pythagorean decomposition (mixed terms may be nonzero), and the weights  $\lambda_i$  need not lie in  $\mathcal{B}_k$ , so applying  $\text{ct}$  to the energy itself is not legitimate. The argument works because an ordinary sum-of-squares map either annihilates the whole positive-exponent part or makes the norm infinite. For closest points the contraction always preserves admissibility; for shortest *nonzero* vectors it need not, since  $x \neq 0$  does not imply  $\text{ct}(x) \neq 0$ ; for unimodular vectors it does (proposition 26.1).

For a definite  $Q$  the contraction gives a Tarski-free proof that omnific points on a finite level of an ordinary definite form are ordinary, the conclusion of `odg:cor:definite` in the omnific Diophantine report.

## 23 Separated positive pencils

**Definition 23.1** (Separated positive pencil [10]). Let  $n \geq 1$ ,  $m \geq 0$ . A separated positive pencil consists of ordinary positive-semidefinite matrices  $Q_0, \dots, Q_m$  and positive surreals  $\varepsilon_0, \dots, \varepsilon_m$  with  $\varepsilon_{j+1} \ll \varepsilon_j$ . Put  $q_j(x) = x^* Q_j x$ ,

$$K_j = \bigcap_{i < j} \ker_k Q_i, \quad L_j = K_j \cap D^n, \quad K_0 = k^n,$$

and require  $K_m \neq 0$  and  $K_{m+1} = 0$ . The energy is  $\mathcal{E}(x) = \sum_{j=0}^m \varepsilon_j q_j(x)$  for  $x \in F^n$ . For an ordinary subspace  $K$ ,  $K_F$  is its scalar extension. Repeated kernels and zero forms are allowed.

The condition  $K_m \neq 0 = K_{m+1}$  says that the last index is the first at which the cumulative kernel vanishes; no finer forms are appended after it, and this convention is part of the scope of the exact minimum formula below. For  $m = 0$  it says  $Q_0$  is positive definite (10).

**Lemma 23.2** (Coercivity at the last scale [10]). *There are ordinary  $c, C > 0$  with  $c\varepsilon_m \|x\|^2 \leq \mathcal{E}(x) \leq C\varepsilon_0 \|x\|^2$  on  $F^n$ , and  $q_m$  is positive definite on  $K_m$ .*

*Proof.*  $S = \sum_j Q_j$  is positive definite; choose  $cI \leq S \leq CI$  over  $k$ , which persists over  $F$  by sum-of-squares factorization, and use  $\varepsilon_m \leq \varepsilon_j \leq \varepsilon_0$ . A vector of  $K_m$  killed by  $q_m$  lies in  $K_{m+1} = 0$ .  $\square$

**Lemma 23.3** (Exact lexicographic comparison [10]). *For ordinary  $a, z, z' \in k^n$ , the sign of  $\mathcal{E}(z - a) - \mathcal{E}(z' - a)$  is the sign of the first nonzero entry of  $(q_j(z - a) - q_j(z' - a))_{j=0}^m$ , and the energies are equal if all entries vanish.*

*Proof.* If the first nonzero entry is  $d_j$ , divide by  $\varepsilon_j$ : the result is  $d_j$  plus the infinitesimal  $\sum_{i > j} (\varepsilon_i / \varepsilon_j) d_i$ .  $\square$

This is exact for fixed ordinary vectors, not uniform in all ordinary integer coordinates; an infinitesimal weight times any one ordinary real is infinitesimal, which is central to the Pell example (10).

## 24 The unrestricted shortest-vector dichotomy

For a nonempty class  $S \subseteq \mathbf{No}^{>0}$  let  $\text{LB}_{\geq 0}(S) = \{b \in \mathbf{No}_{\geq 0} : b \leq s \text{ for all } s \in S\}$ . A subclass  $T \subseteq S$  is coinital if every  $s \in S$  has some  $t \in T$  with  $t \leq s$ . A surreal infimum is a greatest lower bound in  $\mathbf{No}$ , not an adjoined gap. Let  $\mathcal{V} = \{\mathcal{E}(x) : x \in \mathcal{A}^n \setminus \{0\}\}$ .

**Theorem 24.1** (Minimum or exact missing-infimum cut [10]). *For a separated positive pencil exactly one of the following holds.*

- (a) *If  $L_m \neq 0$ , then  $\mu = \min\{q_m(z) : z \in L_m \setminus \{0\}\}$  is an attained positive ordinary real,  $\min \mathcal{V} = \varepsilon_m \mu$ , and the minimizers are the finitely many  $z \in L_m \setminus \{0\}$  with  $q_m(z) = \mu$ .*
- (b) *If  $L_m = 0$ , then  $\mathcal{E}(x)/\varepsilon_m$  is positive infinite for every nonzero  $x \in \mathcal{A}^n$ , and  $\text{LB}_{\geq 0}(\mathcal{V}) = \varepsilon_m(\mathcal{O}_{\text{fin}})_{\geq 0}$ . In particular  $\mathcal{V}$  has neither a minimum nor a surreal infimum.*

*Proof.* If  $x \notin D^n$ ,  $\mathcal{E}(x)/\varepsilon_m \geq c\|x\|^2$  is positive infinite (lemmas 2.1 and 23.2). If  $x \in D^n \setminus K_m$ , some  $q_i(x) > 0$  with  $i < m$  is an ordinary positive real and  $\mathcal{E}(x)/\varepsilon_m \geq (\varepsilon_i/\varepsilon_m)q_i(x)$  is infinite. If  $x \in L_m$ ,  $\mathcal{E}(x) = \varepsilon_m q_m(x)$ .

(a)  $q_m$  is positive definite on  $K_m$ , so the ordinary lattice vectors with  $q_m(z) \leq q_m(z_0)$ , for one  $z_0 \in L_m \setminus \{0\}$ , have bounded norm and are finitely many; their least positive value is  $\mu$ . The classification gives the rest.

(b) The classification makes every nonnegative finite multiple of  $\varepsilon_m$  a lower bound. Conversely let  $b \geq 0$  with  $h = b/\varepsilon_m$  infinite; choose ordinary  $w \in K_m$  with  $q_m(w) = 1$  and  $\gamma > 0$  with  $2\gamma < \deg_\omega h$ . Then  $x = \omega^\gamma w \in \Pi_k^n \setminus \{0\}$ , the earlier forms vanish on it, and  $\mathcal{E}(x) = \varepsilon_m \omega^{2\gamma} \ll \varepsilon_m h = b$ , so  $b$  is not a lower bound. The class  $\varepsilon_m(\mathcal{O}_{\text{fin}})_{\geq 0}$  has no greatest member ( $b + \varepsilon_m$  is larger), and a greatest lower bound would be nonnegative, so no infimum exists.  $\square$

**Theorem 24.2** (Improvement of every set-sized family [10]). *Assume  $L_m = 0$ . For every set-indexed family  $(x_s)_{s \in S} \subseteq \mathcal{A}^n \setminus \{0\}$  there are an ordinary  $w \in K_m$  with  $q_m(w) = 1$  and  $\gamma > 0$  such that  $x = \omega^\gamma w$  satisfies  $\mathcal{E}(x) \ll \mathcal{E}(x_s)$  for every  $s$ . Consequently  $\mathcal{V}$  has no set-sized coinital subclass.*

*Proof.* Each  $h_s = \mathcal{E}(x_s)/\varepsilon_m$  is positive infinite; lemma 2.3(iii) gives  $\gamma > 0$  with  $\omega^{2\gamma} \ll h_s$  for all  $s$ . The construction applies directly to a set of energy values, through their leading exponents, without choosing vector representatives.  $\square$

**Corollary 24.3** (No ordinary-factor approximation [10]). *If  $L_m = 0$ , for every nonzero  $y \in \mathcal{A}^n$  there is a nonzero  $x \in \mathcal{A}^n$  with  $\mathcal{E}(x) < c\mathcal{E}(y)$  for every ordinary  $c > 0$ .*

The witness is a single monomial vector; no sum or limit is taken, and replacing the missing infimum by the limit of a chosen ordinary sequence would be incorrect. In the full surreal field no countable list of improving vectors is coinital in  $\mathcal{V}$  (10). The monomial witness  $\omega^\gamma w$  is the mechanism of the homogeneous cone criterion and the zero Pell fiber of the omnific Diophantine report (odg:thm:cone, odg:prop:zeropell).

## 25 Recovering the kernel flag from low energies

For  $0 \leq j \leq m$  let  $\text{Low}_j = \{x \in \mathcal{A}^n : \mathcal{E}(x) \ll \varepsilon_j\}$ , with  $0 \in \text{Low}_j$ . (Source 10 writes  $V_j$ ; the letter is left to the radical flag of the Hahn–Tate report.)

**Theorem 25.1** (Field span and integral trace of small energies [10]). *For every  $j$ :  $\text{Low}_j \subseteq (K_{j+1})_F$ ,  $\text{span}_F \text{Low}_j = (K_{j+1})_F$ , and  $\text{ct}(\text{Low}_j) = L_{j+1}$ .*

*Proof.* For  $x \in \text{Low}_j$  and  $i \leq j$ ,  $0 \leq q_i(x) \leq \mathcal{E}(x)/\varepsilon_j \in \mathfrak{m}$ ; by the sum-of-squares argument of theorem 22.1,  $q_i(x)$  is an ordinary nonnegative real or positive infinite, so it is 0. This gives the inclusion, and applying ordinary matrices and  $\text{ct}$  gives  $\text{ct}(x) \in L_{j+1}$ . Every  $z \in L_{j+1}$  has  $\mathcal{E}(z)/\varepsilon_j = \sum_{i > j} (\varepsilon_i/\varepsilon_j) q_i(z) \in \mathfrak{m}$ . For the span, take an ordinary basis  $w_\ell$  of  $K_{j+1}$ ; each  $h_\ell = \mathcal{E}(w_\ell)/\varepsilon_j$  is positive infinitesimal, and for  $0 < 2\gamma < -\deg_\omega h_\ell$  (all  $\ell$ ) the vectors  $\omega^\gamma w_\ell$  lie in  $\text{Low}_j$  and span  $(K_{j+1})_F$ .  $\square$

An irrational kernel can have a large field span of small-energy vectors and a small integral trace; in the binary example the trace is 0 (10).

## 26 Unimodular vectors

Let  $\text{Um}(\mathcal{A}^n)$  be the unimodular vectors: those  $x$  with  $\sum_\ell a_\ell x_\ell = 1$  for some  $a_\ell \in \mathcal{A}$ .

**Proposition 26.1** (Unimodular descent [10]). *If  $x \in \text{Um}(\mathcal{A}^n)$  then  $\text{ct}(x) \in \text{Um}(D^n)$ , and  $\text{Um}(D^n) \subseteq \text{Um}(\mathcal{A}^n)$ . Hence  $\mathcal{E}(x) \geq \mathcal{E}(\text{ct}(x))$  on  $\text{Um}(\mathcal{A}^n)$ , strictly unless  $x$  is ordinary; the energies of ordinary unimodular vectors form a countable coinital subset of the unimodular energies; and every unimodular minimizer, if any, is ordinary.*

*Proof.* Apply the ring homomorphism  $\text{ct}$  to a relation  $\sum a_\ell x_\ell = 1$ ; a relation over  $D$  is one over  $\mathcal{A}$ . Then use [corollary 22.2](#) at target 0;  $D^n$  is countable.  $\square$

The converse “ $\text{ct}(x)$  unimodular implies  $x$  unimodular” for arbitrary  $x \in \mathcal{A}^n$  is neither needed nor asserted (10). Descent of unimodularity under  $\text{ct}$  is the mechanism of the omnific Diophantine report’s unimodular Fermat theorem and rational-direction theorem (`odg:thm:unimodularfermat`, `odg:thm:realray`).

**Corollary 26.2** (Countable coinitality [10]). *If the unimodular energy has no minimum, its coinitality is exactly  $\aleph_0$ . In particular [theorem 24.2](#) fails if every vector in it is required to be unimodular.*

*Proof.* [Proposition 26.1](#) gives a countable coinital set; a finite one would contain a minimum.  $\square$

**Lemma 26.3** (Discreteness of a projected lattice). *Let  $V$  be a finite-dimensional real vector space,  $\Lambda$  a full lattice in  $V$  and  $T : V \rightarrow W$  real linear. Then  $T(\Lambda)$  is discrete iff  $\ker T = \text{span}_{\mathbb{R}}(\ker T \cap \Lambda)$ . If  $T(\Lambda)$  is not discrete, it has nonzero points arbitrarily close to 0, it is not closed, and there is  $h \in \overline{T(\Lambda)} \setminus T(\Lambda)$ . The same holds for complex-linear maps and Gaussian lattices, through the underlying real spaces.*

*Proof.* The equivalence is `tate:theta:lem:latticeprojection` of the Hahn–Tate report (for  $\Lambda = \mathbb{Z}^d$ , which is the general case after choosing a basis); source 10 proves it independently by the same saturation and rank count. The remaining assertions are 10’s. A nondiscrete subgroup contains nonzero  $g_j \rightarrow 0$ ; passing to a subsequence with  $g_j / \|g_j\| \rightarrow u$ , for real  $t$  let  $a_j$  be an integer nearest to  $t / \|g_j\|$ ; then  $a_j g_j \rightarrow tu$ , so a closed such subgroup would contain  $\mathbb{R}u$ , contradicting countability. For the complex case, a complex subspace is spanned over  $\mathbb{C}$  by its Gaussian integral vectors iff it is spanned over  $\mathbb{R}$  by them in real coordinates, because the Gaussian lattice and its intersection with a complex subspace are closed under multiplication by  $i$ .  $\square$

Let  $r$  be the largest index in  $\{0, \dots, m\}$  with  $L_r \neq 0$  (it exists since  $L_0 = D^n$ ),  $\Lambda = L_r$ ,  $W = \text{span}_k \Lambda$  and  $W' = W \cap K_{r+1}$  (10’s  $H$ ). By maximality  $K_{r+1} \cap D^n = 0$ , and  $\Lambda$  is a full lattice in  $W$  (in real coordinates in the complex case).

**Theorem 26.4** (Exact unimodular attainment criterion [10]). *For a separated positive pencil the following are equivalent:*

1.  $\mathcal{E}$  has a minimum on  $\text{Um}(\mathcal{A}^n)$ ;
2.  $\mathcal{E}$  has a minimum on  $D^n \setminus \{0\}$ ;
3.  $q_r$  is positive definite on  $W$ ;
4.  $W' = 0$ .

When they hold, the minimizers in (i) and (ii) are the same finite set of ordinary unimodular vectors, obtained by minimizing  $q_r$  on  $\Lambda \setminus \{0\}$  and breaking ties lexicographically by  $q_{r+1}, \dots, q_m$ . When they fail, the unimodular energies have coinitality  $\aleph_0$ .

*Proof.* The kernel of  $q_r$  on  $W \subseteq K_r$  is  $W'$ , so (iii) $\Leftrightarrow$ (iv). A nonzero ordinary vector outside  $K_r$  has a positive earlier coefficient and is worse than every fixed nonzero vector of  $\Lambda$  (lemma 23.3), so ordinary minimization may be restricted to  $\Lambda \setminus \{0\}$ .

If  $W' = 0$ ,  $q_r$  has a positive minimum on  $\Lambda \setminus \{0\}$  attained at finitely many vectors, and lexicographic tie-breaking leaves a finite nonempty set of minimizers on  $D^n \setminus \{0\}$ . Each is unimodular:  $D$  is Euclidean, and dividing a non-unimodular ordinary vector by a nonunit gcd  $g$  gives a nonzero ordinary vector of energy  $\mathcal{E}(z)/|g|^2 < \mathcal{E}(z)$ . So (ii) holds, and proposition 26.1 gives (i) with no extra nonordinary minimizers.

If  $W' \neq 0$ , factor  $q_r|_W = \|T \cdot\|^2$  with an ordinary  $T$  of kernel  $W'$ . Since  $W' \cap \Lambda = 0$ , lemma 26.3 makes  $T(\Lambda)$  nondiscrete, while  $q_r(z) > 0$  on  $\Lambda \setminus \{0\}$  because  $L_{r+1} = 0$ ; so  $q_r$  has infimum 0 and no minimum there. Dividing by the coordinate gcd keeps vectors in  $\Lambda$ , makes them unimodular and does not increase  $q_r$ , so arbitrarily small positive  $q_r$  values are attained by unimodular vectors, and any ordinary candidate in  $\Lambda$  is beaten (lemma 23.3); candidates outside  $K_r$  were already worse. So there is no ordinary nonzero or unimodular minimum, and proposition 26.1 excludes a nonordinary one. The coinitality is corollary 26.2.  $\square$

The criterion uses the span of the actual integral intersection, not the whole deepest kernel, and can hold when theorem 24.1 gives unrestricted nonattainment (example 29.4).

## 27 Closest points

### 27.1 What is already in the Hahn–Tate report

Part II of the Hahn–Tate uniformization report (docs/surcomplex/hahn-tate-uniformization, `tate:part:theta`), present in the collection at 10’s pin, studies lexicographically valued quadratic lattice energies  $\mathcal{E}_j(n) = \frac{1}{2}\mathcal{B}_j(n, n) + b_j(n)$  with Amini–Nicolussi higher-rank forms [1]. It already contains the ordinary backbone of this section:

- the lattice-projection lemma `tate:theta:lem:latticeprojection` (lemma 26.3) and the one-level quadratic criterion `tate:theta:lem:onelevel`;
- the exact flag criterion `tate:theta:thm:flag` and its corollary `tate:theta:cor:domain`: for an AN-admissible form, the compatible linear terms are exactly those giving an energy with a global lattice minimum;
- the parity bound: at most  $2^g$  minimizers, attained (`tate:theta:thm:minimizers`, with the example  $\sum_j n_j(n_j - 1)$ );
- the Pell example `tate:theta:ex:irrational`: the form  $((m - \sqrt{2}n)^2, m^2 + n^2)$  and the Pell vectors  $p_k + q_k\sqrt{2} = (1 + \sqrt{2})^k$ , along which the lexicographic energies strictly decrease.

Source 10 cites the Lean module that formalizes part of that report (`Surreal/Algebra/LatticeEnergyCertificate.lean`) but not the report itself; its repository search for “shortest vector” found no hit, which is literally true and misleading (Appendix A.3). These results are cited here, not reprinted; 10’s own proofs are kept only where they prove more (the non-closedness in lemma 26.3, the converse in theorem 27.2, the Gaussian bounds, arbitrary surreal targets) or where the omnific statement needs them.

The dictionary is as follows. For ordinary PSD forms  $q_j$  on all of  $k^n$  and an ordinary center  $a$ , the profile  $z \mapsto (q_j(z - a))_{j=0}^m$  on  $D^n = \mathbb{Z}^n$  (real case) is, up to additive constants, a lexicographic energy with  $r = m + 1$  coordinates,  $\mathcal{B}_{j+1}(z, z') = 2z^\top Q_j z'$  and  $b_{j+1}(z) = -2a^\top Q_j z$  (the Hahn–Tate forms are indexed from 1). Its radical flag `tate:theta:eq:flag` is  $V_j = K_j$ , ending

with  $V_{m+1} = K_{m+1} = 0$ ; the global radical condition `tate:theta:eq:globalradical` holds automatically because  $K_{j+1} \subseteq \ker Q_j$ ; and the compatibility  $b_{j+1}(V_{j+1}) = -2a^\top Q_j K_{j+1} = 0$  holds for every center for the same reason. AN admissibility then amounts to rationality of every  $K_j$ . The Gaussian case is the real case for the rank- $2n$  lattice  $\mathbb{Z}[i]^n \cong \mathbb{Z}^{2n}$  (a Hermitian PSD form is a real PSD form on the underlying space, with the same kernel).

## 27.2 Reduction to an ordinary lexicographic problem

For  $a \in F^n$  a closest point is a minimizer of  $\mathcal{E}(x - a)$  on  $\mathcal{A}^n$ , with no nonzero or unimodular restriction.

**Proposition 27.1** (Exact reduction for an ordinary target [10]). *Every closest point to  $a \in k^n$ , if any, is ordinary, and it is exactly a lexicographic minimizer over  $z \in D^n$  of  $(q_0(z - a), \dots, q_m(z - a))$ . Equivalently, with  $S_{-1} = D^n$  and  $S_j = \arg \min_{z \in S_{j-1}} q_j(z - a)$ , a closest point exists iff every stage is nonempty, and then  $S_m$  is the finite set of closest points.*

*Proof.* Strict descent ([corollary 22.2](#)) gives ordinarity and [lemma 23.3](#) the lexicographic description. On  $S_m$  all  $q_j(z - a)$  are fixed, their sum is an ordinary positive-definite form, so  $\|z - a\|$  is bounded and  $S_m$  finite.  $\square$

## 27.3 Rational flags

Call an ordinary subspace  $K \subseteq k^n$  *D-rational* if  $K = \text{span}_k(K \cap D^n)$ : ordinary rationality for  $k = \mathbb{R}$ , Gaussian rationality for  $k = \mathbb{C}$ ; no rationality of a displayed basis or matrix is required.

**Theorem 27.2** (Closest points for every ordinary center [10]; sufficiency also in the Hahn–Tate report). *For a separated positive pencil, every  $a \in k^n$  has a closest point in  $\mathcal{A}^n$  iff every prefix kernel  $K_j$  is D-rational. If so, after stage  $j$  the staged sets of [proposition 27.1](#) are finite unions of cosets of  $L_{j+1}$ . If not, an ordinary center can be chosen for which one stage has an unattained ordinary infimum 0.*

*Proof. Sufficiency.* By [proposition 27.1](#) it is the existence of a lexicographic minimum of the profile on  $D^n$ , which is `tate:theta:cor:domain` with `tate:theta:thm:flag` under the dictionary of [subsection 27.1](#). The coset description is 10's induction, which is also the inductive step of the proof of `tate:theta:thm:flag`: on a coset  $v + L_j$  of  $S_{j-1}$ , the stage- $j$  function  $\ell \mapsto \|B_j(v - a) + B_j\ell\|^2$  has kernel  $K_{j+1}$  on  $K_j$ , which is rational, so  $B_j L_j$  is a lattice ([lemma 26.3](#)) and the minimum is attained at finitely many image points, each with preimage one coset of  $L_{j+1}$  (`tate:theta:lem:onelevel`); retaining the smallest value over the finitely many cosets proves the induction, and  $L_{m+1} = 0$  makes the final set finite.

*Necessity* [10]. Let  $K_{j+1}$  be the first irrational kernel, so  $K_j$  is rational and  $L_j$  is a full lattice in  $K_j$ . With  $q_j = \|B_j \cdot\|^2$ ,  $G = B_j L_j$  is not discrete ([lemma 26.3](#)); choose  $h \in \overline{G} \setminus G$ . Since  $B_j K_j$  is closed and contains  $\overline{G}$ , there is ordinary  $a \in K_j$  with  $B_j a = h$ . For  $z \in L_j$  the earlier  $q_i(z - a)$  vanish and  $q_j(z - a) = \|B_j z - h\|^2 > 0$  has infimum 0 and no minimum, while ordinary  $z \notin K_j$  are worse than every fixed candidate in  $L_j$ . So no ordinary lexicographic minimizer exists, and strict descent excludes nonordinary closest points.  $\square$

The converse, an explicit bad center for an irrational flag, is not in the Hahn–Tate report, which proves nonsummability for non-admissible forms rather than the absence of a minimum for some center. Both 10 and that report credit the rational-filtration mechanism to higher-rank Voronoi theory [1].

**Corollary 27.3** (Nearest-point multiplicity [10]). *For an ordinary target with closest points, their number is at most  $2^n$  in the real case and  $4^n$  in the Gaussian case, and both bounds are attained.*

*Proof.* The real bound is the midpoint argument of `tate:theta:thm:minimizers`: distinct minimizers  $z, z'$  congruent modulo  $2D^n$  would have midpoint in  $D^n$  and  $\mathcal{E}(\frac{z+z'}{2} - a) = \frac{\mathcal{E}(z-a) + \mathcal{E}(z'-a)}{2} - \frac{\mathcal{E}(z-z')}{4}$  smaller. The Gaussian bound is the same bound for the rank- $2n$  lattice  $\mathbb{Z}[i]^n$ . For the Euclidean form with one weight, the target  $(1/2, \dots, 1/2)$  has  $2^n$  closest integer points and the target with every coordinate  $(1+i)/2$  has  $4^n$  closest Gaussian points (10).  $\square$

The parity argument is a standard finite-lattice argument; source 10 includes it as a consequence, not as a priority claim.

## 27.4 Arbitrary surreal targets

Taking the standard part of a target and using its ordinary closest points is not enough: infinitesimal displacements can decide between branches before the next energy scale (10). For  $a \in F^n$  split supports coordinatewise:

$$a = a^+ + a^0 + a^-, \quad a^+ \in \Pi_k^n, \quad a^0 \in k^n, \quad a^- \in \mathfrak{m}_k^n, \quad (27.1)$$

so  $b = a^0 + a^-$  is finite with standard part  $a^0$ . This is a support decomposition, not a claim that  $\text{ct}$  is a homomorphism on  $F$ .

**Lemma 27.4** (Matching the target's infinite part [10]). *For  $a \in F^n$ ,  $x \in \mathcal{A}^n$ ,  $y = x - a^+$  and  $z = \text{ct}(y) \in D^n$ ,  $\mathcal{E}(x - a) \geq \mathcal{E}(a^+ + z - a) = \mathcal{E}(z - b)$ , strictly unless  $x = a^+ + z$ . Every closest point to  $a$  has purely infinite part exactly  $a^+$ .*

*Proof.*  $y \in \mathcal{A}^n$ ; write  $y = z + j$  and  $q_i = \|B_i \cdot\|^2$ .  $B_i(z - b)$  is finite and  $B_i j$  purely infinite; if  $B_i j \neq 0$ ,  $\|B_i(y - b)\|^2$  is infinite and exceeds  $\|B_i(z - b)\|^2$ . All weighted differences are nonnegative, and one is positive when  $j \neq 0$ .  $\square$

**Theorem 27.5** (Finite candidates for every target [10]). *Assume every prefix kernel is  $D$ -rational. For every  $a^0 \in k^n$  there is a finite nonempty  $C(a^0) \subseteq D^n$ , depending only on  $a^0$  and  $q_0, \dots, q_m$ , such that for every  $a \in F^n$  with decomposition (27.1) the closest points of  $a$  are exactly*

$$a^+ + \arg \min_{z \in C(a^0)} \mathcal{E}(z - a^0 - a^-).$$

$|C(a^0)| \leq 2^n$  (real) or  $4^n$  (Gaussian), and  $C(a^0)$  does not depend on  $a^-$  or on the separated weights. Consequently closest points exist for every  $a \in F^n$  iff they exist for every  $a \in k^n$  iff every prefix kernel is  $D$ -rational.

*Proof.* Build a finite tree of cosets: the root is  $D^n$ ; at level  $j$ , for each retained coset  $v + L_j$ , minimize  $q_j(z - a^0)$  on it and retain *all* minimizing cosets of  $L_{j+1}$  as children (finitely many and at least one, by the proof of [theorem 27.2](#)). Unlike the staged sets, no parent is discarded because another has a smaller value. At level  $m + 1$  the cosets are singletons;  $C(a^0)$  is the set of leaves.

Every ordinary  $z$  is dominated by a leaf for the target  $b = a^0 + a^-$ : if  $z$  is in a retained parent  $v + L_j$  but not in a minimizing child, pick  $z'$  in one; for  $i < j$ ,  $q_i(z - b) = q_i(z' - b)$  exactly because  $z - z' \in L_j$ ; the difference  $q_j(z - b) - q_j(z' - b)$  has positive ordinary standard part  $q_j(z - a^0) - q_j(z' - a^0)$ ; later differences are finite, so after division by  $\varepsilon_j$  their weighted

sum is infinitesimal and  $\mathcal{E}(z - b) > \mathcal{E}(z' - b)$ . Repeating reaches a leaf, strictly better unless  $z$  was a leaf. [Lemma 27.4](#) handles nonordinary candidates.

For the bound let  $d_j = \dim_k K_j - \dim_k K_{j+1}$ . The children of a parent are closest points in the projected lattice  $B_j L_j$  of real rank  $d_j$  or  $2d_j$ ; two congruent modulo twice that lattice would have a better midpoint, as in [corollary 27.3](#); so each parent has at most  $2^{d_j}$  or  $4^{d_j}$  children, and  $\sum_j d_j = n$ . The equivalences follow from [theorem 27.2](#).  $\square$

**Example 27.6** (The ordinary minimizer set is too small [10]). Let  $0 < \varepsilon \in \mathfrak{m}$ ,  $\eta = \sqrt{\varepsilon}$ , and  $\mathcal{E}(x, y) = x^2 + \varepsilon(y - x/2)^2$ ; the flag is rational. The ordinary target  $a^0 = (1/2, 1/4)$  has the unique closest point  $(0, 0)$ , but the tree retains  $C(a^0) = \{(0, 0), (1, 0), (1, 1)\}$ . For  $a = (1/2 + \eta, 1/4)$ ,  $\mathcal{E}((1, 0) - a) - \mathcal{E}((0, 0) - a) = -2\eta + \varepsilon(1/4 - \eta/2) < 0$  and  $\mathcal{E}((1, 1) - a) - \mathcal{E}((1, 0) - a) = \varepsilon\eta > 0$ , so the unique closest point is  $(1, 0)$ : a branch losing the ordinary comparison must be kept.

This is a finite mathematical reduction, not an algorithm for arbitrary surreal inputs; constructing or comparing unrestricted surreals is a separate representational problem (10).

## 28 A binary model with two different missing cuts

Consider, in both rows,

$$\mathcal{E}_\varepsilon(x, y) = |x - \sqrt{2}y|^2 + \varepsilon(|x|^2 + |y|^2), \quad 0 < \varepsilon \in \mathfrak{m}, \quad (28.1)$$

with matrix  $Q_\varepsilon = \begin{pmatrix} 1+\varepsilon & -\sqrt{2} \\ -\sqrt{2} & 2+\varepsilon \end{pmatrix}$ ,  $\det Q_\varepsilon = \varepsilon(3 + \varepsilon) > 0$ . Here  $K_1 = k(\sqrt{2}, 1)$ ,  $K_2 = 0$ , and  $K_1 \cap D^2 = 0$  since  $\sqrt{2} \notin \mathbb{Q}(i)$ . The form and the Pell vectors below are those of `tate:theta:ex:irrational`, where the lexicographic energy  $((m - \sqrt{2}n)^2, m^2 + n^2)$  is shown to decrease strictly along the Pell vectors, so that its theta series is not Hahn summable. Source 10 uses the same example for a different purpose: to compare two omnific lower cuts.

**Theorem 28.1** (Unrestricted binary cut [10]). *The nonzero-vector energies of (28.1) have nonnegative lower bounds exactly  $\varepsilon(\mathcal{O}_{\text{fin}})_{\geq 0}$ , no surreal infimum and no set-sized coinital family; every set-sized collection of nonzero candidates is improved by some  $x_\gamma = \omega^\gamma(\sqrt{2}, 1)/\sqrt{3}$ , with  $\mathcal{E}_\varepsilon(x_\gamma) = \varepsilon\omega^{2\gamma}$ .*

*Proof.* [Theorems 24.1](#) and [24.2](#) with  $q_1 = |x|^2 + |y|^2$ .  $\square$

Define ordinary integers by  $p_0 = 1$ ,  $q_0 = 0$ ,  $p_{n+1} = p_n + 2q_n$ ,  $q_{n+1} = p_n + q_n$ . By induction

$$p_n \pm q_n \sqrt{2} = (1 \pm \sqrt{2})^n, \quad p_n^2 - 2q_n^2 = (-1)^n, \quad p_n q_{n+1} - q_n p_{n+1} = (-1)^n, \quad (28.2)$$

so consecutive  $u_n = (p_n, q_n)$  form an ordinary unimodular basis and every  $u_n$  is unimodular over  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , with

$$\mathcal{E}_\varepsilon(u_n) = (\sqrt{2} - 1)^{2n} + \varepsilon(p_n^2 + q_n^2). \quad (28.3)$$

These identities come from the recurrence, not from a classification of Pell solutions over the omnific integers (10); Pell rigidity in the omnific Diophantine report (only ordinary solutions at nonzero levels) is consistent with them.

**Theorem 28.2** (Unimodular cut and countable coinitality [10]). *For (28.1) the nonnegative lower bounds of the unimodular energies are exactly  $\mathfrak{m}_{\geq 0}$ ; there is no infimum and no minimum, and  $(\mathcal{E}_\varepsilon(u_n))_{n \in \mathbb{N}}$  is strictly decreasing and coinital in the unimodular energies, in both rows.*

*Proof.* For  $x \in \text{Um}(\mathcal{A}^2)$  and  $z = \text{ct}(x)$ ,  $z$  is a nonzero ordinary unimodular vector and  $r_z = |z_1 - \sqrt{2}z_2|^2 > 0$  is an ordinary real, so  $\mathcal{E}_\varepsilon(x) \geq \mathcal{E}_\varepsilon(z) > r_z$  (theorem 22.1); every nonnegative infinitesimal is a lower bound. A positive noninfinitesimal  $b$  exceeds an ordinary  $r > 0$ ; choose  $n$  with  $(\sqrt{2}-1)^{2n} < r/2$ , and since  $p_n^2 + q_n^2$  is ordinary,  $\mathcal{E}_\varepsilon(u_n) < r < b$ . The class  $\mathbf{m}_{\geq 0}$  has no greatest member. For coinitality apply this with  $r < r_z$ . Consecutive energies differ in the dominant coefficient by a positive ordinary real and in the rest by an ordinary multiple of  $\varepsilon$ , so the sequence decreases (lemma 23.3).  $\square$

For a fixed positive ordinary weight the second term of (28.3) eventually stops the improvement; for infinitesimal  $\varepsilon$  no ordinary index reaches that balance, and substituting an infinite omnific index into the recurrence is not part of its definition and would not justify a limit (10).

With  $\varepsilon = \omega^{-1}$  the two cuts differ visibly:  $\text{LB}_{\geq 0}(\text{all nonzero energies}) = \omega^{-1}(\mathcal{O}_{\text{fin}})_{\geq 0}$  and  $\text{LB}_{\geq 0}(\text{unimodular energies}) = \mathbf{m}_{\geq 0}$ . For example  $\omega^{-1/2}$  bounds every unimodular energy but not  $\mathcal{E}(x_{1/8}) = \omega^{-3/4}$ . Every unrestricted witness with small  $\gamma$  beats the whole Pell sequence, yet it is not unimodular, since its constant term is 0. “A coinital Pell family” is correct for unimodular energies and false for unrestricted ones (10).

## 29 Three attainment properties with strict implications

**Corollary 29.1** (A strict hierarchy [10]). *For separated positive pencils: closest points for every target in  $F^n \Rightarrow$  a shortest nonzero vector  $\Rightarrow$  a shortest unimodular vector; neither implication reverses.*

*Proof.* If every prefix kernel is rational,  $K_m \neq 0$  contains a nonzero integral vector and theorem 24.1 applies; its ordinary minimizers are unimodular, since division by a nonunit gcd would give a shorter nonzero vector. Strictness: the examples below.  $\square$

**Example 29.2** (All three properties [10]).  $\mathcal{E}(x, y) = x^2 + \varepsilon y^2$  on  $\mathbf{Oz}^2$ : all kernels rational, shortest vectors  $\pm(0, 1)$  of energy  $\varepsilon$ ; the Gaussian version has the four unit multiples of  $(0, 1)$ .

**Example 29.3** (A shortest vector without universal closest points [10]). On  $\mathbf{Oz}^3$  let  $\mathcal{E} = (z - \sqrt{2}y)^2 + \varepsilon(x^2 + y^2 + z^2)$ . The last kernel is  $K_1 = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, 1, \sqrt{2})\}$  with  $K_1 \cap \mathbb{Z}^3 = \mathbb{Z}(1, 0, 0)$ , so the shortest vectors are  $\pm(1, 0, 0)$  with minimum  $\varepsilon$ ; but  $K_1$  is not rational. For the center  $(0, 0, 1/2)$  the dominant term on integer candidates is  $(z - \sqrt{2}y - 1/2)^2$ , with infimum 0 never attained:  $\mathbb{Z} + \sqrt{2}\mathbb{Z}$  contains the arbitrarily small nonzero  $(1 - \sqrt{2})^n$ , so it is dense, and  $1/2 \notin \mathbb{Z} + \sqrt{2}\mathbb{Z}$ .

**Example 29.4** (Unimodular attainment without unrestricted attainment [10]). On  $\mathbf{Oz}^3$  let  $\mathcal{E} = (z - \sqrt{2}y)^2 + \varepsilon x^2 + \varepsilon^2(y^2 + z^2)$ . Then  $K_1 = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, 1, \sqrt{2})\}$ ,  $K_2 = \mathbb{R}(0, 1, \sqrt{2})$ ,  $L_1 = \mathbb{Z}(1, 0, 0)$  and  $L_2 = 0$ ; the unrestricted energies have no infimum, with lower cut  $\varepsilon^2(\mathcal{O}_{\text{fin}})_{\geq 0}$ . For the unimodular criterion  $r = 1$ ,  $W = \mathbb{R}(1, 0, 0)$  and  $W \cap K_2 = 0$ : the shortest unimodular vectors are exactly  $\pm(1, 0, 0)$ , of energy  $\varepsilon$ . Directly: an ordinary vector off the  $x$ -axis has positive dominant coefficient, ordinary unimodular vectors on the axis are  $\pm(1, 0, 0)$ , and nonordinary unimodular vectors are worse than their constant terms. The Gaussian analogue has four unimodular minimizers.

The binary form (28.1) has neither minimum (section 28).

| Real example                 | All centers<br>have closest points | Shortest<br>nonzero | Unimodular<br>shortest |
|------------------------------|------------------------------------|---------------------|------------------------|
| $x^2 + \varepsilon y^2$      | yes                                | yes                 | yes                    |
| <a href="#">Example 29.3</a> | no                                 | yes                 | yes                    |
| <a href="#">Example 29.4</a> | no                                 | no                  | yes                    |
| Form (28.1)                  | no                                 | no                  | no                     |

The table is not a classification of arbitrary surreal forms; it records all patterns allowed by the implication chain within the pencil class (10).

### 30 Reduction obstructions

Nonattainment can rule out the existence of the reduced basis itself, not only the termination of an algorithm. A basis of  $\mathcal{A}^n$  means the columns of a matrix in  $\text{GL}_n(\mathcal{A})$ ; an invertible matrix over  $F$  is not a change of basis of the same  $\mathcal{A}$ -module (10).

**Lemma 30.1** (What Gauss inequalities would imply [10]). *Over  $\mathbf{Oz}$ , with  $\beta(u, v) = u^\top Qv$  the bilinear form of the energy  $\mathcal{E}$ , a basis with  $\mathcal{E}(b_1) \leq \mathcal{E}(b_2)$  and  $2|\beta(b_1, b_2)| \leq \mathcal{E}(b_1)$  has  $b_1$  as a shortest nonzero vector.*

*Proof.* For  $s, t \in \mathbf{Oz}$  not both 0,  $\mathcal{E}(sb_1 + tb_2) \geq (s^2 - |st| + t^2)\mathcal{E}(b_1)$ . If one coefficient vanishes the other has modulus at least 1 ([lemma 2.1](#)); otherwise  $s^2 - |st| + t^2 = (|s| - |t|)^2 + |st| \geq 1$ .  $\square$

**Corollary 30.2** (No Gauss-reduced basis [10]). *A binary separated positive pencil with  $L_m = 0$ , for instance (28.1), has no Gauss-reduced  $\mathbf{Oz}$ -basis.*

For a real basis  $b_1, \dots, b_n$  perform finite Gram–Schmidt orthogonalization,  $b_j = b_j^* + \sum_{i < j} \mu_{ji} b_i^*$  with  $\Delta_i = \mathcal{E}(b_i^*) > 0$  (10 writes  $D_i$ ; all divisions are in  $\mathbf{No}$ ). The  $\delta$ -LLL conditions [7] are

$$|\mu_{ji}| \leq \frac{1}{2} \quad (i < j), \quad \Delta_j + \mu_{j,j-1}^2 \Delta_{j-1} \geq \delta \Delta_{j-1} \quad (j \geq 2). \quad (30.1)$$

**Theorem 30.3** (No ordinary-parameter LLL basis [10]). *Let  $\mathcal{E}$  be a separated positive pencil on  $\mathbf{Oz}^n$  with  $L_m = 0$ . For no ordinary  $\delta \in (1/4, 1]$  does an  $\mathbf{Oz}$ -basis satisfy (30.1).*

*Proof.* With  $c = \delta - 1/4 \in (0, 3/4]$  ordinary, the size bound and the Lovász inequality give  $\Delta_j \geq c \Delta_{j-1} \geq c^{j-1} \Delta_1$ . For nonzero  $v = \sum a_i b_i$ ,  $a_i \in \mathbf{Oz}$ , with  $a_k$  the last nonzero coefficient, the  $b_k^*$ -coefficient of  $v$  is  $a_k$ , so  $\mathcal{E}(v) \geq a_k^2 \Delta_k \geq \Delta_k \geq c^{n-1} \mathcal{E}(b_1)$ , using  $|a_k| \geq 1$  and  $c \leq 1$ . The factor  $c^{n-1}$  is ordinary, contradicting [corollary 24.3](#) for  $b_1$ .  $\square$

This is an existence obstruction. No assertion is made about a modified reduction notion with different constants or an infinitesimal gap  $\delta - 1/4$  (10). It applies also to [example 29.4](#), which has a shortest unimodular vector: LLL conditions control every nonzero module vector, including the non-unimodular monomial vectors.

For  $F = \mathbf{No}[i]$  use  $\langle u, v \rangle = u^* Qv$ , conjugate-linear in the first argument,  $\mu_{ji} = \langle b_i^*, b_j \rangle / \Delta_i$ , and the explicitly stated conditions

$$|\Re \mu_{ji}| \leq \frac{1}{2}, \quad |\Im \mu_{ji}| \leq \frac{1}{2} \quad (i < j), \quad \Delta_j + |\mu_{j,j-1}|^2 \Delta_{j-1} \geq \delta \Delta_{j-1} \quad (j \geq 2), \quad (30.2)$$

stated to avoid relying on a convention for complex LLL parameters (10).

**Theorem 30.4** (Gaussian reduction obstruction [10]). *For a separated positive pencil on  $\mathbf{Oz}[i]^n$  with  $L_m = 0$ , no  $\mathbf{Oz}[i]$ -basis satisfies (30.2) for an ordinary  $\delta \in (1/2, 1]$ .*

*Proof.* The size conditions give  $|\mu_{j,j-1}|^2 \leq 1/2$ ; with  $c = \delta - 1/2 \in (0, 1/2]$ ,  $\Delta_j \geq c^{j-1}\Delta_1$ , and  $|a_k| \geq 1$  gives  $\mathcal{E}(v) \geq c^{n-1}\mathcal{E}(b_1)$ , contradicting corollary 24.3.  $\square$

## 31 Closed, uniformly separated Euclidean lattices

For ordinary real  $\alpha$  and positive  $\varepsilon$  let

$$T_{\alpha,\varepsilon} = \begin{pmatrix} \sqrt{1+\varepsilon} & -\alpha/\sqrt{1+\varepsilon} \\ 0 & \sqrt{\varepsilon(1+\alpha^2+\varepsilon)/(1+\varepsilon)} \end{pmatrix}, \quad T_{\alpha,\varepsilon}^* T_{\alpha,\varepsilon} = \begin{pmatrix} 1+\varepsilon & -\alpha \\ -\alpha & \alpha^2+\varepsilon \end{pmatrix},$$

with positive surreal square roots and determinant  $\sqrt{\varepsilon(1+\alpha^2+\varepsilon)} \neq 0$ . The module  $\mathcal{L}_\varepsilon = T_{\sqrt{2},\varepsilon}\mathcal{A}^2 \subseteq F^2$  satisfies  $\|T_{\sqrt{2},\varepsilon}x\|^2 = \mathcal{E}_\varepsilon(x)$ , so the binary example is a literal Euclidean lattice problem in surreal coordinates (10).

**Proposition 31.1** (Uniform separation without a shortest vector [10]). *For positive infinitesimal  $\varepsilon$ ,  $\mathcal{L}_\varepsilon$  is uniformly separated and closed in the  $\mathbf{No}$ -valued Euclidean topology, yet its nonzero distances to 0 have neither a minimum nor a surreal infimum.*

*Proof.*  $\mathcal{E}_\varepsilon(x) \geq \varepsilon \|x\|^2$  and a nonzero  $x \in \mathcal{A}^2$  has a coordinate of modulus at least 1, so distinct points are at distance at least  $\rho = \sqrt{\varepsilon}$ . For  $p \notin \mathcal{L}_\varepsilon$  the ball  $B(p, \rho/3)$  contains at most one lattice point  $\ell$ , and the ball of radius  $\min(\rho/3, \|p - \ell\|/2)$  contains none; so the complement is open, without any set-indexed limit. Squaring is an order isomorphism of the nonnegative surreals, so theorem 28.1 transfers.  $\square$

Lemma 23.2 gives uniform separation for every pencil; the missing ingredient for an attained minimum is ordinary compactness and finiteness, not positive definiteness. No contradiction to a Minkowski theorem with a surreal volume is claimed, since no such volume theory is constructed;  $\mathcal{L}_\varepsilon$  is a proper-class module, and its two generators are a basis over  $\mathcal{A}$ , not over  $\mathbb{Z}$  (10).

## 32 Scope and sharp boundaries

**The full surreal class matters.** The failure of an infimum and the failure of set-sized coinitality are distinct; the latter uses a positive exponent below an arbitrary set and must not be transferred to a fixed Hahn field (10). In the Hahn field  $k((t^{\mathbb{R}}))$  ( $t$  positive infinitesimal, well-ordered supports) let  $A^{\text{Hahn}} = D + k((\mathbb{R}_{<0}))$ , the series with nonpositive exponents and constant coefficient in  $D$  (here  $k((\mathbb{R}_{<0}))$  denotes series supported in  $\mathbb{R}_{<0}$ , not a field). For the binary form with  $\varepsilon = t^\eta$ ,  $\eta > 0$  real, the vectors  $v_n = t^{-1/n}(\sqrt{2}, 1)/\sqrt{3}$  have energies  $t^{\eta-2/n}$ ; every nonzero energy divided by  $\varepsilon$  is still positive infinite, and if its leading term is  $ct^{-d}$ ,  $d > 0$ , then  $n$  with  $2/n < d$  gives  $\mathcal{E}(v_n)/\mathcal{E}(x)$  infinitesimal. So  $(v_n)$  is a countable coinital family there; the energies still have no infimum. In  $\mathbf{No}$  the exponent  $\gamma = 1/\omega$  lies below every  $1/n$ , and  $\omega^{1/\omega}(\sqrt{2}, 1)/\sqrt{3}$  defeats the whole sequence under  $t = \omega^{-1}$ .

**Ordinary coefficient forms are essential.** For the positive-definite surreal form  $q(x, y) = (x - \omega y)^2 + y^2$  and  $v = (\omega, 1) \in \mathbf{Oz}^2$ ,  $q(v) = 1$  but  $q(\text{ct } v) = q(0, 1) = \omega^2 + 1$ : contraction fails as strongly as possible. This form is even reduced by the unimodular  $\mathbf{Oz}$ -basis  $(1, 0), (\omega, 1)$

(both energies 1, orthogonal), so it is not asserted that all positive-definite surreal forms lack reduced bases. A general asymptotic expansion of a positive-definite matrix need not have positive-semidefinite coefficients and cannot be inserted into the hypotheses (10).

**What is and is not uniform.** [Theorem 27.5](#) handles every target, but it does not say that infinitesimal perturbations leave the nearest point unchanged ([example 27.6](#)). The unrestricted theorem improves every set-sized family, not a family indexed by the whole class, and in the unimodular problem even a countable coinital family cannot be improved simultaneously by another unimodular vector. The LLL obstructions need an ordinary parameter strictly above the critical value (10).

## Relations, questions, and records

### 33 Relations to neighbouring reports

- `docs/surreal/omnific-diophantine-geometry` (`odg:`). Part I answers its question `odg:q:affine` for closed subgroup schemes of  $GL_N$  over  $\mathbb{Z}$  (and over  $\mathbb{Z}[i]$  for  $\mathbf{Oz}[i]$ ) and contains `odg:cor:unipotent` and `odg:cor:orthogonal` as special cases ([remark 4.6](#)); [theorem 3.3](#) and `odg:thm:bounded` use the same transfer and neither contains the other. The anisotropic example  $x^2 + y^2 - 3z^2$  is covered by `odg:cor:unipotent` and is printed here only for its rational part ([section 5](#)). Part III's constant-term contraction gives `odg:cor:definite` without Tarski transfer, its monomial witnesses are the mechanism of `odg:thm:cone`, and its unimodular descent is that of `odg:thm:unimodularfermat` and `odg:thm:realray`. The word *primitive* keeps that report's meaning; 10's primitive vectors are called unimodular here.
- `docs/surreal/set-sized-quotients-of-omnific-integers` (`osq:`). [Theorem 9.1](#) strengthens `osq:thm:universal`, `osq:cor:faithful` and `osq:prop:matrices` from ring and module maps to abstract group maps and implies the ring theorem for  $K \in \{\mathbb{R}, \mathbb{C}\}$  ([remark 9.6](#)); its first two proofs use that report's collision lemmas `osq:lem:division` and `osq:lem:collision`. The finite-support boundary ([proposition 12.2](#)) is the group form of its support threshold `osq:thm:support`, and [proposition 11.1](#) is the linear-group form of its finite-side theorem `osq:thm:finite`. Source 11's third proof derives [theorem 9.1\(iii\)](#) from that report's `osq:thm:universal(iii)` ([remark 9.6](#)); 11's set-sized fragments are that report's ring  $R_\kappa^{11}$  (`osq:prop:lex`, `osq:rel:lem:embedding`), and its exact ring threshold `osq:prop:card11`, with `osq:lem:collision`, gives the exact group thresholds of [proposition 21.12](#); the five set-sized models of `osq:thm:thresholds` give [corollary 21.13](#). Two manuscripts of the batch of `f4c9504` have since been merged into that report: manuscript 05, on countable-domain representations detecting the extended ideal, is a ring-level relative of the countable detection of [section 15](#); manuscript 01, on relation modules  $\ker(\omega^\gamma C)$ , is a relative of the rational kernel flags of [Part III](#). Neither is used here.
- `docs/surreal/euclidean-three-space` (`e3:`), Part III (`e3:cut:part`). The standard-part theorems for  $SO(n, \mathbf{No})$  and  $SU(n, \mathbf{No}[i])$  are parallel to [theorem 9.1](#) but neither implies the other ([remark 11.2](#)). Its open question on a uniform reductive classification (`e3:cut:sub:open`) concerns the surreal field and is not answered here.
- `docs/surcomplex/hahn-tate-uniformization` (`tate:`), Part II (`tate:part:theta`). The ordinary backbone of [section 27](#) and the Pell example are there and are cited,

not reprinted ([subsection 27.1](#)); `Surreal/Algebra/LatticeEnergyCertificate.lean` formalizes part of it. The omnific statements of Part III are not in that report.

- `docs/surcomplex/surcomplex-field-automorphisms (saut:)`. The exponent rescaling  $\Phi_h$  of [lemma 8.3](#) is its dilation (`saut:eq:dilation`).
- `docs/surreal/omnific-preserving-automorphisms`, written concurrently from the same batch. Its manuscript on algebraic-parameter rigidity treats algebraic group actions on  $\mathbf{Oz}$ , not points of groups over  $\mathbf{Oz}$ ; it is cited by directory only.

## 34 Questions

The sources call these questions suggested by their proofs, not previously published named open problems (06, 07, 10, 11).

**Question 34.1** (The full special linear groups [06, 07, 11]). For  $n \geq 3$ , is  $\mathrm{SL}_n(\mathbb{Z})$  the universal set-sized quotient of  $\mathrm{SL}_n(\mathbf{Oz})$ , and  $\mathrm{SL}_n(\mathbb{Z}[i])$  that of  $\mathrm{SL}_n(\mathbf{Oz}[i])$ ? More generally, what can set-sized groups detect on  $\mathrm{SL}_n(\mathcal{A}_{D,k})$  outside  $E_n(\mathcal{A}_{D,k})$ , and can the cusp-residue obstruction be organized into a classification of elementary orbits or of extensions of two-row representations?

*Status: open.* Non-elementary matrices and relative algebraic  $K$ -theory are the additional obstruction (06). [Theorem 18.2](#) shows that the issue cannot be bypassed by declaring all unimodular matrices elementary, and [theorem 18.4](#) shows only that one family becomes invisible after one stabilization; it does not classify the remaining obstructions (07). Source 11 asks the first part again (its Question 11.1) and notes that elementary generation, or an independent elimination of the unstable obstruction, would settle it; neither step is hidden in any proof here.

**Question 34.2** (Rank two [06, 11]). Does an analogue of kernel invisibility hold for  $E_2$ ?

*Status: answered negatively by 07* ([theorems 15.3](#) and [16.5](#)): every nonidentity element of  $P_2$  is detected by a countable group, although no finite group detects it. Source 11 asks for the set-sized images of  $E_2(\mathbf{Oz})$ ,  $E_2(\mathbf{Oz}[i])$  and their kernels (its Question 11.2), at a pin where this answer was already in the report: there is no universal set-sized quotient ([corollary 15.5](#)), finite images are arithmetic ([proposition 16.2](#)), and one countable group separates finite sets ([theorem 15.3](#)).

**Question 34.3** (Other root systems [07, 11]). For which elementary Chevalley-type groups over  $\mathcal{A}_{D,k}$  does a collision force the whole relative kernel to have no set-sized images, and which low-rank systems keep additive specializations like [proposition 15.2](#)?

*Status: open.* Other root systems have extra root factors and integer structure constants in their commutator formulas, which a formal analogy does not control (07). Source 11 suggests the same for elementary symplectic and other split Chevalley groups through root-kernel reconstruction, and notes that root lengths and commutator constants must be addressed; it also leaves the kernel of  $\mathrm{St}_n(\mathcal{A}_{D,k}) \rightarrow E_n(\mathcal{A}_{D,k})$  uncomputed ([remark 20.3](#)).

**Question 34.4** (Thresholds at set-sized stages [07; 06]). For a given exponent group and support restriction, what is the least target cardinal detecting a prescribed nonzero relative elementary matrix?

*Status: partly answered [merge].* [Propositions 9.7](#) and [12.1](#) give lower bounds; a matching upper bound needs an actual detecting representation at that stage (06, 07). The ring analogue

is studied in the sibling report’s fixed-group thresholds. Source 11 proves the lower bound  $\kappa$  for the rings of supports of size  $< \kappa$  over  $\bigoplus_{\alpha < \kappa} \mathbb{Q}$  ([theorem 21.3](#)) and an abstract criterion for exactness ([proposition 21.11](#)); combined with the sibling report’s field bound, the least detecting cardinal of every nonidentity relative elementary matrix,  $n \geq 3$ , is exactly  $\kappa^{<\kappa}$  there, and the size of the ring in each of the sibling report’s five set-sized models ([proposition 21.12](#) and [corollary 21.13](#)). Other exponent groups and support restrictions remain open, and nothing is said here about  $n = 2$  at set-sized stages.

**Question 34.5** (Arbitrary positive-definite surreal matrices [10]). Classify shortest-vector existence for arbitrary positive-definite surreal matrices on  $\mathbf{Oz}^n$ , for instance by classifying which changes of coordinates in  $\mathrm{GL}_n(\mathbf{Oz})$  bring a form into a separated positive pencil.

*Status: open.* The form  $(x - \omega y)^2 + y^2$  shows that the kernel-flag criterion does not apply to arbitrary decompositions ([section 32](#)).

**Question 34.6** (Reduction toward cuts [10]). Is there a theory of reduced bases whose targets are cuts rather than attained successive minima, distinguishing changes of basis over  $\mathbf{Oz}$  from scalar-field diagonalization, and not assuming that a shortest unimodular vector controls every nonzero module vector?

*Status: open.*

**Question 34.7** (Fixed exponent groups [10]). In a fixed Hahn field, what is the least size of a coinital family of energies, in terms of the coinitality of the positive exponent group?

*Status: open.* The real-exponent comparison of [section 32](#) suggests the question; an exact theorem must state divisibility and coinitality hypotheses (10).

The question `odg:q:affine` of the omnific Diophantine report is answered here for closed subgroup schemes of  $\mathrm{GL}_N$ . *Status* (batch 31): that report now answers it for geometrically integral curves, except over  $\mathbf{Oz}$  for real curves with normalization  $\mathbb{A}_{\mathbb{R}}^1$  whose integer points all lack a real normalization preimage (`odg:sg:q:real`); it stays open for general affine varieties of dimension at least two ([remark 4.6](#)).

## A Provenance

### A.1 Section by section

Table 2: Provenance of each part of the report.

| Part of this report              | Sources and choices                                                                                                                                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Section 2</a>        | 06 §2, 07 §2, 10 §2, printed once; also in the omnific reports.                                                                                                                                                                                           |
| <a href="#">Sections 3 and 4</a> | 06 §§3–4. The answer to <code>odg:q:affine</code> and the comparison with the omnific Diophantine report are the merge’s ( <a href="#">remark 4.6</a> ).                                                                                                  |
| <a href="#">Section 5</a>        | 06 §5. $\mathrm{O}_n(\mathbf{Oz})$ printed by citation of <code>odg:cor:orthogonal</code> ; the existence of the anisotropic family printed by citation of <code>odg:cor:unipotent</code> ; 06’s rational anisotropy argument and explicit matrices kept. |

| Part of this report                | Sources and choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Section 6</a>          | 06 §6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">Section 7</a>          | 07 §2.3 ( <a href="#">lemma 7.1</a> ); 06 §8.1 (its Propositions 8.1, 8.2: <a href="#">propositions 7.2</a> and <a href="#">7.3</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">Section 8</a>          | 06 §7 (absorption); 07 §3 (compression); both identified with the sibling report's lemmas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">Section 9</a>          | The $n \geq 3$ theorem: 06's Theorem C with its Theorem 8.3 and Corollaries 8.4, 8.5, and 07's Theorem A with its Theorem 4.1, Corollary 4.2 and Proposition 4.3, printed once in 06's generality for $D$ ; both proofs kept. 11's Theorem A (its Theorem 5.2, Lemma 5.3) is the same theorem and is not reprinted; its proof is kept as the third route, with its §4 (Theorem 4.1, Corollaries 4.3, 4.4, Proposition 4.5, Lemma 4.6: <a href="#">theorem 9.2</a> , <a href="#">corollary 9.3</a> , and <a href="#">lemma 9.4</a> ); its §3 (the ring obstruction) is cited as <code>osq:thm:universal</code> and <a href="#">lemma 8.1</a> . 11's Theorem 7.1 and Corollary 7.2 are credited at <a href="#">corollary 9.5</a> . <a href="#">Remark 9.6</a> is the merge's, with 11's converse. <a href="#">Proposition 9.7</a> is 07 §11.1. |
| <a href="#">Section 10</a>         | 06 §9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">Section 11</a>         | The merge ( <a href="#">proposition 11.1</a> ), from an observation of the placement dossier; the comparison with the rotation groups is the merge's.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">Section 12</a>         | 06 §10, 07 §§11.2–11.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">Sections 13 to 18</a>  | 07 §§5–10 (07's Theorems B, C, D). The Mirzaii–Torres and Cohn citations were checked for this report. 11's Theorem 9.1 is credited at <a href="#">theorem 16.4</a> ; its Proposition 11.3 at <a href="#">proposition 12.2</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <a href="#">Section 19</a>         | 11 §6 (Theorems 6.2, 6.3, 6.4, Proposition 6.5, Corollary 6.6; its Lemma 6.1 is <a href="#">lemma 2.1(ii)</a> ), 11 Proposition 7.4, 11 Corollary 9.2 (with the Bass–Milnor–Serre input). 11's Theorem B, printed as these statements about $P_n$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <a href="#">Section 20</a>         | 11 §8 (Lemma 8.1, Theorem 8.2 and its remark).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <a href="#">Section 21</a>         | 11 §10 (Lemma 10.1, Theorem 10.2 as <a href="#">lemma 21.2</a> , Theorem 10.3, Corollary 10.4, Propositions 10.5–10.7, Corollaries 10.8, 10.9, the remark on $\aleph_1$ , Proposition 10.11); 11's Theorem C. The ring is cited from the sibling report. <a href="#">Proposition 21.12</a> and <a href="#">corollary 21.13</a> are the merge's.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">Part III</a>           | 10, all sections. The lattice-projection equivalence, the sufficiency half of the closest-point criterion, the real parity bound and the Pell example are printed by citation of the Hahn–Tate report ( <a href="#">subsection 27.1</a> ); 10's proofs are kept for its converse, the non-closedness, the Gaussian bounds, arbitrary targets and all omnific statements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <a href="#">Sections 33 and 34</a> | The merge, with the sources' questions (06 §10.3, 07 §12.4, 10 §14.1, 11 §11).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Section numbers of the sources refer to their own manuscripts, which are not shipped. Where the sources overlap, the statement with the weakest hypotheses is printed. No source proof needed a hypothesis that the printed statement lacks: 06's proof of the  $n \geq 3$  theorem uses only single-product divisibility, gaps and the geometric inverse, and 07's and 11's use  $D$  only through the split inclusion, so all three apply to every unital  $D \subseteq k$ . 11's structural, Steinberg and fragment results are printed for  $(D, k) \in \{(\mathbb{Z}, \mathbb{R}), (\mathbb{Z}[i], \mathbb{C})\}$ , as 11 states them.

## A.2 Renamed symbols

Table 3: Symbols renamed from the sources.

| Source | Source symbol                                                       | Here, and why                                                                                                                             |
|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 06     | $\mathcal{I}_k$                                                     | $\Pi_k$ , the collection's letter.                                                                                                        |
| 06     | $\mathcal{B}_k, A_{D,k}$                                            | $\mathcal{B}_k, \mathcal{A}_{D,k}$ (unchanged meaning).                                                                                   |
| 06     | $P_n(k)$                                                            | $P_n$ .                                                                                                                                   |
| 06     | $N_a$ (congruence subgroups)                                        | $P_n^{(a)}$ , since 07's kernel was $N_n$ .                                                                                               |
| 06     | $F_r$ (filtration), $U_n$                                           | $\text{Fil}_r$ , $U_n = \text{UT}_n(\Pi_k)$ ; $F$ is the ambient field.                                                                   |
| 06     | $\mathcal{I}_k(\Gamma)$                                             | $\Pi_k(\Gamma)$ .                                                                                                                         |
| 07     | $\mathcal{J}_k, c_0$                                                | $\Pi_k$ , ct.                                                                                                                             |
| 07     | $R_D$                                                               | $\mathcal{A}_{D,k}$ .                                                                                                                     |
| 07     | $N_n$                                                               | $P_n$ .                                                                                                                                   |
| 07     | $K = \text{Frac } D$                                                | $K_D$ , since $K_j$ are 10's kernels.                                                                                                     |
| 07     | $G = \text{SL}_2(D)$                                                | $G_D$ , since $G$ is an algebraic group in Part I.                                                                                        |
| 07     | $T$ (monomial $\omega^{2\delta}$ or $\omega^{2a}$ )                 | $\tau$ , or written out.                                                                                                                  |
| 07     | $T$ (transversal)                                                   | $\mathcal{T}$ .                                                                                                                           |
| 07     | $X, Y$ (finite and infinite points)                                 | $P_{\text{fin}}, P_{\infty}$ ; $X$ is a scheme in Part I.                                                                                 |
| 07     | $L(x)$ (lower root)                                                 | $e_{21}(x)$ ; $L$ is the compressed field.                                                                                                |
| 07     | $P$ (coefficient map), $A, B$ (commutator factors), $P$ (index set) | $\mathfrak{p}, X, Y, \mathcal{P}$ .                                                                                                       |
| 10     | $\mathcal{J}, \mathcal{B}, \mathcal{A}, \mathcal{F}$                | $\Pi_k, \mathcal{B}_k, \mathcal{A}, F$ .                                                                                                  |
| 10     | $E, \mathcal{E}$                                                    | $\mathcal{E}$ (energy), $\mathcal{V}$ (its values), since $E_n$ is the elementary group.                                                  |
| 10     | $\mathcal{O}, \mathfrak{m}$                                         | $\mathcal{O}_{\text{fin}}, \mathfrak{m}$ , as in the omnific Diophantine report.                                                          |
| 10     | primitive, $U(\mathcal{A}^n)$                                       | unimodular, $\text{Um}(\mathcal{A}^n)$ ; <i>primitive</i> keeps its meaning of <code>odg:thm:realray</code> , and $U(x)$ is $e_{12}(x)$ . |
| 10     | $V_j$                                                               | $\text{Low}_j$ ; $V_j$ is the Hahn–Tate radical flag.                                                                                     |
| 10     | $H = W \cap K_{r+1}$                                                | $W'$ ; $H$ is a target group in Part II.                                                                                                  |
| 10     | $D_i$ (Gram–Schmidt)                                                | $\Delta_i$ ; $D$ is the constant ring.                                                                                                    |
| 10     | $B(u, v)$ (Gauss)                                                   | $\beta(u, v)$ ; $B$ is a factor matrix.                                                                                                   |

| Source | Source symbol                                                       | Here, and why                                                                                                                  |
|--------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 11     | $\Pi_k, A_D, \deg_\omega$                                           | $\Pi_k, \mathcal{A}_{D,k}, \deg_\omega$ (unchanged meaning).                                                                   |
| 11     | $N_n(A), U_n(I)$                                                    | $P_n; \langle e_{ij}(I) \rangle$ , equal to $P_n$ for $I = \Pi_k$ ( <a href="#">proposition 7.3</a> ); $U(x)$ is $e_{12}(x)$ . |
| 11     | $K_n(A)$ (Steinberg kernel)                                         | $P_n^{\text{St}}$ ; $K_D$ is $\text{Frac } D$ and $K_j$ are prefix kernels.                                                    |
| 11     | $q = 1 + \omega, N^{(m)}$                                           | written out; $P_n \langle (1 + \omega)^m \rangle$ , since $q_j$ are Part III's forms and $P_n^{(a)}$ is the tower.             |
| 11     | $G, K$ (root-generated group, normal closure)                       | $\Sigma, N'$ ; $G$ is an algebraic group, $K_D$ a field.                                                                       |
| 11     | $T(a), \rho_\varphi$ (root tuple)                                   | $\vartheta_\varphi(a)$ ; $\mathcal{T}$ is a transversal and $\rho_\ell$ a specialization.                                      |
| 11     | $F$ (finite subgroup)                                               | $\Theta$ ; $F$ is the ambient field.                                                                                           |
| 11     | $H$ (exponent bound in its Proposition 6.5)                         | $h_0$ ; $H$ is a target group.                                                                                                 |
| 11     | $B = D \oplus I$ (split ring)                                       | $B$ in <a href="#">lemma 20.1</a> , $A_0 = D_0 \oplus I$ in <a href="#">proposition 21.11</a> .                                |
| 11     | $A_{\kappa,D}, \Pi_{\kappa,k}, N_{n,\kappa,D}$                      | $R_{<\kappa}, \Pi_{<\kappa}, P_{n,<\kappa}$ ; the sibling report's $R_\kappa^{11}, \Pi_\kappa^{11}$ .                          |
| 11     | $\omega^{\iota(\gamma)}$ (outer monomials),<br>$q_\kappa = 1 + t_1$ | $X^\gamma$ , as in the sibling report; $1 + t_1$ written out.                                                                  |

### A.3 Stale repository statements

Sources 06, 07 and 10 pin commits (220784a, fb5c4b5) at which the omnific directories held only the placed base manuscripts of the two omnific reports, with bare labels. The merged reports were written afterwards (bbdd536, 7494d53, 0240140). The following statements are corrected here; the pins are kept as provenance.

1. 06 cites “the geometric collision construction in Section 3” of the quotient article, and 07 its “scaled-field collision”: at the pins these were Section 3 of the placed base manuscript; the collection now prints them as `osq:lem:division` and `osq:lem:collision`.
2. 06 read only the opening lines of the Diophantine article and the main statements of the quotient article at its pin. Its accounts are accurate for the material it read; the collection now contains the merged reports, in which, in particular, the question `odg:q:affine` answered by Part I appears.
3. 06 describes the euclidean-three-space theorem from the catalogue only, as a universal set-sized quotient of  $\text{SO}(3, \mathbf{No})$ ; the report also proves it for  $\text{SO}(n, \mathbf{No})$ ,  $n \geq 3$ , and  $\text{SU}(n, \mathbf{No}[i])$ ,  $n \geq 2$  (`e3:cut:thm:allranks`). 06's statement that the catalogue describes 51 reports was still what `docs/README.md` said when this report was first written; the catalogue has since grown, and lists 56 reports at the time of the addition of source 11.
4. 06 and 07 leave rank two and  $\text{SL}_n$  open; within this report rank two is settled by 07 and  $\text{SL}_2(\mathbf{Oz}) \neq E_2(\mathbf{Oz})$  ([questions 34.1](#) and [34.2](#)).
5. 10 reports that a repository search for “shortest vector” returned no hit and cites the Lean module `LatticeEnergyCertificate.lean` but not the report it formalizes. The search

result is literally true at the pin and today, but the Hahn–Tate report’s Part II, present at the pin, already contains the lattice-projection lemma, the rational-flag minimum criterion, the  $2^n$  parity bound and the same Pell example (subsection 27.1).

6. 07’s remark that the formalization ledger should not be read as covering its results remains true: `docs/FORMALIZATION.md` maps no label of this report and no omnific statement to Lean.

Source 11 pins 9693b28, at which this report had already been written (1e54d5a); its label `ogl:el:thm:invisible` occurs 16 times in the article at that pin. The following statements of 11 are corrected here, the pin being kept as provenance.

7. 11 says that its searches “did not locate these elementary-group or Steinberg-group statements in the inspected material”. For the elementary-group statements this is not so: its main theorem, the finite residual, the finite-support counterexample and its rank-two question were all in this report at the pin. The report was then missing from the catalogue, which 11 inspected: `docs/README.md` at the pin describes 51 reports while the tree had more report directories, and has no row for this one. For the Steinberg statements the sentence is accurate; no Steinberg group appears elsewhere in the collection.
8. 11’s question on rank two (its Question 11.2) was answered within this report at its pin (question 34.2).
9. 11 deliberately does not identify the root-generated subgroup with the kernel or assert that it is normal; propositions 7.2 and 7.3 (06), present at the pin, prove both. 11’s proofs never depend on the identification, so nothing in them changes.
10. 11 says that its fragments are “related to the set-sized support constructions already studied” in the sibling report. They are the same ring with the same exponent group (`osq:prop:lex`, present at the pin); the embedding in **Oz** that 11 gives was added to that report after the pin (`osq:rel:lem:embedding`).
11. 11 says that the sibling report’s exact ring-threshold constructions “motivate” its exact-threshold criterion. One of them, `osq:prop:card11`, present at the pin, satisfies its hypothesis after the collision argument of proposition 21.12.
12. 11’s descriptions of the quotient report (an unrefereed AI-assisted merge dated 22–23 September 2026 whose ring theorem factors every set-sized ring image through `ct`) and of the catalogue (51 reports) are accurate for the pin.

## B What this report does not claim

Every limitation, disclaimer and priority caveat stated by a source is kept here, numbered per source.

### B.1 Source 06 (23 items)

- 06.1 An AI-assisted, proof-bearing research draft; not peer reviewed and not verified in Lean.
- 06.2 The combined main statements are proposed contributions, not claims of verified historical priority; they were not located in the inspected sources, which is a qualified novelty assessment.

- 06.3 No named longstanding conjecture is claimed settled.
- 06.4 Normal forms, real closedness, torus structure, nilpotent exponentials, Cartan theory and elementary commutator identities are classical.
- 06.5 The constant-term formalism, common monomial divisibility and geometric collision identities already occur in the repository; the needed forms are reproved, not imported as correct from an unreviewed draft.
- 06.6 The unitriangular and Lie-algebra descriptions are specializations of standard machinery, not independent major discoveries.
- 06.7 The geometric-series identity is a formal identity of normal forms, not convergence in the fine topology.
- 06.8 The only transfer is between real closed fields; no elementary equivalence of  $\mathbb{Z}$  and  $\mathbf{Oz}$ , and no transfer in a language with an omnific predicate, is asserted.
- 06.9 Only the characteristic-zero fiber of an integral model is used; smoothness at the primes is not needed, and ordinary integral points depend on the model.
- 06.10 A conjugate-transpose unitary group is a different real group from the complex orthogonal group of the Gaussian example.
- 06.11 The anisotropic example does not contradict the rigidity of nonzero binary norm fibers.
- 06.12 The comparison table concerns constant-term rigidity, not finite generation of arithmetic subgroups or elementary generation of all omnific matrices.
- 06.13 No convergence of an exponential series at an infinite argument is asserted; all displayed matrix functions are finite polynomials.
- 06.14 The kernel theorem is not a statement about all purely infinite matrix groups:  $\mathrm{UT}_n(\Pi_k)$  has nontrivial set-sized characters.
- 06.15  $E_n(D) = \mathrm{SL}_n(D)$  is used only for ordinary  $D$ ; no Euclidean algorithm in  $\mathbf{Oz}$  is assumed.
- 06.16 The theorem does not make  $P_n$  trivial; its tautological proper-class representation is faithful.
- 06.17 No simplicity claim: the tower shows  $P_n$  is not simple, and no numerical size is assigned to  $P_n$ .
- 06.18 Invisibility fails in every nontrivial fixed set-sized Hahn workspace; the local cardinal form is a conditional lower bound, and Archimedean exponent groups supply no suitable scales.
- 06.19 Finite supports admit a visible image onto  $\mathrm{SL}_n(k)$ .
- 06.20 Nothing is asserted about rank two or about  $\mathrm{SL}_n(\mathcal{A})$ ; no stable-range theorem is used, and the two questions left are not designated open problems in the literature.
- 06.21 Neither an unproved omnific factorization conjecture nor a global analytic theory on  $\mathbf{No}[i]$  is used; published factorization work is context, not a hidden hypothesis.
- 06.22 The repository was read at its pin through its hosted interface, not modified, cloned or built; the inspection was targeted, and the literature search cannot exclude an earlier result in other terminology.

06.23 The finite checks do not verify the Hahn support lemma, support gaps, the proper-class collision, Cartan's theorem or the universal quantifiers; the integer filtration instances are not representations of arbitrary elements of  $\Pi_k$ .

## B.2 Source 07 (20 items)

- 07.1 The results concern  $E_n$  (finite products of elementary matrices), not all of  $\mathrm{SL}_n$ ; the questions for  $\mathrm{SL}_n(\mathcal{A}_{D,k})$  beyond the explicit non-elementary matrices are open.
- 07.2 The two-row presentation is a specialized form of Cohn's classical phenomena; the broad distinction between two rows and larger sizes is not new.
- 07.3  $E_2(\mathbf{Oz})^{\mathrm{ab}} \cong \mathbb{Z}/12 \oplus \Pi$  is a classical specialization, not a novelty claim.
- 07.4 The Cohn matrix pattern and the block identity are not claimed as new.
- 07.5 The proposed contribution is the omnific detection package; priority is not established; the literature and repository check was limited; no named historical factorization conjecture is claimed resolved.
- 07.6 No peer review and no Lean verification.
- 07.7 The scaled-field collision is a prior repository ingredient; no unverified repository theorem is imported as an axiom.
- 07.8 There is no single faithful countable representation, only elementwise separation by varying maps; the separating maps are not a set-indexed family with faithful product.
- 07.9 The coefficient functional is an existence construction, not a computability claim.
- 07.10 No congruence subgroup property is asserted or used.
- 07.11 No numerical description of  $\mathrm{SL}_2(\mathbb{Z}[i])^{\mathrm{ab}}$  is given.
- 07.12 The stabilization identity itself does not depend on surreals, and one-step stabilization does not classify the remaining obstructions.
- 07.13 In a fixed set-sized Hahn workspace the compressed field need not lie in a preassigned subfield or birthday-bounded fragment; the optimal threshold at a set-sized stage is not determined.
- 07.14 No classification of homomorphisms from  $\mathrm{SL}_2(\mathcal{A}_{D,k})$ , higher  $\mathrm{SL}_n$ , or unstable  $K$ -theory obstructions.
- 07.15 The class statements concern class homomorphisms existing in the chosen foundational setting, not externally described maps in a larger metatheory.
- 07.16 No analytic assumption: no continuity, fine-topology limits, global complex exponential, analytic uniqueness principle or differential equation.
- 07.17 The tautological representation on the proper-class space  $F^n$  remains faithful and is not covered.
- 07.18 The finite checks validate finite identities and examples only, not the normal-form theorem, the foundations, compression or the group arguments; the PDF build checks typesetting only.
- 07.19 The formalization ledger does not cover these results.
- 07.20 Global choice is part of the stated framework but is not used.

### B.3 Source 10 (21 items)

- 10.1 The results concern separated positive pencils only, not arbitrary positive-definite surreal matrices;  $(x - \omega y)^2 + y^2$  defeats contraction and even has a reduced basis.
- 10.2 The strongest coinitality assertion uses the full surreal class and fails in fixed Hahn fields such as  $k((t^{\mathbb{R}}))$ .
- 10.3 The ordinary higher-rank Voronoi background (Amini–Nicolussi) is not a discovery; the rational-flag sufficiency is a specialization of it.
- 10.4 A bounded literature comparison, not an exhaustive certification of novelty.
- 10.5 No named factorization conjecture is resolved; the primality of  $\omega^{\sqrt{2}} + \omega + 1$  is L’Innocente–Mantova’s Theorem B.
- 10.6 Normal forms and the constant-term retraction are classical, not new.
- 10.7 The converse “ $\text{ct}(x)$  unimodular implies  $x$  unimodular” is not asserted.
- 10.8 The simultaneous improvement is not a sequential limit statement.
- 10.9 The parity bound is a standard finite-lattice argument, not a priority claim.
- 10.10 The finite candidate reduction is not an algorithm for arbitrary surreal inputs.
- 10.11 The Pell identities come from the recurrence, not a classification of Pell solutions over  $\mathbf{Oz}$ ; an infinite index in the recurrence justifies no limit.
- 10.12 The hierarchy table is not a classification of arbitrary surreal forms.
- 10.13 The LLL obstruction concerns an ordinary parameter above the critical value; modified reduction notions and infinitesimal gaps are not analyzed.
- 10.14 No surreal volume or Minkowski theory; the module is a proper class, and its generators are a basis over  $\mathcal{A}$ , not over  $\mathbb{Z}$ .
- 10.15 Set-sized families are improved, not class-indexed ones; in the unimodular problem even a countable coinital family cannot be improved simultaneously.
- 10.16 Ordinary coefficient forms are essential; general asymptotic expansions need not have positive-semidefinite coefficients.
- 10.17 Infinitesimal target perturbations can change the nearest point.
- 10.18 A research draft, not peer reviewed or proof-assistant verified; no Lean code, build or audit for these theorems; related Lean modules are not a certificate.
- 10.19 The finite checks verify exact identities, not the class-theoretic or minimization theorems; no floating-point number stands in for an infinitesimal.
- 10.20 The repository comparison was targeted; a search without hits is not proof of absence.
- 10.21 The further directions are not named open problems from the literature.

## B.4 Source 11 (24 items)

- 11.1 An unrefereed, AI-assisted, proof-bearing manuscript; its proofs have not been independently certified or formalized in Lean, and existing Lean coverage in the repository is not verification of it.
- 11.2 Its principal group-theoretic results are proposed as original extensions of the cited repository work; priority is not certified.
- 11.3 No named longstanding conjecture is claimed solved.
- 11.4 The repository comparison was limited to the catalogue, the introduction and abstract of the quotient report and targeted searches; not all 51 reports or all source manuscripts were audited, the repository was not modified, and the absence of a search result is not a priority certificate.
- 11.5 The small-ring-image theorem, including its nonunital form, belongs to the repository's quotient report; it is reproved, not claimed, and citing it does not certify the rest of that report.
- 11.6 Recovering coefficient algebra from root commutators has substantial precedent (Voronetsky, including the earlier type- $A$  case), and the normal-generation machinery is standard; neither is claimed as new, and the finite-tuple lemma does not import that paper's deeper results.
- 11.7 The Bass–Milnor–Serre theorem is imported, not reproved; only the preceding factorization is proved.
- 11.8 The main theorems concern  $E_n(A)$ , not  $\mathrm{SL}_n(A)$ ;  $E_n(\mathbf{O}\mathbf{z}) = \mathrm{SL}_n(\mathbf{O}\mathbf{z})$  is not established, nor its Gaussian counterpart, and the question on  $\mathrm{SL}_n(A)$  is generated by the work, not a verified literature problem.
- 11.9 Rank two is left separate: neither failure nor a solution is asserted, and  $n \geq 3$  is a proved range, not a claim that  $n = 2$  is a sharp negative boundary.
- 11.10 The Steinberg constant-term kernel is not  $K_2$ ; no  $K_2$  is calculated, no universal central extension in any unstable rank and no excision theorem is claimed, and nothing about algebraic  $K$ -theory in general is implied.
- 11.11 The Steinberg kernel is not claimed centerless or torsion-free, and it differs from the kernel of  $\mathrm{St}_n(A) \rightarrow E_n(A)$ , which is not computed.
- 11.12  $\mathrm{St}_n(D)$  must not be replaced by  $\mathrm{SL}_n(D)$  in the Steinberg theorem.
- 11.13 Triviality of all class-sized representations is not claimed: the inclusion into  $\mathrm{GL}_n(\mathbf{No})$  or  $\mathrm{GL}_n(\mathbf{No}[i])$  is faithful.
- 11.14 “Arithmetic” means only factorization through  $\mathrm{SL}_n(D)$ ; the representations of  $\mathrm{SL}_n(D)$  are not classified or asserted to be algebraic.
- 11.15 It is not claimed that every finite subgroup is conjugate to a constant subgroup.
- 11.16 Invisibility does not pass to arbitrary subgroups: a root subgroup has nonzero coefficient maps to  $\mathbb{R}$ .
- 11.17 Other root systems do not follow by a change of notation; commutator constants and root lengths must be addressed.

- 11.18 Exact image thresholds are not claimed for the explicit fragments, neither  $\kappa$  nor  $\kappa^{<\kappa}$ ; any application of the exact-threshold criterion must verify its ring hypothesis. (The merge supplies this for  $R_{<\kappa}$  in [proposition 21.12](#); the non-claim describes 11.)
- 11.19 The fragment theorems are not extended to singular cardinals or to finite supports.
- 11.20 At  $\aleph_1$  nothing excludes an action on a countably infinite set or every finite-dimensional complex representation.
- 11.21 No isomorphism of  $E_n(\widehat{\mathbf{Oz}[i]})$  with  $\mathrm{SL}_n(\widehat{\mathbb{Z}[i]})$  is asserted.
- 11.22 The congruence chain says nothing about intersections over the class of all principal ideals, and no quotient ring  $A/aA$  is claimed to be a set.
- 11.23 No compactness theorem for proper classes, no set of all class homomorphisms, no fine-topology limit (strongly summable sums are coefficientwise Hahn sums) and no identification of an arbitrary integer-part matrix group with its elementary subgroup is used; class homomorphisms are handled one at a time, and the hom-set bijections abbreviate factorization principles.
- 11.24 The finite checks test signs, indices, finite matrix identities and finite geometric remainders; they do not verify arbitrary Hahn supports, cardinal statements, class algebra or the universal assertions. Its proof audit is the manuscript's own review, not an independent referee report, and its build report is a rendering report, not a correctness certificate.

## B.5 The merge (10 items)

- M.1 The results marked [merge] ([remarks 4.6](#) and [9.6](#), [propositions 11.1](#) and [21.12](#), and [corollary 21.13](#)) are no more refereed than the sources.
- M.2 [Remark 9.6](#) is a logical implication, not an independent proof of the ring theorem, and covers  $K \in \{\mathbb{R}, \mathbb{C}\}$  only.
- M.3 Part I answers `odg:q:affine` only for closed subgroup schemes of  $\mathrm{GL}_N$ , not for general affine varieties or curves.
- M.4 In the nonrigid case of Part I, no description of all omnific points is claimed; only injective families through the integral points are constructed.
- M.5 Neither [theorem 9.1](#) nor [proposition 11.1](#) implies the rotation-group theorems of the euclidean-three-space report, or conversely; nothing is claimed for  $n = 2$  on the finite side.
- M.6 Nothing in this report is formalized in Lean; `docs/FORMALIZATION.md` maps none of its labels.
- M.7 The verification of the Mirzaii–Torres and Cohn citations and of the Milne theorem numbers checks citations, not proofs; 06's reference to real rank and relative root groups in Milne's *Reductive Groups*, Sections 22–23, was not checked.
- M.8 [Proposition 21.12](#) and [corollary 21.13](#) answer [question 34.4](#) only for  $n \geq 3$ , only for the rings named there, and only for the least cardinality of a detecting group; they rest on statements of the sibling report (`osq:prop:card11`, `osq:lem:collision` and, for [corollary 21.13](#), the properties established in the proof of `osq:thm:thresholds`), whose review status is that report's.

- M.9 Source 11’s third proof of [theorem 9.1](#)(iii) is not independent of the collision identities of the other two: it imports the ring theorem. The equivalence recorded in [remark 9.6](#) is an equivalence of statements, not a new proof of either.
- M.10 The material added from source 11 was not reviewed beyond this integration: the proofs were re-derived for the write, 11’s finite checks were rerun, and its Bass–Milnor–Serre and Voronetsky citations were not checked against the originals for this report.

## C Verification suites

Four finite programs accompany the report, as shipped (byte-identical to the deliveries). The first three need Python 3 and SymPy (the recorded runs used Python 3.13.5 and SymPy 1.14.0); source 11’s needs only the Python standard library (version 3.10 or later). All four were rerun for this report on copies in a scratch directory with Python 3.14.4 (and SymPy 1.14.0).

- `code/06-algebraic-groups-verification.py` (06): 1,108 exact checks in ten families (908 elementary-identity checks in ranks 3–6, 18 for the two orthogonal examples [\(5.1\)](#) and [\(5.2\)](#), 139 unitriangular instances, 43 geometric remainders and finite-support augmentations). It prints to standard output and writes a file only with `-output`; the recorded run is `data/06-algebraic-groups-verification-results.txt`.
- `code/07-matrix-dichotomy-checks.py` (07): 23 checks, including 120 exact reduced alternating words in the countable target, the Steinberg identities, [\(17.1\)](#), the commutator of [example 15.4](#), the determinant and square-zero identities of  $M_\alpha$ , and the stabilized commutator. It always writes `checks-results.json` next to itself; the recorded run is `data/07-matrix-dichotomy-checks-results.json` with `data/07-matrix-dichotomy-checks-output.txt`. The rerun reproduced it except for the recorded Python version.
- `code/10-shortest-vectors-verify.py` (10): six groups of checks (the Cholesky identity, the Pell recurrence and 81 norms, determinants and gcds for indices 0–80, the Gram–Schmidt identity, the three-dimensional examples, and the perturbed-target differences of [example 27.6](#)); standard output only, recorded in `data/10-shortest-vectors-verification-output.txt`.
- `code/11-matrix-shadows-verify.py` (11): 2,111 deterministic exact checks (seed 20260923) in twelve families, with exact fractions and finite Laurent polynomials over the lexicographic group  $\mathbb{Q} \oplus \mathbb{Q}$ : root additivity (272), root inverses (272), root commutators (840), constant-term multiplication and nonnegative supports (200 each), word inverses, conjugated commutators, constant terms of conjugated roots and matrix constant-term multiplication (40 each), exact geometric remainders and positivity of the quotient supports (81 each, the identity  $(X^{2e_1} - X^{e_1}) \sum_{r=0}^M X^{e_0 - (r+2)e_1} = X^{e_0} - X^{e_0 - (M+1)e_1}$ ), and the Whitehead diagonal identity over rational units (5). It prints its results and always overwrites `verification_results.json` next to itself (under the shipped name, `code/verification_results.json`); the recorded run is `data/11-matrix-shadows-verification_results.json`, and the rerun reproduced its content exactly.

These programs check finite identities and examples. They do not verify Hahn summability, support gaps, class-size or cardinal arguments, algebraic-group structure, the reduced-word arguments, or any minimization or coinitality theorem, which are proved in the text.

## References

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